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CliniCause: Constructing Clinically Grounded Observational Testbeds for Causal Analysis from Irregular ICU Time Series

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תקציר

בנתונים תצפיתיים אין אמת־מידה נצפית להערכת אומד סיבתי, משום שהתוצאה הנגד־עובדתית אינה נצפית. לעומת זאת, סביבות הערכה מבוקרות משיגות תשובה ידועה באמצעות פישוט או יצירה של חלקים מתהליך יצירת הנתונים. CliniCause בונה סביבות מבחן תצפיתיות משלימות לניתוח סיבתי, המשמרות מדידות לא־סדירות המתועדות במקור ביחידות טיפול נמרץ, דפוסי חסר, משתנים נצפים ותמותה, ומציגות במפורש את הייצוגים וההנחות שהוגדרו במחקר.

מודל שפה גדול, שעוגן בסכמות הנתונים ובספרות, סייע רק בשלב התכנון בהצעת מצבי־פרוקסי, משפחות כללים וגרפים סיבתיים; הוא לא נחשף לרשומות מטופלים ולא אמד ניגודים. הצעות שנבחרו במסגרת הפרויקט קודדו בכללים ובגרפים דטרמיניסטיים. שני המשאבים הנפרדים כוללים 26,845 רשומות ותשעה מצבי־פרוקסי המשמשים כחשיפות אנליטיות ב־MIMIC-III, ו־7,993 רשומות ועשרה מצבים כאלה ב־PhysioNet 2012. בכל משאב נבחנו ארבעה מודלי סדרות זמן—STraTS, GRU, GRU-D ו־TCN—ושלושה אומדים סיבתיים: CausalForestDML כאומד ראשי, LinearDML כאומד השוואה משני ו־CausalPFN כאומד חקרני, ללא איגום בין המאגרים.

בחיזוי תוויות הפרוקסי היה טווח AUROC במבחן 0.867–0.918; STraTS הוביל במדדי MIMIC הארכיוניים, ו־GRU-D הוביל במדדי PhysioNet. CausalForestDML ו־LinearDML הסכימו בכיוון הניגוד שנאמד במודל בכל 19 ההשוואות בין מאגר לחשיפה, ושלושת האומדים הסכימו ב־18 מתוך 19. הלם ב־PhysioNet היה החריג: שני אומדני ה־DML היו שליליים, ואומדן CausalPFN היה חיובי מעט. השוואה הקשרית לספרות הקלינית הצביעה הן על חפיפה רחבה והן על פערים מהותיים. הסכמת האומדים התקיימה לצד מגבלות התאמה ותמיכה, שינויים בעקבות דגימת־חסר מבוססת־תוצאה ורגישות לערבול עקב משתנים שהושמטו.

לאף אחת מסביבות המבחן אין אמת סיבתית ידועה, ומצבי־הפרוקסי, הגרפים והניגודים שנאמדו במודל לא עברו תיקוף קליני. התוצאות אינן מבססות השפעות טיפול, המלצות טיפול או מוכנות להטמעה. אינטגרציה, חוזי נתונים מפורשים, שכבת ייצוג דטרמיניסטית, בדיקות אבחון ותיעוד מקור ועקיבות מאפשרים לבחון את המשאבים. תרומתם היא בהשלמת סביבות הערכה מבוקרות באמצעות חשיפת יציבות האומדים, אי־הסכמות, מגבלות תמיכה התלויות בייצוג ורגישות, תוך שימור המדידות, דפוסי החסר, עקבות תהליכי הטיפול, המשתנים הנצפים והתמותה המתועדים במקור.

Abstract

Real-world causal estimators lack an observed counterfactual answer key, while controlled benchmarks obtain known answers by simplifying or generating parts of the data-generating process. CliniCause constructs complementary observational causal-analysis testbeds that preserve source-recorded irregular intensive-care measurements, missingness, covariates, and mortality while making researcher-defined representations and assumptions explicit.

A schema- and literature-grounded large language model assisted only at design time by proposing dataset-specific proxy constructs, rule families, and causal graphs; it did not access patient records or estimate effects. Project-selected proposals were encoded as deterministic rules and graphs. Separate MIMIC-III and PhysioNet 2012 resources contain 26,845 records with nine analyzed proxy exposures and 7,993 records with ten, respectively. Four time-series models—STraTS, GRU, GRU-D, and TCN—and three causal estimators—primary CausalForestDML, secondary LinearDML, and exploratory CausalPFN—were evaluated without pooling the datasets.

Proxy-label test AUROC ranged from 0.867 to 0.918; STraTS led the archived MIMIC metrics and GRU-D led the PhysioNet metrics. CausalForestDML and LinearDML agreed in the direction of model-estimated contrasts in all 19 dataset–exposure comparisons, and all three estimators agreed in 18 of 19. PhysioNet shock was the exception: the two DML estimates were negative and CausalPFN was slightly positive. Contextual clinical-literature comparison showed both broad overlap and material discrepancies. Estimator agreement coexisted with matching and support limitations, outcome-downsampling changes, and omitted-confounding sensitivity.

Neither testbed has known causal ground truth, and the proxy constructs, graphs, and model-estimated contrasts have not been clinically validated. The results do not establish treatment effects, treatment recommendations, or deployment readiness. Integration, explicit data contracts, deterministic representation, diagnostics, and provenance make the resources inspectable; their contribution is to complement controlled benchmarks by exposing estimator stability, disagreement, representation-dependent support limitations, and sensitivity while retaining source-recorded measurements, missingness patterns, care-process traces, covariates, and mortality.

Keywords

English Keywords

Observational causal-analysis testbeds; causal-estimator evaluation; representation-layer construction; irregular ICU time series; proxy states; LLM-assisted knowledge elicitation; causal machine learning; double machine learning; causal forests; omitted-variable sensitivity; MIMIC-III; PhysioNet 2012.

מילות מפתח

סביבות מבחן תצפיתיות לניתוח סיבתי; הערכת אומדים סיבתיים; בניית שכבת ייצוג; סדרות זמן לא־סדירות מטיפול נמרץ; מצבי־פרוקסי; הפקת ידע בסיוע LLM; למידת מכונה סיבתית; למידת מכונה כפולה; יערות סיבתיים; רגישות למשתנים שהושמטו; MIMIC-III; PhysioNet 2012.

Contents

Abbreviations

Abbreviation	Meaning in this thesis
ICU	Intensive Care Unit.
EHR	Electronic Health Record.
DAG	Directed Acyclic Graph.
LLM	Large Language Model; used in this thesis for design-time knowledge elicitation, not runtime patient-level inference.
STraTS	Self-Supervised Transformer for Sparse and Irregularly Sampled Multivariate Clinical Time Series.
GRU	Gated Recurrent Unit.
GRU-D	Gated Recurrent Unit with Decay.
TCN	Temporal Convolutional Network.
DML	Double Machine Learning.
CATE	Conditional Average Treatment Effect; reported here only as a model-estimated quantity under the stated assumptions.
HTE	Heterogeneous Treatment Effect.
AUROC	Area Under the Receiver Operating Characteristic Curve.
AUPRC	Area Under the Precision–Recall Curve.
minRP	The maximum over thresholds of the minimum of precision and recall.
SOFA	Sequential Organ Failure Assessment.
ARDS	Acute Respiratory Distress Syndrome.
CausalPFN	Exploratory prior-fitted causal estimator whose primary method source is cited; the exact producing package version and checkpoint remain unresolved.

Notation and Symbols

Symbol	Meaning in this thesis
i	Index of a normalized analysis record.
n_i	Number of observed measurement events for record i.
t_{ij}	Elapsed time in minutes for event j of record i.
v_{ij}	Observed variable name for event j of record i.
x_{ij}	Numeric value recorded for event j of record i.
$\mathcal{V}$	Set of observed measurement-variable names.
$\mathcal{O}_i$	Long-format measurement-event object for record i.
L_{ik}^{rule}	Rule-derived binary proxy label for record i and proxy state k.
$\hat{p}_{ik}^{(m)}$	Probability for proxy state k exported by model m for record i.
$\hat{L}_{ik}^{(m)}$	Binary predicted proxy label corresponding to $\hat{p}_{ik}^{(m)}$.
$\widetilde{L}_{ik}$	Aggregated predicted proxy-label representation; in the archived final causal runs, deterministic voting combined one rule-derived source with four aligned binary model-prediction sources.
A_i	Selected binary proxy-state exposure for record i.
Y_i	In-hospital mortality outcome for record i.
W_i	Observed adjustment variables selected from the project-specified DAG and available columns.
X_i	Effect-modifier features used to describe heterogeneity.
$Y_i(a)$	Conceptual potential outcome under exposure value a; its interpretation requires the assumptions stated in the methods.

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Chapter 1

Introduction

1.0 Motivation: From Randomized Trials to Observational Data

Clinical causal questions ask whether an exposure causes a change in an outcome, not merely whether the two co-occur. For example, do ICU patients who develop an acute physiological illness state such as shock have higher mortality because of the state itself, or because more severely ill patients are both more likely to develop shock and more likely to die regardless? This confounding by indication makes a naive comparison between exposed and unexposed patients misleading.

Randomized controlled trials address baseline confounding through their assignment mechanism. In expectation, randomization makes treatment assignment independent of baseline potential outcomes and thereby removes baseline confounding by design. Under the trial protocol and the required consistency, no-interference, adherence/estimand, and outcome-observation conditions, causal contrasts can then be identified without invoking the observational assumption of no unmeasured confounding [30]. Randomization does not make those other conditions disappear. Moreover, trials can be expensive and slow, can face ethical restrictions and narrow eligibility criteria that limit external validity, and can be infeasible when prospective assignment must occur during an acute life-threatening episode.

Large retrospective collections of routinely collected clinical data offer a complementary opportunity: they preserve detailed physiological trajectories, care-process measurements, and outcomes without the cost and ethical constraints of trials. But retrospective causal interpretation requires assumptions about exposure assignment, measurement, selection, and support that are neither guaranteed nor directly testable from the observed data alone. Abundant records improve finite-sample estimation without resolving these identification conditions. Association in observational data is not causation, and knowing that two variables co-occur does not tell us what would happen if we intervened on one of them.

A further difficulty arises even for researchers who accept the identification assumptions and fit a causal estimator: there is no straightforward way to know whether the estimator worked. In supervised prediction, held-out labels reveal prediction error. In causal estimation, the held-out data never reveal the counterfactual outcome—the outcome a patient would have had under the exposure condition they did not experience. Causal estimators therefore cannot be evaluated the way predictive models can, and the field has not converged on a substitute.

The dominant response has been to evaluate causal estimators on synthetic or semi-synthetic benchmarks, where the ground truth is known by construction. Fully synthetic studies specify the entire data-generating process; semi-synthetic studies such as IHDP retain real covariate distributions while simulating assignments and outcomes [16]. These benchmarks enable direct error measurement, but they also determine what difficulty is present: smooth confounding surfaces, fully observed covariates, and well-behaved propensity scores may not reproduce the irregular measurement, informative missingness, care-process confounding, and near-boundary support of real ICU data. Performance on a benchmark does not automatically predict performance in a real clinical setting.

The exposures studied in this thesis are proxy states of this kind: illness states derived from available ICU measurements rather than directly manipulable treatment interventions. The causal analyses concern adjusted outcome contrasts under a proxy-state exposure, not the effect of assigning a treatment. This distinction recurs in the methodological definitions and limitations because different physiological mechanisms can produce the same observed proxy state.

In this thesis, *answer-key causal benchmark* denotes an evaluation resource that provides a known or externally anchored causal target against which estimation error can be assessed. *Observational testbed* denotes a structured resource for exercising and characterizing causal-analysis procedures without claiming a known causal answer. CliniCause is an observational testbed rather than an answer-key causal benchmark: it retains source-recorded ICU structure while making the representation layer—proxy constructs, DAGs, adjustment sets, cohort contracts—explicit and inspectable. The specific datasets, two-dataset design rationale, and evaluation framework are introduced in Sections 1.1–1.3 and Chapter 2.

1.1 The Causal-Estimator Evaluation Problem

For each unit i, let $Y_i(0)$ and $Y_i(1)$ denote the potential outcomes under the two exposure conditions. Only the outcome under the condition that occurred is observed. Population targets such as the average treatment effect, $\mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)]$, and the conditional average treatment effect, $\mathrm{CATE}(w) = \mathbb{E}[Y(1) - Y(0) \mid W = w]$, aggregate this unobserved individual contrast and require identification assumptions; they do not reveal the missing individual counterfactual [31].

Three distinct questions follow. *Identification* asks whether the causal estimand is a function of the observed-data distribution under the assumed causal model. *Estimation* asks how well a procedure estimates that identified functional from finite data. *Evaluation* asks how one can know whether the estimator performed well without ever observing counterfactual truth. Modern causal machine learning addresses the second problem once identification is assumed.

CliniCause is concerned with the third; it does not solve the first.

This evaluation problem is structurally different from ordinary supervised prediction. A held-out prediction set supplies observed labels against which predictions can be scored, but a held-out causal-analysis set still does not reveal $Y_i(1 - A_i)$. Observational records therefore provide neither a direct individual answer key nor the true exposure-assignment process and extent of unmeasured confounding. Evaluation must either construct a synthetic answer key by controlling the data-generating process or characterize estimator behavior through indirect evidence.

Section 2.1 surveys how existing evaluation designs navigate this trade-off; the benchmark gap they leave motivates the approach taken in this thesis.

1.2 Observational Fidelity in Irregular ICU Records

Intensive-care records combine laboratories, vital signs, treatments, administrative fields, and outcomes observed at nonuniform times and for different purposes. Variable-specific sampling rates, pervasive missingness, and measurement intensity shaped by clinical concern or workflow make these records high dimensional and heterogeneous. Irregular medical time-series methods must therefore represent timing, absence, and changing observation availability while preserving a defensible link to the analytical question [1].

Regular resampling, imputation, and summary aggregation choose a temporal resolution, which observations count, and how absence is represented. Sparse-event or missingness-aware models preserve different aspects of the observation process, but exploiting masks, gaps, or measurement intensity predictively does not remove their causal implications. Missingness and irregularity can affect the representation, cohort, rule-derived labels, and adjustment; this thesis therefore treats them as motivations for explicit design and limitations, not as local causal findings.

Two complementary needs follow. First, an analysis must accommodate irregular longitudinal measurements rather than force all observations into an unqualified regular grid. Architectures for sparse clinical series and missingness-aware recurrent models illustrate different ways to use values, time intervals, and observation patterns [2, 3]. Second, the analysis needs interpretable variables that can connect a rich event stream to later questions about prognosis and adjusted mortality contrasts. In this thesis, that intermediate layer consists of *clinically inspired proxy states*: rule-derived analytical constructs that summarize evidence for selected clinical patterns from available measurements.

This wording is deliberate. A rule-derived proxy label is not a chart-adjudicated diagnosis, ground truth, or validated phenotype: a positive tag records that project-specific rules fired on available data, while a negative tag does not establish absence of a condition. The transparent proxy layer supplies multi-label prediction targets and documented observational exposures while retaining a visible link to their measurements and rules. Electronic-phenotyping research provides context for this design choice but does not validate the local constructs [4].

Prediction and causal analysis then answer different questions. Predicting a rule-derived proxy label asks whether an irregular-time-series model can reproduce that label from the available inputs. Estimating an adjusted mortality contrast for a proxy-state exposure asks a separate observational question about modeled differences in outcome under explicit assumptions. Strong performance on the first task does not establish the clinical validity of the label, causal validity of the second analysis, or actionability of the exposure. The latter task requires a stated causal model, project-specified directed acyclic graphs (DAGs), exposure-specific adjustment logic, attention to support and overlap, sensitivity and permutation diagnostics, and traceable evidence. Even with these elements, adjusted estimates remain conditional on the adequacy of the proxy measurement, graph, temporal ordering, observed covariates, and model specification [5, 6, 7].

The framework is instantiated separately on PhysioNet 2012 and MIMIC-III [8, 9]. Their eras, scales, variables, data contracts, measurement processes, proxy definitions, and project graphs differ, precluding automatic construct equivalence or numerical pooling. Explicit handoffs from source-specific contracts through proxy construction and prediction to adjusted analysis expose whether a difference entered through the record, representation, export, or estimator.

1.3 Representation-Centered Observational Testbeds

The central move in CliniCause is to shift controlled researcher intervention away from simulating the patient-level data-generating process and toward specifying an inspectable *representation layer*. That layer comprises proxy constructs, operational rules and temporal cutoffs, analytical cohorts, aggregation, project-specified DAGs, exposure-specific adjustment assumptions, and provenance. Patient records contribute source-recorded measurements, availability and missingness patterns, care-process traces, covariates, and in-hospital mortality; empirical support and proxy prevalence emerge only after the representation and cohort are defined. CliniCause instantiates proxy labels deterministically and does not simulate records, exposure-assignment mechanisms, or outcomes. Source recording does not imply error-free or causally sufficient data.

The design and instantiation roles are deliberately separated. The LLM received schema-level information—available column names and the mortality outcome—together with instructions to use relevant literature. It did not receive patient records, empirical distributions, exposure prevalences, estimated effects, or execution access. Its output consisted of literature-grounded design proposals for proxy constructs, operational rules, missingness considerations, and dataset-specific graphs. Project-selected rules and graphs were subsequently encoded as deterministic source. The LLM was therefore a design assistant, not a clinical or causal expert, an executed patient-level component, or a source of ground truth.

This representation is instantiated independently on MIMIC-III and PhysioNet 2012. The resulting resources have a common analytical role but are not pooled replicas of one object. Similarly named proxy constructs can differ in measurement availability, operational definition, prevalence, missingness, support, graph structure, adjustment, and population. Each resource is a *testbed* when used to study causal-analysis behavior; the term does not imply a benchmark with known causal truth.

Figure 1.1 summarizes the separation among design, deterministic instantiation, and evaluation.

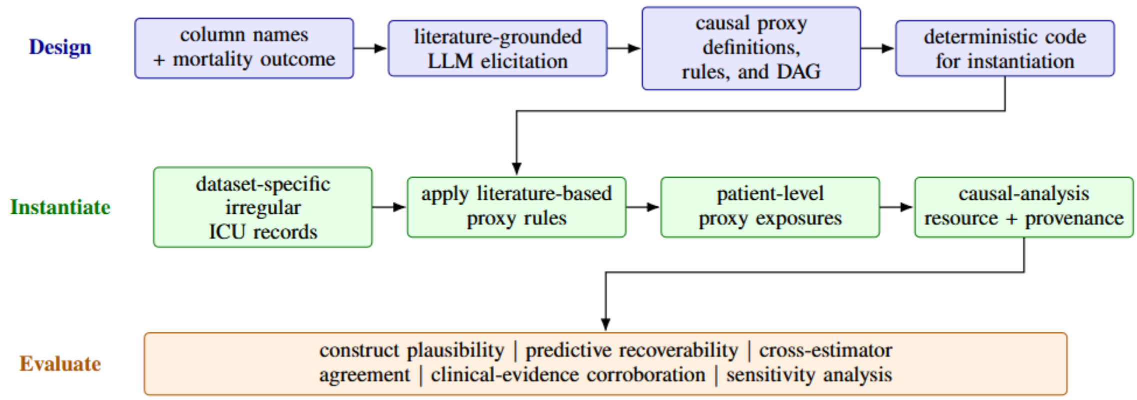

Figure 1.1: The Design, Instantiate, and Evaluate stages of CliniCause. Schema and literature information support LLM-assisted representation proposals, which project-selected deterministic code instantiates at patient level on source-recorded ICU records and outcomes. The resulting dataset-specific resources are characterized through converging evidence without a known causal answer key; the LLM receives schema information rather than patient records.

The controlled element in CliniCause is the analytical representation and its documentation, not the clinical data-generating mechanism. Deterministic construction improves inspectability and reproducibility of the representation; it does not make the represented exposure randomized, remove confounding, validate the DAG, or create causal ground truth. Standard answer-key benchmarks obtain a known causal answer by controlling the outcome or treatment-assignment layer of the data-generating process. CliniCause inverts this: it fixes the representation layer—proxy constructs, DAGs, adjustment sets, and cohort contracts—while leaving source-recorded outcomes and treatments intact. The gain is observational fidelity; the cost is the absence of a ground-truth answer key.

Constructing the testbed independently in two ICU resources provides a second axis of characterization: whether workflow behavior and qualitative estimator patterns persist when the measurement system, available variables, proxy definitions, support structure, and patient population change. MIMIC-III and PhysioNet 2012 differ in scale, era, variable coverage, and coding practices; similarly named proxy constructs are not assumed equivalent across datasets. Cross-resource agreement is therefore not replication of a common causal effect. Instead, disagreement exposes representation- and context-dependence, while agreement provides qualified additional evidence. These considerations motivate the formal causal framework in Section 2.5 and the distinct identification, estimation, and evaluation problems it introduces.

1.4 Thesis Gap and Objective

Prior work supplies the components rather than the complete evaluation object: electronic phenotyping and data programming motivate inspectable analytical labels [4, 10]; medical-LLM and graph-prior studies motivate bounded design proposals [11, 12]; and causal-inference work supplies DAG, DML, forest, and ICU-design principles [13, 14, 15]. We did not identify prior work that treats this particular combination as a single observational causal-analysis testbed constructed from source-recorded irregular ICU data with explicit representation contracts, project-specified DAGs, and multi-estimator diagnostic characterization. None of the component literatures validates the local proxy rules, graphs, or estimates.

The thesis objective is to construct and characterize clinically grounded observational causal-analysis testbeds from irregular, source-recorded ICU data. Operationally, the framework preserves source-recorded ICU outcomes and measurement structure; provides a deterministic, inspectable proxy representation with documented rules; makes a project-specified DAG and adjustment layer explicit; supports multi-estimator and diagnostic characterization without known causal truth; and tracks provenance and evidence boundaries. Evidence tracking distinguishes design proposals, selected source behavior, checked numerical artifacts, and unresolved validation or lineage requirements.

At design time, a structured LLM-assisted protocol elicited proxy ontologies, rule families, missingness considerations, and dataset-specific DAG proposals; project-selected designs were encoded as deterministic source. Execution then applies source-specific preprocessing, rule-label construction, multi-label prediction, normalization and fixed aggregation, graph-guided adjustment, matching and effect estimation, and robustness/provenance evaluation. The LLM is not invoked on patient records or during estimation.

These interfaces make the testbeds inspectable without upgrading their evidence. Prediction metrics concern rule-derived targets; matching is a descriptive comparison; mean model-estimated CATE is conditional on the implemented representation and assumptions; and support, permutation, and sensitivity diagnostics address particular failure modes. The framework provides neither clinical validation nor unconditional causal effects, recommendations, deployment readiness, or full historical rerun reproducibility.

1.5 Research Questions and Contributions

The main research question is:

> How can clinically grounded observational causal-analysis testbeds be constructed from source-observed irregular ICU records—without simulating patient records, treatment or exposure-assignment mechanisms, or outcomes—and what converging evidence can characterize their usefulness and limitations for evaluating causal estimators?

The words “observational” and “characterize” are central. The thesis investigates how a linked workflow can construct and qualify retrospective evaluation resources; it does not claim to score estimates against causal truth, prove that proxy-state exposures cause mortality, or show that manipulating a proxy would improve outcomes. The main question is addressed through five secondary research questions:

SRQ-1: How can clinically and literature-grounded representation designs be instantiated deterministically and separately in MIMIC-III and PhysioNet while retaining source-specific measurement and missingness processes?

SRQ-2: Which data contracts, normalization, aggregation, DAG, adjustment, and provenance mechanisms make the constructed resources explicit, auditable, and reusable?

SRQ-3: How recoverable are the deterministic proxy labels from irregular time series, and what mortality-relevant predictive information is retained by the constructed proxy representations?

SRQ-4: What converging evidence—including estimator sign, rank, and magnitude agreement, clinical-literature comparison, matching and support, permutation disruption, and omitted-variable sensitivity—characterizes each resource?

SRQ-5: Which limitations prevent the resources from being interpreted as clinical validation, known causal truth, or proof of estimator correctness?

The primary contribution is a representation-centered framework and two concrete, dataset-specific observational causal-analysis testbeds constructed from source-recorded MIMIC-III and PhysioNet 2012 records. The framework keeps proxy definitions, operational rules, cohort choices, DAGs, adjustment assumptions, and provenance explicit and replaceable; converging, non-equivalent evidence characterizes the resources because no causal answer key is available.

Enabling methods span schema-bounded LLM-assisted elicitation, deterministic proxy and DAG source, dataset-specific contracts, proxy prediction and aggregation, adjustment, matching, estimation, diagnostics, and provenance. CausalForestDML is primary, LinearDML secondary, and CausalPFN exploratory because it lacks the DML estimators’ diagnostic support.

Empirically, four prediction models characterize proxy recoverability; original-cohort analyses retain all 19 dataset–exposure combinations; and outcome-downsampling and cross-resource comparisons remain separate, without pooling. Forest and Linear agree in direction for 19 of 19 combinations, while all three estimators agree for 18 of 19 and retain PhysioNet shock as the exception. Matching/support, mortality prediction, clinical comparison, disruption, omitted-confounding sensitivity, population perturbation, and provenance supply additional qualified evidence. Checked tables, manifests, hashes, and decision records support numerical traceability but not a demonstrated clean-checkout historical rerun.

These contributions are cumulative, not interchangeable: workflow integration makes hand-offs inspectable, empirical comparisons characterize the resources, and evidence records separate numerical traceability from reproducibility. Agreement does not establish estimator accuracy, graph correctness, identification, or clinical validity, and no layer upgrades a proxy label to a diagnosis or a graph-guided contrast to a well-defined intervention effect. STraTS led the archived MIMIC metrics and GRU-D the PhysioNet metrics, a resource-dependent result whose detailed qualifications appear in Chapter 9.

1.6 Thesis Organization

The remainder of the thesis follows construction and characterization of the two testbeds from context to interpretation. Background and study design establish the evaluation spectrum, resource contracts, questions, and assumptions. Chapters 4–8 construct and characterize the resources through source-specific contracts, proxy instantiation and prediction, DAG-guided adjustment and estimation, diagnostics, and provenance. The final chapters report the checked evidence, interpret its limitations, and state the conclusions and future-work priorities.

Chapter 2

Background and Related Work

This chapter connects several bodies of work needed to construct an observational causal-analysis testbed: causal-method evaluation, critical-care data and irregular time series, electronic phenotyping and programmatic labeling, bounded LLM-assisted design, and observational causal inference. The emphasis is comparative: each supplies a component or boundary, but none alone defines the complete resource constructed here. Project-specific implementation begins in Chapter 3.

2.1 Causal-Estimator Evaluation: Control and Observational Fidelity

Evaluating a causal estimator differs from evaluating a factual-outcome predictor because the individual counterfactual outcome is not observed. Evaluation designs therefore differ in how they obtain an answer key and in how much of the observed system they preserve. The resulting spectrum is not a ranking in which one setting dominates the others: control supports direct error measurement, whereas observational fidelity preserves difficulties that a specified mechanism may omit.

2.1.1 Controlled answer-key designs

Fully synthetic studies specify the complete data-generating process. The evaluator chooses the covariate distribution, exposure-assignment mechanism, potential-outcome surfaces, effect function, confounding strength, and regions of support. Because both potential outcomes and the target effect are known by construction, bias, root-mean-squared error, interval coverage, and heterogeneous-effect error can be measured directly. The same control is also the main limitation: the benchmark contains only the difficulties encoded by its designer. Canonical examples include the evaluator-specified simulation settings used to study double/debiased machine learning and causal forests [13, 14]. Such settings can vary nonlinearity, heterogeneity, and overlap cleanly, but need not reproduce irregular sampling, informative missingness, care-process confounders, or changing measurement vocabularies found in clinical records.

Semi-synthetic designs retain selected empirical components while specifying the causal mechanism. The IHDP family begins with covariates from the Infant Health and Development Program, a randomized intervention for low-birth-weight premature infants. A commonly reused construction creates an observational treatment configuration from those experimental data and simulates outcome surfaces, so the covariate distribution is empirical while the potential outcomes and target effect are evaluator-defined [32, 16]. This mixture of real covariates and synthetic causal structure has made IHDP-style data one of the most frequently reused benchmark families in causal machine-learning evaluation. The Twins benchmark uses a different construction: paired twin birth records provide real covariates and observed outcomes, and one twin's outcome is treated as a counterfactual analogue for the other. It therefore preserves a distinctive matched clinical population, but its answer key depends on the co-twin construction rather than observation of both potential outcomes for one infant.

Experimentally anchored evaluation obtains its reference from a randomized study rather than from a wholly specified outcome model. In the LaLonde/National Supported Work style, the randomized experiment supplies a reference effect; the randomized controls are removed and replaced with observational comparison units, after which an observational estimator is judged by whether it recovers the experimental benchmark. This design gives the comparison strong scientific relevance for the specific intervention, population, outcome, and follow-up represented by the experiment [17]. It does not provide a universal individual counterfactual key, and differences in sampling or harmonization between experimental and observational controls can become part of the measured difficulty.

Empirically anchored generation moves further toward observed distributional structure while retaining an evaluator-specified answer. For example, an evaluator can sample from a real multivariate covariate distribution, preserve empirical dependence and support, and then apply a known assignment or outcome model to generate potential outcomes. More elaborate variants can calibrate those models to observed data before generating a known target. These resources can look closer to the source population than a stylized synthetic design, but their causal mechanism remains controlled; fitted or chosen mechanisms do not become the unknown mechanism that generated the original observations.

Across all of these designs, control enables error measurement, but results remain conditional on construction, sampling, population, and harmonization choices and may not encode the structural features of real ICU data.

2.1.2 Observational testbeds from source-recorded data

A source-recorded observational testbed preserves a different object. Its measurements, covariates, outcomes, missingness patterns, and care-process traces come from routine recording rather than an evaluator-specified patient-level data-generating process. This retains irregular timing, clinician-driven observation, heterogeneous availability, selection structure, and empirical regions of weak support that answer-key benchmarks may smooth away or omit. Cohort selection, proxy definitions and prevalence, temporal cutoffs, aggregation, and empirical support nevertheless remain representation-dependent; source recording does not make those choices neutral or the measurements error-free.

What such a testbed does not provide is a known causal truth. The unobserved counterfactual remains unobserved, and neither estimator agreement nor a successful diagnostic can be converted into causal estimation error. Estimator behavior must instead be characterized through indirect, non-equivalent evidence such as support, sensitivity, perturbation, cross-estimator comparison, and contextual external comparison. These diagnostics can reveal fragility or dependence on a particular representation, but they cannot jointly manufacture a ground-truth effect.

Controlled benchmarks and observational testbeds therefore answer different questions. A benchmark asks how accurately an estimator recovers a target under a specified causal mechanism; a testbed asks how an estimator and its surrounding workflow behave under source-recorded empirical structure whose causal mechanism is not known. Both are needed because performance under specified mechanisms and behavior under real measurement structure are complementary, not substitutes. CliniCause is positioned in the testbed category: its contribution is not a new answer-key benchmark, but a structured observational environment built from real ICU records with explicit proxy, cohort, graph, adjustment, and provenance contracts.

2.2 ICU EHR Datasets and Irregular Sampling

2.2.1 Irregular measurement and observation

ICU records combine physiology, laboratories, treatments, observations, and administrative information without a common sampling clock. Irregular spacing within variables, heterogeneous rates across variables, and patient-specific measurement density challenge complete fixed-dimensional sequence models [1]. Absence can reflect a decision not to measure, device or table availability, an analysis window, or preprocessing; masks and elapsed time can carry predictive information without identifying the process that generated it [3]. Identifiers, time zero, cohort entry, repeated stays, and summary windows likewise determine the analytical unit and temporal interpretation.

Regular resampling and imputation choose a grid and a rule for unobserved cells. Sparse-event representations preserve observed values and times, while missingness-aware recurrence encodes masks and gaps. Neither removes the need to distinguish value, observation, and care processes, so irregular sampling remains both a representation problem and an interpretation boundary.

2.2.2 PhysioNet 2012 and MIMIC-III

PhysioNet 2012. The PhysioNet/Computing in Cardiology Challenge 2012 is a bounded early-ICU mortality-prediction resource drawn from the single-center MIMIC-II database. The challenge selected 12,000 adult patients and provided data from the first available ICU stay;

each included source stay lasted at least 48 hours. The challenge divided these records into three sets of 4,000 for development and evaluation [8]. Those source totals are not the CliniCause analysis population. After the thesis-specific downstream inclusion and representation steps, the validated original-cohort causal-analysis artifact contains 7,993 model rows (Table 9.3); Chapter 4 documents the implemented preprocessing contract and the limits on reconstructing its exact artifact lineage.

Each challenge record covers the first 48 hours after ICU admission. The published resource contains five general descriptors recorded at admission—age, gender, height, ICU type, and initial weight—and 36 time-series variables. The longitudinal variables span invasive and non-invasive blood pressure, heart rate, respiratory rate, temperature, oxygen saturation, Glasgow Coma Score, urine output, ventilation status, and laboratory measurements including creatinine, glucose, bilirubin, platelet count, and white-cell count. SAPS-I and SOFA are supplied among the outcome-related descriptors for the released training set rather than as additional admission descriptors, and in-hospital death is the challenge outcome used by this thesis [8].

Longitudinal observations carry elapsed hour-and-minute timestamps, and multiple measurements may occur for a variable within one record. Their availability and timing vary by variable and patient: some streams are repeatedly charted, whereas others have sparse or absent observations. The 48-hour boundary therefore standardizes the outer observation horizon without imposing a regular sampling grid inside it. That bounded structure, documented variable list, and challenge-defined mortality outcome constrain preprocessing more tightly than MIMIC-III. For CliniCause, this improves visibility of the source-to-analysis transformation but limits which proxy constructs and DAG nodes are feasible; the source paper does not establish the later project-specific inclusion rules, missing-value handling, splits, or scores.

MIMIC-III. MIMIC-III is a deidentified, single-center relational database of critical-care admissions at Beth Israel Deaconess Medical Center. It spans 2001–2012 and contains more than 53,000 distinct adult ICU stays involving 38,597 adult patients, in addition to neonatal data [9]. As with PhysioNet, these database totals are distinct from the thesis analysis population. The validated original-cohort causal-analysis artifact contains 26,845 model rows (Table 9.2); the extraction contract and its unresolved raw-snapshot and artifact-provenance limitations are described in Chapter 4.

The relational schema covers nurse-verified bedside vital signs, laboratory and microbiology results, fluid inputs and outputs, medications, procedures, ICD-9 diagnosis and procedure codes, free-text clinical notes and reports, demographics, admission and discharge information, and survival data [9]. In-hospital mortality is directly represented, while recorded death dates permit investigators to derive fixed-horizon outcomes such as 30-day mortality when the study design calls for them; this thesis uses source-observed in-hospital mortality. Physiological observations are documented approximately hourly in the source description, while other events arrive according to laboratory, treatment, and documentation workflows. Local item identifiers create a broader, noisier vocabulary than PhysioNet's fixed list; related observations can use different identifiers or source systems.

Unlike the challenge resource, the MIMIC-III database does not impose one analysis window. Investigators must define the analytical unit, time zero, eligibility, observation and follow-up windows, variables, and outcome; a patient may contribute multiple linked hospital or ICU identifiers. Measurement density varies across patients, variables, systems, and time. Missingness is therefore pervasive and potentially informative because whether and when a value is recorded can reflect clinical concern, monitoring, treatment, and workflow. Benchmark work shows how standardized prediction tasks can be extracted, but such a benchmark is one particular transformation rather than the database itself or the local proxy task [18].

For CliniCause, this breadth enables a richer proxy vocabulary and more possible graph nodes, but it also creates more representation choices. The active preprocessing contract defines ICU-relative time, an early-ICU restriction, adult-stay filtering, source-table mappings, unit and range handling, identifier normalization, duplicate-event aggregation, and the exported event/outcome schema. Chapter 4 records those choices together with the remaining lineage and source-level reproducibility risks. Explicit documentation is part of the testbed contract because a different time-zero rule, item mapping, cohort restriction, or outcome construction could produce a different analytical resource.

Comparison and two-dataset rationale. PhysioNet's bounded challenge role and MIMIC-III's broader relational role differ in scale, era, variable coverage, identifier structure, measurement systems, and preprocessing flexibility. They are not interchangeable: similarly named clinical constructs may be defined differently, use different measurement variables, and yield different prevalences and support structures across the two datasets. Pooling records or estimates across the two is therefore not appropriate. Constructing the testbed independently in each dataset provides a second characterization axis: whether workflow behavior and qualitative estimator patterns persist when the measurement system, available variables, proxy definitions, support structure, and patient population change. Cross-resource agreement is not replication of a common causal effect—it is qualified additional evidence. Disagreement exposes representation- and context-dependence. This is the scientific motivation for the two-dataset design.

Whether and when an ICU variable is measured may depend on latent severity, clinician concern, prior treatment, and workflow. Missingness indicators or elapsed-time features can improve prediction precisely because they encode care-process information correlated with outcomes. This is useful for predictive modeling and problematic for causal inference: predictively modeling the observation process is not equivalent to causally adjusting for why observations occurred. Informative observation is therefore not only a representation problem but also a source of confounding and potential selection that standard covariate adjustment may not fully address. This connection between the representation challenges in Sections 2.2–2.3 and the identification assumptions below motivates treating irregular sampling and missingness as causal design considerations, not only as preprocessing choices.

2.3 Irregular Time-Series Representation Models

2.3.1 Representation challenges and design choices

Predictive models for irregular clinical time series differ first in what they consider an input. One family regularizes observations onto a sequence of time bins and supplies imputed values, masks, or time gaps. Another works directly with observed events, representing measurement identity, value, and time without filling every unobserved cell. Within these representations, temporal dependence can be modeled by recurrent state updates, temporal convolution, or attention. These choices encode distinct assumptions about which temporal relationships should be easy to learn and how the observation process should enter the model [1].

No representation removes the need to define the task and preprocessing contract. Dense sequences require a bin width, aggregation rule, and treatment of missing cells. Sparse events require a variable vocabulary, continuous or discrete encodings of time and value, and a policy for very long event streams. Positional or temporal encodings provide order information to attention mechanisms, while masks can prevent padded or future positions from entering a computation. Self-supervised pretraining can exploit unlabeled records, but its utility depends on the pretext task and the relation between pretraining and supervised populations. The four learned models reviewed below are the predictive families compared by the thesis; their published forms are conceptual precedents rather than proof that every local adaptation is identical.

2.3.2 Recurrent baselines: GRU and GRU-D

The gated recurrent unit (GRU) updates a hidden state sequentially using reset and update gates. Conceptually, the update gate balances retention of the previous state against incorporation of a candidate state, while the reset gate controls how strongly earlier information contributes to that candidate. This gating provides a flexible general sequence baseline with fewer gating components than a conventional long short-term memory unit [19]. A standard GRU can learn from a suitably prepared clinical sequence, but it does not inherently know how much time elapsed between observations or why a variable is absent. Its handling of irregularity therefore depends on the input representation supplied to it.

GRU-D modifies recurrent modeling for multivariate series with missing values. It uses observation masks and elapsed-time information, and learns decay mechanisms that move missing inputs toward empirical reference values and attenuate hidden states as gaps increase [3]. This formulation makes the missingness pattern and time since observation explicit rather than treating imputation as an invisible preprocessing step. It is one principled response to irregular sampling, not a universal solution: learned decay imposes its own structure, the observation process may differ across sites, and predictive use of missingness does not correct causal bias from measurement decisions.

2.3.3 Temporal convolution: TCN

Temporal convolutional networks apply one-dimensional convolution in temporal order. Causal convolution prevents future positions from contributing to an earlier output, dilation expands the receptive field without requiring a kernel at every lag, and residual blocks stabilize deeper models. Because convolutions for many positions can be computed in parallel, a TCN offers a different computational and inductive bias from a recurrent state update [20]. The canonical TCN study is a general sequence-modeling comparison rather than an ICU-specific method paper. In an irregular ICU setting, the model still requires an appropriate regular sequence representation, so resampling, aggregation, masks, and time features remain part of its effective design.

2.3.4 Sparse-event transformer modeling: STraTS

STraTS is designed for sparse and irregular multivariate clinical series. It treats an observed measurement as a triplet of time, variable, and value; continuous embeddings of time and value are combined with a variable embedding, and transformer-style attention contextualizes the observed events. This avoids constructing a fully imputed variable-by-time matrix and preserves fine-grained observation times. The method also introduces a self-supervised forecasting objective for pretraining representations before supervised prediction [2].

This design is well matched to sparse event streams, but it does not make preprocessing irrelevant. Event selection, normalization, truncation, static features, and the variable vocabulary still determine what the transformer sees. Moreover, the published method and the thesis's supervised multi-label adaptation are distinct objects. The paper supports the triplet representation, attention mechanism, and pretraining rationale; Chapter 6 defines the local target heads, training, and export contract.

2.3.5 Relationship among the four included model families

GRU supplies a general recurrent baseline; GRU-D makes missingness and elapsed time explicit within recurrence; TCN provides multiscale temporal filters; and STraTS applies attention to observed event triplets with optional pretraining. Their suitability can vary with density, preprocessing, labels, sample size, and site-specific observation patterns. The comparison therefore tests distinct inductive biases rather than a universal sophistication ranking; Chapters 6 and 9 define the local implementations and results.

2.4 Proxy Phenotyping and Weak Supervision

2.4.1 Electronic phenotyping and rule-derived labels

Electronic phenotyping identifies patients or records with characteristics of interest from routinely collected EHR data. Phenotypes may combine diagnosis or procedure codes, laboratory and vital measurements, medications, text, temporal criteria, deterministic rules, statistical models, or machine learning. The field has consequently developed from manually specified rule sets toward hybrid and learned approaches, while retaining the need to validate the construct for its intended use [4].

Rule-based phenotypes remain attractive when transparency matters: reviewers can trace labels to variables, thresholds, and time conditions and identify implausible clauses. That inspectability does not make a rule clinically correct.

Transparency is not equivalence to clinical truth. Thresholds can be sensitive to units and preprocessing, required variables may be missing, and a rule developed at one site may not transport to another. Temporal ambiguity can cause measurements before, during, and after a clinical episode to be collapsed into one label. Codes and treatments may reflect billing or care processes as well as physiology. Rule outputs can therefore be reproducible yet noisy, and a negative label can mean insufficient evidence rather than absence of the condition. Validation must be tied to the intended use: a cohort-retrieval rule, prediction target, and causal exposure may require different evidence.

2.4.2 Clinical-definition sources and approximation

Clinical consensus documents provide bounded conceptual grounding for organ dysfunction and acute syndromes. Sepsis-3, KDIGO acute-kidney-injury criteria, the ISTH DIC framework, and the Berlin ARDS definition are multicomponent and context dependent [21, 22, 23, 24].

These sources cannot upgrade a partial project rule into a diagnosis. Local proxies may omit duration, baseline change, imaging, interventions, or context and are only clinically informed by the cited definitions; construct validity requires rule-level review and chart-adjudicated evaluation.

2.4.3 Weak supervision and label aggregation

Weak supervision addresses settings with scarce hand labels and imperfect programmatic sources. Data programming expresses heuristics as labeling functions and models noisy, conflicting sources to create training data [10]. The relevant lesson is that source quality and correlation matter; adding sources does not guarantee cancellation of shared bias.

A deterministic majority vote is not a learned source-quality model: it applies a fixed rule to binary sources. Its output is algorithmic aggregation, not clinical consensus, and correlated voters trained on the same rule-derived labels can preserve common error.

2.4.4 Implications for this thesis's proxy states

The thesis uses *clinically inspired proxy state*, *rule-derived proxy label*, *proxy phenotype*, and *weak clinical label* to distinguish analytical constructs from verified diagnoses. Repository identifiers beginning with `LAT_` are historical software names. They do not establish that a formally identified latent clinical variable exists. This distinction is essential because the same column interface is used across several stages without making the generating mechanisms equivalent.

Proxy-state evidence spans distinct questions: whether code executes the documented rule, identifiers and columns align, models learn the rule-derived labels, labels correspond to the intended clinical construct, and proxies are temporally coherent causal exposures. Repository evidence supports implementation, schema, and learnability more directly than clinical construct or causal measurement validity.

The shared proxy interface supplies prediction targets, aggregation inputs, and downstream exposure representations. It makes handoffs auditable but also links rule, prediction, aggregation, and causal-measurement error; predictive accuracy against rule labels is learnability, not clinical or causal validity.

Beyond learnability, proxy-label error has direct causal consequences. When a rule-derived exposure label misclassifies the true underlying state, causal estimates may be biased even if the identification assumptions hold for the true exposure. Even nondifferential misclassification—error that does not depend on the outcome—can bias effect estimates; its direction and magnitude depend on the estimand, the exposure–outcome structure, and the error mechanism and are not predictable in general. Differential misclassification is less constrained still. Predicted proxy labels compound rule-label error with model-prediction error; models trained on the same rule-derived target are correlated sources and do not independently validate one another's outputs or the underlying clinical construct. Misclassification propagates into prevalence, overlap structure, adjustment relationships, and downstream causal estimates. Proxy recoverability and causal validity are therefore separate axes: accurately reproducing a deterministic rule from irregular time-series data demonstrates that the representation is learnable, not that the underlying clinical construct is valid or that a causal contrast based on it is correctly identified [30].

2.4.5 LLM-Assisted Clinical and Causal Knowledge Elicitation

Advanced LLMs can encode and express substantial medical knowledge under structured evaluation. In MultiMedQA evaluations, PaLM-family models produced medically relevant answers and reasoning, and instruction tuning improved several measured capabilities [11]. The same study's human evaluation exposed clinically important gaps, and Med-PaLM remained inferior to clinicians. This evidence supports the feasibility of using an LLM to generate design proposals; it does not establish reliability for a particular proxy-design task, correctness of every rule, suitability as an autonomous medical expert, or clinical validation of CliniCause outputs.

LLM semantic judgments have also been studied as prior information for causal-graph construction. Darvariu and colleagues evaluate prompt designs and soft LLM-derived priors for causal discovery, reporting the greatest benefit for edge orientation in selected common-sense benchmark settings while also finding that standalone LLM priors are not uniformly beneficial [12]. This is a methodological precedent for treating model output as candidate relations or orientation information, not evidence that the CliniCause DAGs are correct. The local graphs were not learned from patient-level data, the study did not implement the paper's discovery algorithm, and formal adjustment validity still depends on the correctness and completeness of the project-specified assumptions.

A separate precedent is programmatic labeling: inspectable heuristics generate weak labels for discriminative learning [10]. CliniCause is only conceptually related; its selected decision trees generate deterministic proxy targets, and its later aggregate is a fixed vote rather than a learned noise model.

Within this thesis, the LLM-assisted elicitation protocol is consequently treated as a substantive design-time component with bounded authority. It produced proposals for proxy-state ontologies, binary decision rules, missingness considerations, and dataset-specific DAGs; the project selected designs and encoded them in deterministic source files. The prompt artifacts document that process, whereas active tagger and graph scripts define implemented behavior. Neither the literature above nor the local prompt record establishes clinical validity, complete human review, or causal identification.

2.5 Causal Diagrams, Identification, HTE, Overlap, and Sensitivity

2.5.1 Observational causal questions and target-trial thinking

For each unit i with adjustment covariates W_i and binary exposure $A_i \in \{0, 1\}$, define potential outcomes $Y_i(0)$ and $Y_i(1)$, the outcomes that would be observed under each exposure condition. The observed outcome satisfies

$$Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0).$$

Because only one potential outcome is realized, the individual treatment effect $Y_i(1) - Y_i(0)$ is never observed. Population estimands—$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$, $\text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid A = 1]$, and $\text{CATE}(w) = \mathbb{E}[Y(1) - Y(0) \mid W = w]$—aggregate this unobservable individual contrast and require identification assumptions before they can be expressed as functions of the observed-data distribution [31, 30].

Three assumptions are most directly at stake here, although they are not exhaustive. First, *consistency/SUTVA* requires the observed-data link $A_i = a \Rightarrow Y_i = Y_i(a)$, no interference between units, and no hidden versions of the exposure, so that $Y_i(a)$ is well defined. A shock proxy can conflate septic, cardiogenic, and hemorrhagic mechanisms with different mortality relationships; if those versions are combined in one binary label, $Y_i(1)$ is not unique. The thesis therefore treats proxy states as explicitly rule-derived analytical constructs and restricts estimand language accordingly [7].

Second, *exchangeability* or unconfoundedness requires

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp A_i \mid W_i.$$

Conditional on the observed adjustment covariates, exposure assignment must be independent of the potential outcomes. In an ICU, more severely ill patients are more likely both to activate a rule-defined illness state and to die regardless of any causal pathway from that state. If relevant severity is unrecorded or only imperfectly captured by available EHR variables, exchangeability fails. This no-unmeasured-confounders condition is the central unverifiable threat in the observational analyses.

Third, *positivity* or overlap requires, for every w in the support of W_i and both $a \in \{0, 1\}$,

$$\Pr(A_i = a \mid W_i = w) > 0.$$

Certain patient strata can have near-deterministic proxy exposure when thresholds, deterministic rules, or missingness patterns fire reliably for particular covariate profiles. Support failure can therefore arise from rule design and cohort construction as well as clinical practice. The thesis tracks this threat through explicit overlap and matching diagnostics [33].

In an ideal randomized trial, the assignment mechanism removes baseline confounding by design and supplies treatment-assignment support within the randomized eligible population. Consistency, interference, adherence/estimand, missing-outcome, and other design conditions still require attention. In observational data, exchangeability and support are not supplied by randomization and must be justified through design, measurement, and substantive assumptions, none of which is directly testable from the observed-data distribution alone [30].

Identification, estimation, and evaluation are distinct problems. Exchangeability, consistency, positivity, temporal ordering, and an appropriate causal model determine whether the target causal contrast is identified from the observed-data distribution—that is, whether it is a well-defined functional of what can in principle be observed. DML, causal forests, and related methods address estimation after identification is assumed: flexible nuisance estimation can reduce modeling bias and accommodate high-dimensional functions, but it cannot recover a causal quantity that is not identified. Evaluation then asks how one can know whether the estimator performed well, a question that real observational outcomes cannot answer by comparison with counterfactual truth. This third problem is the one CliniCause is designed to study; it does not solve identification. Valid uncertainty quantification and transport to another population require still further assumptions.

ICU databases condition on ICU entry and commonly impose additional cohort requirements such as survival to a measurement window, availability of particular variables, or sufficient observation density. When inclusion is influenced by causes of both exposure and outcome, conditioning on inclusion induces selection bias. Cohort construction is consequently a causal design decision rather than merely a preprocessing step, and the explicit cohort contracts used in this thesis make that decision auditable [30].

Causal analysis begins with a question, not an estimator. Target-trial thinking makes the desired comparison explicit by specifying eligibility, treatment or exposure strategies, assignment, time zero, follow-up, outcome, causal contrast, and analysis plan before observational data are analyzed [6]. This discipline exposes mismatches such as using post-baseline information to define eligibility, comparing prevalent with incident exposure, or allowing eligibility and follow-up to begin at different times. ICU EHR data make these issues acute because clinical state, treatment, measurement, and selection evolve rapidly and are documented through care processes.

The exposure must also correspond to a sufficiently well-defined intervention or contrast. Different ways of producing the same measured state can lead to different outcomes, so a contrast on a state such as body mass index—or, here, an illness-state proxy—may violate consistency when the underlying intervention is unspecified [7]. This concern is not repaired by choosing a flexible estimator. It requires clarity about what changes, when it changes, and which versions of the exposure are being compared.

Retrospective ICU studies additionally face confounding by indication, changing severity, informative measurement, treatment–confounder feedback, selection, and limitations in outcome measurement. A review of observational causal inference in intensive care emphasizes careful question formulation, assumptions, data preparation, and reporting across the analysis workflow [15]. Marginal structural models provide important background for time-varying treatment and confounder processes, but they are not the point/binary exposure estimator implemented in this thesis. Nor does this thesis claim a complete target-trial emulation.

The Causal Evaluation Problem and the Benchmark Gap

The identification–estimation–evaluation chain introduced in Section 1.1 makes evaluation especially difficult on real observational data. As established in Section 2.1, evaluation designs range from high control to high observational fidelity. The remaining gap is that even well-developed causal machine-learning methods are assessed primarily on resources that construct a known causal answer by design, and the constructed mechanism may omit failure modes present in real ICU data.

Because counterfactuals are unobserved, causal estimation error cannot be computed directly from real data. A held-out test set supplies factual outcomes but not the counterfactual outcomes needed to score individual or average treatment-effect estimates. This evaluation paradox distinguishes causal-estimator assessment from supervised-prediction evaluation.

Building on the taxonomy in Section 2.1, fully synthetic data-generating processes construct covariates, assignment, potential outcomes, and effects from a known model: the evaluator gains control, but realistic-looking data are not guaranteed to reproduce real measurement, missingness, or care-process structure. Semi-synthetic IHDP-style designs retain real baseline covariates from a randomized program evaluation while simulating treatment assignment and an outcome surface; covariate structure is empirical, but the causal mechanism is synthetic [32]. Experimentally anchored LaLonde/NSW-style settings use a randomized reference effect to assess recovery from an observational analogue, providing scientific relevance for a narrow setting rather than a universal patient-level counterfactual. Natural- or quasi-experimental anchors likewise obtain design-based identification in a particular empirical context but do not create a general answer key.

Causal machine-learning benchmarks are concentrated in the first two categories. Their mechanisms can emphasize smooth confounding surfaces, fully observed baseline covariates, and well-behaved propensity scores without reproducing the irregular sampling, informative missingness, noisy rule-derived proxy exposures, near-boundary support, and treatment–confounder feedback of longitudinal ICU EHR data. Strong performance on an IHDP-style simulation therefore does not by itself establish performance under the irregular measurement, care-process confounding, missingness, and support structure of source-recorded ICU records.

In the absence of causal ground truth, diagnostics answer narrower questions than benchmark error. Agreement across estimators exposes dependence on estimator choice but not estimator accuracy. Overlap diagnostics reveal unsupported regions; perturbation analyses reveal instability; sensitivity analysis quantifies the strength of omitted confounding required to alter a conclusion; and external clinical literature supplies contextual comparison. None of these establishes causal correctness individually or collectively. Their value is diagnostic: non-equivalent evidence can characterize robustness and expose failure modes without converting an observational testbed into an answer-key benchmark.

2.5.2 DAGs, backdoor reasoning, and adjustment

A directed acyclic graph (DAG) represents assumed causal relations through nodes and directed edges. Paths into an exposure can encode confounding routes, while colliders require special care because conditioning on them can open otherwise blocked associations. Backdoor reasoning uses the graph to identify covariates that block noncausal paths without adjusting for inappropriate descendants or colliders [5]. A DAG thus makes assumptions inspectable and connects scientific reasoning to adjustment-set selection.

Formally, let W be a set of observed covariates containing no descendant of A and blocking every path between A and Y that has an arrow into A. When such a sufficient adjustment set exists and is observed, the backdoor adjustment functional is

$$\mathbb{E}[Y(a)] = \mathbb{E}_W[\mathbb{E}[Y \mid A = a, W]] \,.$$

Here W is the sufficient set selected by the project-specified DAG helper, not every observed covariate. More adjustment is not automatically safer: conditioning on a common effect of exposure and outcome opens a path through that collider, and a clinically plausible post-exposure variable may be such a collider. Temporal ordering is therefore required before a variable is eligible for adjustment.

Graphical transparency must not be mistaken for empirical proof. A DAG does not establish that its arrows are correct, and a set that blocks paths in the drawn graph does not guarantee exchangeability in the clinical population. Required confounders may be unavailable in the observed data; measurement can be an imperfect proxy for a graph node; and temporal order must be justified scientifically rather than inferred from an arrow. In this thesis, project-specified DAGs make the assumed adjustment layer inspectable and replaceable and guide observed-variable adjustment at a high level; they do not prove that the structure is correct, that all confounders are captured, or that adjustment identifies the contrast in the clinical population. Their dataset-specific nodes, mappings, and implementation are deferred to Chapter 7.

2.5.3 Double machine learning

Double machine learning (DML) separates the causal parameter of interest from flexible nuisance functions, including outcome regression $g(W) = \mathbb{E}[Y \mid W]$ and exposure propensity $e(W) = \Pr(A = 1 \mid W)$. Machine-learning estimators regularize these functions to control variance in high-dimensional settings, but naively substituting regularized predictions into a causal estimating equation can transfer regularization and overfitting bias to the target parameter. DML addresses this problem through Neyman-orthogonal scores, which reduce first-order sensitivity to nuisance-estimation error, and cross-fitting, which estimates nuisance functions on data separate from the observations used to evaluate the score [13].

As an illustration rather than a description of the implemented estimator, consider the partially linear model $Y = \theta A + g(W) + \varepsilon$ with $\mathbb{E}[A \mid W] = m(W)$. Define residuals $\widetilde{A} = A - \widehat{m}(W)$ and $\widetilde{Y} = Y - \widehat{g}(W)$. The residualized estimator takes the form

$$\widehat{\theta} = \frac{\widehat{\mathbb{E}}[\widetilde{A}\widetilde{Y}]}{\widehat{\mathbb{E}}[\widetilde{A}^2]}.$$

This special case illustrates orthogonality. The thesis instead uses LinearDML and CausalForestDML, with a causal-forest final stage for the latter; Chapter 7 defines the exact implementation.

In a residualization interpretation, covariate-explained components are removed from the outcome and exposure, and the residual association enters the orthogonal score. Cross-fitting rotates training and evaluation folds so observations can contribute efficiently without using the same fitted nuisance error twice. These devices broaden the nuisance models that can be used while retaining inference under stated regularity and rate conditions. They do not eliminate the substantive assumptions of causal inference. DML cannot correct an omitted confounder that was never measured, create overlap where none exists, turn a proxy into an accurately measured exposure, repair an incorrect DAG, or define an otherwise ambiguous intervention.

EconML provides software implementations for heterogeneous-effect estimators, including DML-related workflows, and is useful for documenting implementation context [25]. The software reference is distinct from the theoretical basis: library availability does not validate local nuisance models, cross-fitting behavior, or identification assumptions.

2.5.4 Causal forests and heterogeneous effects

Outcome prediction estimates how outcomes vary with covariates under the observed data distribution; effect estimation asks how an outcome contrast varies between exposure conditions under causal assumptions. A strong prognostic variable need not be an effect modifier, and a subgroup with high baseline risk need not receive greater benefit from an intervention. Keeping prognostic and treatment-effect heterogeneity separate is therefore fundamental.

Causal forests adapt random-forest ideas to heterogeneous treatment-effect estimation. Recursive partitioning seeks regions with different treatment contrasts, while aggregation stabilizes local estimates. Under unconfoundedness and appropriate forest construction, the method supports conditional-effect estimation and inference [14].

The theoretical causal-forest construction adapts honest random forests to CATE estimation. Honesty uses separate observations for tree splitting and leaf-level effect estimation, reducing overfitting as local treatment–control contrasts are formed in covariate neighborhoods. Under regularity conditions, asymptotic normality supports pointwise inference, and generalized random forests include causal forests as a special case. These are principles from the causal-forest literature; this thesis uses EconML's CausalForestDML with the archived configuration defined in Chapter 7, and not every detail of the theoretical construction carries over automatically.

A fitted CATE function $\widehat{\tau}(w)$ is a model-estimated conditional contrast. Averaging its fitted values over an analysis sample gives a sample-dependent aggregate. Calling that aggregate a population ATE requires the estimator, target population, weighting, and identification conditions jointly to support the interpretation. This distinction applies directly to the mean model-estimated CATE reported in the thesis, whose language is correspondingly restricted.

These methods are attractive when effect modification is nonlinear or involves interactions, but flexibility does not validate discovered subgroups. Estimated conditional effects can be unstable in sparse regions, depend on nuisance models and tuning, and identify patterns that do not transport. Forest feature importance describes the fitted model rather than a biological mechanism. High-dimensional individual estimates are not automatically clinically actionable.

Critical-care heterogeneity, confounding, informative observation, treatment selection, and covariate shift make conditional effects difficult to validate; a held-out factual outcome cannot reveal an individual's counterfactual error. ICU causal-analysis guidance therefore emphasizes question formulation, assumptions, data preparation, and reporting rather than estimator flexibility alone [15].

CausalPFN is a prior-fitted transformer that amortizes causal-effect estimation through in-context inference after training on simulated data-generating processes satisfying ignorability [26]. It remains exploratory here because the archive does not identify the exact producing package version or checkpoint and provides no DML-equivalent uncertainty, sensitivity, or permutation diagnostics.

2.5.5 Overlap and empirical support

Positivity requires that the exposure conditions being compared are possible within relevant covariate strata. Its empirical counterpart, overlap or common support, asks whether comparable exposed and unexposed observations exist in the analyzed sample. When propensity scores approach their boundaries, estimates rely on a small number of observations or extrapolation, producing instability and changing the population for which a comparison is credible. Work on limited overlap formalizes the variance and target-population consequences and motivates explicit attention to the region of adequate support [27].

Structural nonpositivity occurs when one exposure condition is genuinely impossible within a covariate stratum. Practical nonpositivity occurs when both conditions are possible but one is extremely rare in the analyzed sample, making estimation unstable; the distinction affects whether the response should redefine the target population, gather support, or accept that a contrast is not identified there [33]. For deterministic proxies, support is also representation-dependent: thresholds, aggregation rules, cohort restrictions, and missingness patterns can create near-deterministic strata. Apparent non-overlap is therefore partly a property of the chosen representation layer rather than clinical practice alone.

Overlap-weighted estimands offer an alternative to trimming observations or truncating inverse-propensity weights. Treated units receive weights proportional to $1-e(W)$, and control units receive weights proportional to $e(W)$, concentrating the contrast in the population where propensities are intermediate and overlap is strongest [34]. This changes the target population while improving stability in weak-support regions. The thesis does not implement overlap weighting; it is included as conceptual context.

Overlap is especially important for heterogeneous-effect estimation because local estimates divide the sample into increasingly specific covariate neighborhoods. A global mixture of exposed and unexposed records can coexist with local support failures. Matching rates, failures, and distances can provide indirect evidence about available comparisons, but they depend on the matching representation and algorithm. They do not replace propensity-score distributions, common-support plots, or covariate-balance assessment. Conversely, empirical overlap does not establish exchangeability or validate the DAG. This thesis therefore treats support diagnostics as necessary qualification and does not claim that positivity has been established.

2.5.6 Omitted-variable sensitivity and diagnostic workflows

Even after adjustment, observational estimates can be biased by variables that affect both exposure and outcome but are absent from the model. Omitted-variable-bias sensitivity analysis asks how strongly such a variable would need to relate to the residualized exposure and outcome to change a conclusion. For a chosen statistical-significance level q, the robustness value RV_q is the minimum equal strength an omitted confounder would need to have, expressed through its partial R^2 with the residualized exposure and its partial R^2 with the outcome residuals, to move the conclusion to the significance boundary. It is a threshold on two residual associations rather than a probability that confounding exists. Partial R^2 values for observed covariates such as SOFA score or age can provide a benchmark scale for asking whether a hypothetical unmeasured confounder would need to be stronger than measured factors [28].

Analogous bias bounds and inference for flexible estimators and broader target parameters exist in causal machine learning [29]. Such benchmarking is meaningful only when the observed comparison covariate is scientifically relevant to the exposure–outcome relationship. A robustness value does not show that unmeasured confounding is absent or that the DAG is correct; it characterizes the departure from exchangeability needed to overturn a conclusion, not whether that departure exists.

Sensitivity results remain model- and scale-dependent, and no observed benchmark necessarily bounds every unmeasured process. Sensitivity analysis also does not create support or repair exposure misclassification, so it must be interpreted alongside overlap, perturbation, and measurement diagnostics.

2.5.7 Positioning of the present study

As defined in Chapter 1, CliniCause is an observational testbed rather than an answer-key causal benchmark: it exposes estimator behavior and fragility under real ICU structure without providing a causal answer key.

The literature supplies complementary components rather than the observational evaluation object itself. Irregular-series models address representation; phenotyping and programmatic labeling create inspectable but imperfect analytical variables; bounded LLM work motivates design proposals; DAGs expose assumptions; DML and forests estimate flexible contrasts; and overlap and sensitivity methods reveal fragility. CliniCause integrates these components while keeping representation decisions, handoffs, estimator interfaces, diagnostics, and provenance explicit. Integration makes construction auditable but can also compound rule, prediction, aggregation, graph, support, confounding, and lineage errors. Chapters 3–8 define the implemented objects and boundaries; none of the cited literature validates the local proxies, graphs, estimates, or causal truth.

Chapter 3

Problem Definition and Study Design

This chapter defines CliniCause as a framework for constructing two dataset-specific observational causal-analysis testbeds. Each resource packages source-observed ICU records and outcomes together with an explicit analytical representation, project-specified causal assumptions, estimator interfaces, diagnostics, and provenance. MIMIC-III and PhysioNet 2012 are instantiated separately: their shared contract identifies corresponding analytical roles, but it neither pools records nor makes similarly named proxies, graphs, adjustment sets, or estimands identical.

3.1 Dataset-Specific Observational Testbed Contract

For dataset d, the constructed resource links a source-specific record identity; source-observed longitudinal measurements; observation and missingness information; a deterministic proxy-state representation; model-generated proxy annotations where used; fixed aggregation outputs where used; source-observed in-hospital mortality; observed covariates; a project-specified DAG; exposure-specific adjustment sets; an analytical cohort definition; and provenance records that distinguish the evidence class of each component. The resource is an observational *testbed* when used to examine causal-estimator behavior. It is not a benchmark with a known counterfactual answer.

These components do not have a common evidential status. Record identifiers, recorded measurement events, their availability patterns, observed covariates, and mortality originate in the source data subject to extraction and data-quality limitations. Proxy definitions, cutoffs, temporal rules, cohort definitions, and covariate roles are representation-defined research choices. Rule-derived proxy labels are deterministic derived objects instantiated from those choices and the available observations. Learned proxy annotations are model-generated estimates of the rule-derived labels, while the archived majority-vote interface is a fixed derived analytical output. DAGs and exposure-specific adjustment sets are project-specified causal assumptions. Provenance and evidence-class records link these objects without implying that source observation, deterministic derivation, model estimation, and causal assumption provide equivalent evidence.

The controlled intervention in this construction therefore occurs at the representation layer. Researchers specify which observed evidence defines a proxy, which records form an analytical cohort, which variables serve as covariates and outcomes, and which graph and adjustment assumptions govern each exposure analysis. Deterministic source code then instantiates the selected proxy rules on patient records. CliniCause does not simulate those patient records, a treatment or exposure-assignment mechanism, or mortality outcomes, and deterministic proxy-label construction does not make the underlying clinical state known. The causal answer remains unknown because the data and outcomes are observational and the representation choices do not establish identification.

3.2 Units, Time Horizons, and Data Objects

This thesis studies retrospective intensive-care records represented as irregular clinical time series. The operational study unit is a time-series record or ICU stay, depending on the source dataset and preprocessing path. PhysioNet 2012 source files use record identifiers, whereas MIMIC-III source tables use ICU-stay and admission identifiers. The implemented pipelines normalize these source-specific identifiers to the canonical join key `ts_id`. This identifier should be read as the analysis-record identifier used by the software; it is not assumed to be a globally unique person identifier across repeated admissions or stays.

For a study unit i, the observed longitudinal data are represented as a finite collection of measurement events

$$\mathcal{O}_i = \{(i, t_{ij}, v_{ij}, x_{ij}) : j = 1, \ldots, n_i\},$$

where i is the normalized `ts_id`, t_{ij} is elapsed time in minutes from record start or ICU admission, $v_{ij} \in \mathcal{V}$ is the observed variable name, and x_{ij} is the numeric value recorded for that variable. In the causal repository this long-format object is stored in `ts` with the required columns `ts_id`, `minute`, `variable`, and `value`. Static or baseline quantities such as age, sex or gender coding, height, weight, and ICU-type indicators are represented within the same event schema as variables observed at or near the beginning of the record when the preprocessing implementation provides them.

The event representation is intentionally irregular. Observations occur at nonuniform times, different variables are measured with different frequencies, and most variables are absent at most possible times. A missing measurement is therefore not interpreted as a normal value. It may reflect a clinical decision not to measure, an unrecorded observation, a source-table limitation, or a preprocessing exclusion. The thesis treats informative measurement and missingness processes as important threats to interpretation, not as causal findings established by the repository.

The primary patient- or stay-level outcome used by the downstream causal repository is `in_hospital_mortality`. In the causal processed artifact, this outcome is stored in `oc`, the outcome table keyed by `ts_id`. The third object in that processed artifact, `ts_ids`, is the sorted collection of normalized identifiers represented in `ts`. Later predictive processing uses a distinct split-aware artifact and must not be conflated with this causal contract.

3.3 Design-Time Knowledge Construction and Executed Analysis

The framework distinguishes a design-time representation-construction layer from deterministic patient-level instantiation and subsequent analysis. At design time, the official dataset-description link and instructions to research relevant clinical, causal, and time-series literature were supplied to a structured LLM-assisted elicitation protocol. The protocol requested a dataset audit, proxy-state ontology, binary decision-rule families, reasoning about missingness and measurement processes, and a dataset-specific DAG. The LLM received schema-level information and the mortality outcome, not patient records, empirical distributions, exposure prevalences, estimated effects, or execution access. The resulting proposals were project-selected and encoded in deterministic tagger and graph source files. Available evidence does not establish that every proposal was implemented unchanged, a complete line-by-line prompt-output-to-code mapping, or a formal clinician-review chain.

The analytical layer begins only after those designs have been encoded. It preprocesses raw ICU data, applies deterministic decision-tree rules to generate rule-derived proxy labels, trains deep irregular-time-series models to predict those labels, normalizes predicted outputs, and forms the archived final aggregate from one rule-derived source and four predicted-label sources before DAG-guided matching, effect estimation, sensitivity analysis, and permutation diagnostics. No LLM is invoked for patient-level preprocessing, label generation, prediction, adjustment selection, or effect estimation during this executed layer. Thus prompts document design proposals, active source defines implementation, and every downstream inference remains conditional on proxy, graph, support, and model assumptions. The two datasets follow this factorization separately; source-specific measurement availability and rule semantics are retained rather than forced into a common pooled representation.

3.4 Prediction Task

The prediction task is a multi-label proxy-state prediction problem over irregular ICU time series. For each study unit i and proxy state k, let $L_{ik}^{rule} \in \{0, 1\}$ denote a rule-derived proxy label. These labels are clinically inspired analytical constructs generated from observed data by the deterministic source-code rules selected and encoded after LLM-assisted design elicitation. They are weak clinical labels or proxy phenotypes, not chart-adjudicated diagnoses, manually verified clinical states, or validated phenotypes unless separate validation evidence is supplied in a later chapter.

The supervised predictive workflow estimates a vector of proxy-state labels from the observed time-series representation. Its target matrix is loaded from a latent-label CSV keyed by `ts_id`; the supervised proxy-label targets are not taken from the mortality column in `oc`. For model m, the exported prediction object may contain probabilities $\hat{p}_{ik}^{(m)}$ and corresponding binary predicted labels $\hat{L}_{ik}^{(m)}$. These outputs are model estimates of rule-derived proxy labels rather than direct measurements of patient physiology.

Within construction and characterization of each testbed, five workflow functions are distinguished. First, deterministic proxy-state construction derives binary rule-derived proxy labels from observed clinical measurements. Second, multi-label proxy-state prediction estimates those labels from irregular time-series inputs and measures recoverability rather than clinical accuracy. Third, the archived final causal runs aggregate one rule-derived table with four aligned binary model-prediction tables. Fourth, mortality prediction or association analysis evaluates prognostic information in the proxy representation. Fifth, observational causal-effect estimation treats selected aggregate columns as modeled exposures and estimates adjusted contrasts under explicit assumptions. These functions create or characterize different resource components; none supplies known causal truth. Detailed rules, architectures, estimators, and results follow in later chapters.

3.5 Causal Question, Exposures, Outcome, Estimands, Assumptions

The downstream causal-analysis setting is retrospective and observational. No intervention is assigned by this study. The repository uses software parameters named `treatment` for selected binary `LAT_*` columns, but many such columns represent illness-state or organ-dysfunction proxy states rather than well-defined administered therapies. In the thesis narrative these variables are therefore described as proxy-state exposures or modeled treatment variables unless a later section explicitly defines a manipulable intervention.

Let $A_i \in \{0, 1\}$ denote a selected proxy-state exposure, Y_i denote the binary outcome `in_hospital_mortality`, W_i denote observed adjustment variables selected from a project-specified DAG and available columns, and X_i denote effect-modifier features used to describe heterogeneity. The dataset-specific DAGs were developed through a project design process that included LLM-assisted clinical and causal elicitation and were then encoded in source code; the implemented graph scripts, not the prompt outputs alone, define the DAG artifacts used by the pipeline. Potential-outcome notation such as $Y_i(a)$ is useful as conceptual language, but causal interpretation requires more than the presence of a graph or an estimator. It depends on assumptions about the causal graph, consistency, exchangeability conditional on observed covariates, positivity or overlap, outcome and exposure measurement, and unmeasured confounding [5, 6, 7].

Accordingly, later estimates should be interpreted as adjusted effect estimates under project assumptions, not as unconditional clinical effects. The existence of a DAG, matching script, or CATE estimator does not prove identification. Proxy states may be useful for structured analysis while still carrying label noise, temporal ambiguity, and intervention-definition limitations.

Chapter 4

Data and Preprocessing

Preprocessing instantiates the source-specific data contract for each CliniCause resource. It does not generate synthetic patients or mortality outcomes: it selects, normalizes, aligns, summarizes, masks, imputes, or transforms source-observed measurements and outcomes according to the implemented pipeline. These operations distinguish source-observed events from extraction choices, cohort restrictions, preprocessing transformations, analysis-ready representations, and the downstream derived labels introduced in Chapter 5.

The contract is scientifically consequential because it determines which records enter, which measurements are available, how time and absence are represented, which identifiers survive, and which later proxy and causal-analysis objects can be instantiated. MIMIC-III and PhysioNet 2012 remain separate resources throughout. Their source-recorded measurements, availability and missingness patterns, and care-process traces pass through different extraction and transformation paths; empirical support is assessed only after cohort and representation construction. This heterogeneity defines the two observational testbeds and does not authorize pooling or treating similarly named fields as equivalent.

4.1 PhysioNet 2012 Pipeline

The PhysioNet/Computing in Cardiology Challenge 2012 dataset provides ICU time-series records and associated in-hospital mortality outcomes for a mortality-prediction challenge setting [8]. In this thesis it supplies the source-observed measurements and outcomes from which one dataset-specific resource is instantiated for proxy-state prediction and downstream causal analysis. The local repository does not track the raw challenge files or a verified processed-data pickle, so this section describes the implemented preprocessing contract rather than asserting a raw preprocessing cohort total. Chapter 9 separately reports the validated causal-analysis model-row count.

The causal-repository PhysioNet preprocessing implementation traverses the raw `set-a`, `set-b`, and `set-c` folders and reads one patient-record file at a time. It removes rows with missing parameter names, skips records with at most five retained measurement rows, and removes observations with negative values, which the source code treats as missingness indicators. It converts source times from `HH:MM` strings into elapsed minutes, renames `RecordID` to `ts_id`, `Parameter` to `variable`, and `Value` to `value`, and reads outcome files by mapping `In-hospital_death` to `in_hospital_mortality`. After concatenating the source sets, it drops duplicate events and converts the categorical `ICUType` measurement into one-hot-style variables `ICUType_1` through `ICUType_4`.

The resulting causal processed artifact is serialized as three objects: `ts`, `oc`, and `ts_ids`. The `ts` object is a long event table with `ts_id`, `minute`, `variable`, and `value`. The `oc` object contains `ts_id`, `length_of_stay`, `in_hospital_mortality`, and a source subset indicator. The `ts_ids` object is derived from identifiers observed in `ts`. This artifact is the expected input to rule-based tagging, mortality-from-proxy analysis, matching, and CATE estimation in the causal repository.

4.2 MIMIC-III Pipeline

MIMIC-III is a critical-care electronic health-record database used here to instantiate the second, separate ICU resource [9]. The implemented workflow follows a clinical time-series benchmark style in that it transforms raw ICU events into stay-level time-series examples suitable for prediction, while adapting the output contract for this thesis pipeline [18]. The resulting analysis-ready representation is constructed from source observations rather than being a synthetic patient or outcome generator. The repository does not establish a tracked raw-data snapshot, extraction date, or final processed artifact hash.

The active causal-repository MIMIC preprocessing implementation reads broad categories of MIMIC-III source tables: ICU-stay timing from `ICUSTAYS`, demographics from `PATIENTS`, bedside measurements from `CHARTEVENTS`, laboratory measurements from `LABEVENTS`, fluid outputs from `OUTPUTEVENTS`, administered inputs from `INPUTEVENTS_CV` and `INPUTEVENTS_MV`, and mortality/timing information from `ADMISSIONS`. Large event tables are processed in chunks and written to temporary shards before feature rows are collected. The implementation filters to adult ICU stays, removes chart events marked with an error flag, requires valid chart times and numeric or textual values, applies item-ID mappings and range filters for selected variables, converts selected units, and assigns events without direct ICU-stay identifiers to an ICU interval when possible.

For the causal contract, event times are converted to elapsed minutes relative to ICU admission, events are restricted to the early ICU window used by the implementation, age and gender are added as static rows at minute zero, and duplicate (`ts_id`, `minute`, `variable`) events are collapsed by averaging. The helper contract canonicalizes stay identifiers, normalizing numeric-looking identifiers such as `12345.0` to `12345`. It defines the canonical `ts` columns as `ts_id`, `minute`, `variable`, and `value`, defines the canonical `oc` columns as `ts_id`, `length_of_stay`, `in_hospital_mortality`, and `subset`, and intentionally excludes internal MIMIC identifiers such as `HADM_ID` and `SUBJECT_ID` from the exported outcome table.

Inspection found a source-level issue that should be resolved before relying on local MIMIC preprocessing execution: the active script reads `ADMISSIONS.csv` with `DEATHTIME` for early-death filtering, while the canonical outcome helper requires `HOSPITAL_EXPIRE_FLAG` to construct `in_hospital_mortality`. This does not change the documented target contract, but it remains a source-level reproducibility risk.

4.3 STraTS Split-Aware Artifacts Versus Causal Artifacts

The predictive STraTS repository uses a different processed-data contract from the causal repository. Its loader expects a five-object pickle consisting of `events`, `oc`, `train_ids`, `val_ids`, and `test_ids`, where the final three objects explicitly encode train, validation, and test identifiers. The loader accepts stay identifiers named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and validates the split arrays after the same normalization. This split-aware contract is therefore not interchangeable with the causal repository's unsplit `[ts, oc, ts_ids]` artifact.

In supervised STraTS runs, the latent-label CSV is joined to the selected split identifiers after ID normalization. The loader filters labels to the supervised IDs, raises an error when labels are missing for any supervised `ts_id`, raises an error for duplicate normalized labels, and uses all non-`ts_id` columns in the latent CSV as proxy-label targets. The mortality column in `oc` is not used as the supervised proxy-label target. Prediction export writes `ts_id`, one probability column per proxy state, and one thresholded binary column per proxy state; the inspected wrapper scripts export the `all` split for their prediction CSVs, although final archive-copy provenance remains incomplete.

The predictive loader also implements split-aware leakage controls. It keeps only variables observed in the training split, derives normalization statistics from the training split, computes static-feature normalization from training rows, and validates label alignment before training. For MIMIC-III supervised prediction it restricts events to the first 24 hours. The pretraining path removes test data, uses training-derived variable statistics, uses a 48-hour maximum window for PhysioNet 2012, and permits MIMIC-III observations up to five days while sampling 24-hour input windows. These controls reduce some sources of leakage, but they do not guarantee absence of leakage, nor do they resolve missing raw-data and split-provenance limitations.

The parent router can bridge contracts by reading a causal `[ts, oc, ts_ids]` artifact and constructing a STraTS-style `[ts, oc, train_ids, val_ids, test_ids]` payload through a seeded identifier shuffle. That bridge is an orchestration convenience and does not prove how every archived final STraTS artifact was generated.

Several provenance limitations remain. Raw PhysioNet and MIMIC-III data are not tracked in Git. The processed pickles used by final runs are external or absent from the audited repository. Some run records contain machine-specific absolute paths, and final cohort counts require verified processed artifacts or manifests. Clean-checkout reproduction therefore requires

Table 4.1: Processed artifact contracts used by the causal and STraTS repositories.

Property	Causal repository	STraTS repository
Processed artifact	`[ts, oc, ts_ids]`	`[events, oc, train_ids, val_ids, test_ids]`
Split representation	Unsplit ID collection	Explicit train/validation/test IDs
Main use	Tagging and causal analysis	Predictive training and export
Identifier convention	Canonical `ts_id`	Loader-normalized `ts_id`
Proxy-label source	Separate latent CSV downstream	Separate latent CSV joined to split IDs

authorized data access, artifact hashes, and a documented split-generation record.

Accordingly, the two preprocessing paths instantiate two resource contracts rather than one pooled clinical table. They preserve the distinction between source-observed records and mortality outcomes, extraction and cohort decisions, transformed analysis objects, and downstream representation-defined constructs. The observational data-generating and measurement processes remain unknown; explicit preprocessing makes the transformations applied to their source-observed traces inspectable without claiming that every raw event or missingness pattern survives unchanged.

Chapter 5

Clinical Proxy-State Construction

This chapter defines the representation layer and its deterministic instantiation within each testbed. The construction sequence is schema and literature review, LLM-assisted candidate representation, project selection, deterministic source encoding, patient-level proxy instantiation from available observations, optional model-generated proxy annotations, and the fixed downstream aggregate where used. The LLM proposed designs but did not execute on patient records; active tagger source, rather than prompt text, defines every implemented proxy rule. Project-selected ontologies, decision rules, missingness considerations, and DAG proposals were encoded in tagger and graph source files.

The taggers instantiate representation-defined analytical constructs from source-observed measurements. A positive label means that the encoded rule was satisfied by available evidence. A zero label means that the rule did not fire on that evidence; because a missing measurement can prevent a clause from firing, zero does not establish clinical absence. Model predictions are learned annotations of the rule-derived labels, and the aggregate is a deterministic analytical interface. None of these objects is a diagnosis, expert adjudication, clinical consensus, or causal truth. Interpretability comes from inspectable rules and stable `LAT_*` interfaces, not from chart-adjudicated reference labels.

The chapter retains the scientific meaning of each proxy family and the active aggregation rule. Exhaustive prompt inventories, artifact hashes, prompt-to-source matrices, and clause-level rule tables are maintained in the repository evidence package rather than repeated in the main narrative. Predictive architectures appear in Chapter 6, and numerical results in Chapter 9.

5.1 Rationale and Terminology

Electronic phenotyping uses structured measurements, codes, and rules to construct analytical variables when direct adjudication is unavailable [4]. Here a *proxy state* or *proxy phenotype* is a binary construct summarizing available evidence for a clinically meaningful condition or physiologic pattern. A *rule-derived proxy label* is assigned by deterministic source rules; a *predicted proxy label* is a supervised model estimate of that rule-derived label; and an *aggregated* *predicted proxy-label representation* is the deterministic mixed-source vote used downstream. A binary proxy-state tag is the realized 0/1 value for one `ts_id` and proxy-state column.

Repository names such as `latent`, `latent tag`, and `LAT_*` remain stable implementation vocabulary, but they do not show that a hidden biological state was discovered. The proxy layer converts selected designs into source-coded constructs, supplies multi-label prediction targets, and provides a common schema for aggregation, mortality analysis, matching, and CATE estimation. No chart-adjudicated gold standard was found; missing measurements can prevent clauses from firing; thresholds approximate rather than reproduce complete clinical definitions; and dataset-specific states are not automatically construct-equivalent. The later aggregate is a fixed vote, not a learned source-quality model.

5.1.1 LLM-Assisted Design of Proxy States, Rules, and DAGs

The structured elicitation protocol requested a dataset audit, proxy-state ontology, binary decision-rule families, missingness and measurement-process reasoning, and a dataset-specific DAG for PhysioNet 2012 and MIMIC-III. The final dataset prompts instantiate a common template with dataset-specific names and official description links. Their exported runs preserve the resulting design proposals, but the exact system instructions, decoding settings, hosted-model configuration, follow-up interactions, and export procedure are incompletely documented.

The outputs were proposals subsequently selected and encoded by the project, not clinical validation or executable authority. The LLM did not inspect patient records at runtime, generate final labels, aggregate voters, select adjustment variables dynamically, or estimate effects. Active tagger and graph source defines implemented behavior; prompts document design provenance. Deterministic code then evaluates the encoded clauses against the available observations for each patient-level record. The final outputs align at schema level with the eleven active PhysioNet states and ten active MIMIC states, but complete accepted/rejected decisions and line-level prompt-output-to-code correspondence have not been established.

The LLM prompt-provenance audit retains the prompt inventory, version comparison, hashes, execution-metadata boundary, and prompt-to-source assessment. The active taggers and Step 4 proxy-rule citation matrix retain exact clauses and citation status. These repository records form the technical evidence package; the main text below summarizes the constructs and behaviors needed to understand the method.

Figure 5.1 illustrates this design-to-instantiation distinction for the MIMIC-III shock rule. Its observed-evidence clauses correspond to the active operational definition in Table 5.2; the figure does not define an additional rule.

5.2 PhysioNet Proxy-State Rules

The active PhysioNet tagger pivots the long event table by `ts_id` and minute, computes per-record summaries, and applies one decision function per proxy column. Its eleven-state ontology comprises chronic baseline risk, global severity, shock, respiratory failure, renal dysfunction,

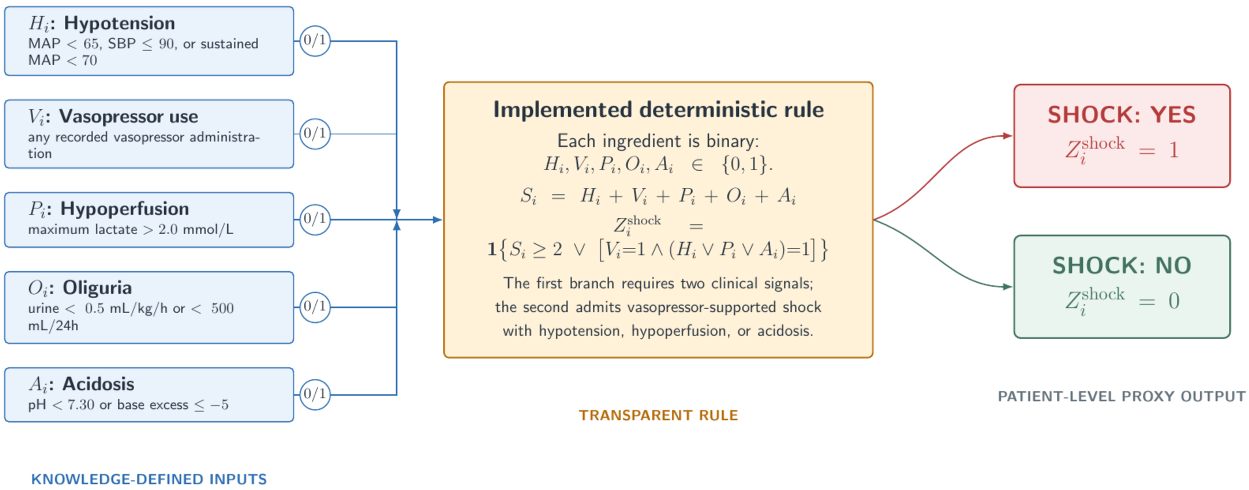

Figure 5.1: Illustrative deterministic instantiation of the project-selected MIMIC-III shock proxy. Observed-evidence clauses feed the encoded rule, and the patient-level label is derived from the available observations. A non-firing clause can reflect absent qualifying evidence; proxy value zero therefore does not prove absence of clinical shock. The figure illustrates the representation contract rather than validating the clinical construct.

hepatic dysfunction, coagulation/hematologic dysfunction, inflammation/sepsis burden, neurologic dysfunction, cardiac injury/strain, and metabolic derangement. The archived rule-derived CSV matches this schema and contains binary tags without duplicate normalized identifiers. The producing command, input hash, source version, and threshold-mode provenance remain incomplete.

Numeric comparisons contribute positive evidence only when a non-missing value exists, so missing measurements are not treated as normal or abnormal. The active summarizer does not impose new time windows beyond the processed record. Urine clauses require a compatible precomputed urine summary, and oxygenation uses an available PaO2/FiO2 summary or an approximation from component summaries. Default source thresholds can be replaced by an explicit optimized-threshold file, but the archive does not establish that this option produced the final CSV.

Table 5.1: Operational PhysioNet 2012 rule-derived proxy-state definitions. Thresholds are active source defaults; the archive does not establish use of the optional optimized-threshold mode.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_CHRONIC_BASELINE_RISK`: baseline vulnerability or limited reserve.	First age; BMI calculated from first height and weight; minimum albumin; first ICU type.	Positive when at least 2 of age ≥ 75, BMI < 18.5 or ≥ 40, albumin < 3.0, and ICU type 1 or 3 are present.	Admission-wide summaries; BMI requires both inputs. The score and ICU-type clause are project specific.
`LAT_GLOBAL_SEVERITY`: multi-domain acute physiologic severity.	Seven domains: hemodynamic pressure; ventilation/oxygenation/respiratory rate; GCS; creatinine/BUN/urine; bilirubin/platelets; lactate/pH/bicarbonate; troponin/heart rate.	Positive for at least 3 abnormal domains, or at least 2 plus lactate ≥ 4.0, pH < 7.20, mechanical ventilation, GCS ≤ 8, or MAP < 60. Main domain cutoffs include MAP < 70, systolic pressure ≤ 100, PF < 300, SaO2 < 92, respiratory rate ≥ 22, GCS < 15, creatinine ≥ 2.0, BUN ≥ 40, urine < 500, bilirubin ≥ 2.0, platelets < 100, lactate > 2.0, and HR ≥ 130.	A project domain count, not SOFA; missing measurements suppress their domains.
`LAT_SHOCK`: circulatory shock or hemodynamic instability.	Minimum invasive/noninvasive MAP and systolic pressure; maximum lactate and HR; compatible urine summary; minimum pH and bicarbonate.	Positive when at least 2 of low MAP (< 70), low systolic pressure (≤ 90), lactate > 2.0, HR ≥ 110, urine < 500, or acidosis (pH < 7.30 or bicarbonate < 18) are present; low MAP with lactate ≥ 4.0 is also an explicit sufficient pair.	No vasopressor input in this dataset rule; urine evidence exists only when precomputed.
`LAT_RESPIRATORY_FAILURE`: hypoxemia, ventilatory failure, or ventilatory support.	Mechanical-ventilation flag; PF ratio; minimum SaO2 and PaO2; extreme respiratory rate; maximum PaCO2 with pH.	Positive for at least 2 of ventilation, PF < 300, SaO2 < 92 or PaO2 < 60, respiratory rate ≥ 22 or < 8, and PaCO2 ≥ 50 or PaCO2 ≥ 45 with pH < 7.30; ventilation with PF < 300 is an explicit sufficient pair.	PF may be approximated from component summaries; not adjudicated ARDS or respiratory failure.
`LAT_RENAL_DYSFUNCTION`: acute or clinically meaningful renal dysfunction.	First and maximum creatinine; maximum BUN; compatible urine summary; maximum potassium; minimum bicarbonate.	Positive for at least 2 of creatinine ≥ 2.0 or rise from first value ≥ 0.3, BUN ≥ 40, urine < 500, and potassium ≥ 5.5 or bicarbonate < 18; creatinine ≥ 3.5 is sufficient.	Not full KDIGO staging; timing is the available record and urine may be unavailable.
`LAT_HEPATIC_DYSFUNCTION`: hepatic injury or poor reserve.	Maximum bilirubin, AST, ALT, and ALP; minimum albumin and platelets.	Weighted score ≥ 2: bilirubin ≥ 2.0 contributes 2; AST or ALT ≥ 200, ALP ≥ 250, albumin < 2.5, and platelets < 100 each contribute 1.	Mixes injury, nutritional/reserve, and platelet evidence; project-specific construct.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_COAG_HEME_DYSFUNCTION`: coagulation or hematologic stress.	Minimum platelets and HCT; maximum HCT and WBC; first-to-minimum platelet change.	Weighted score ≥ 2: platelets < 100 contributes 2; platelets < 150, HCT < 25 or > 55, WBC < 4 or > 20, and platelet drop ≥ 50 each contribute 1.	Not a complete DIC or SOFA score; anemia and leukocyte extremes broaden the construct.
`LAT_INFLAMMATION_SEPSIS_BURDEN`: systemic inflammatory or sepsis-like burden.	Temperature and WBC extremes; maximum HR, respiratory rate, and lactate; minimum PaCO2 and platelets.	Positive for at least 3 of temperature > 38.3 or < 36, WBC > 12 or < 4, HR > 90, respiratory rate > 20 or PaCO2 < 32, lactate > 2.0, and platelets < 150; temperature or WBC abnormality together with lactate stress and score ≥ 2 is also sufficient.	No infection anchor; explicitly not confirmed sepsis or a Sepsis-3 implementation.
`LAT_NEUROLOGIC_DYSFUNCTION`: depressed consciousness or neurologic/metabolic stress.	Minimum GCS, sodium, glucose, SaO2, and pH; maximum sodium, glucose, and PaCO2.	Weighted score ≥ 2: GCS ≤ 12 contributes 2; GCS < 15, sodium < 130 or > 150, glucose < 70 or > 300, and PaCO2 ≥ 50, SaO2 < 90, or pH < 7.25 each contribute 1.	Sedation and ventilation are not separated from neurologic impairment.
`LAT_CARDIAC_INJURY_STRAIN`: myocardial injury, ischemia, or severe strain.	Maximum troponin I/T and HR; minimum HR, MAP, and systolic pressure; maximum lactate; first ICU type.	Weighted score ≥ 2: troponin I > 0.1 or troponin T > 0.01 contributes 2; ICU type 1 or 2, HR ≥ 130 or < 50, MAP < 70 or systolic pressure ≤ 90, and lactate > 2.0 each contribute 1.	Troponin alone can make the rule positive; assay/context harmonization is not encoded.
`LAT_METABOLIC_DERANGEMENT`: acid–base, lactate, electrolyte, or glucose derangement.	Minima/maxima of pH, bicarbonate, sodium, potassium, magnesium, glucose, and PaCO2; maximum lactate.	Positive for at least 2 abnormal domains: pH < 7.30 or > 7.50; bicarbonate < 18 or > 32; lactate > 2.0; sodium < 130 or > 150, potassium < 3.0 or > 5.5, or magnesium < 0.6 or > 1.2; glucose < 70 or > 250; PaCO2 < 32 or > 50. pH < 7.20, lactate ≥ 4.0, or potassium ≥ 6.0 is sufficient.	Project-specific composite; an isolated critical marker can determine the label.

The rules combine clinically recognizable evidence domains with project-specific Boolean or score thresholds. Global severity aggregates organ domains; shock combines hypotension, perfusion, urine, and acid-base evidence; and the remaining families combine their named physiologic domains. Table 5.1 states the active definitions. Sepsis-3, KDIGO, ISTH DIC, and Berlin ARDS sources provide limited conceptual grounding [21, 22, 23, 24]. The implementation does not reproduce those formal definitions, and several clauses remain project specific.

5.3 MIMIC Proxy-State Rules and Validation Artifacts

The active MIMIC tagger defines ten states: chronic burden, inflammation/sepsis, global severity, shock, respiratory failure, renal dysfunction, combined hepatic/coagulation dysfunction, neurologic dysfunction, metabolic derangement, and cardiac strain. Relative to PhysioNet, it uses a chronic-burden construct, combines hepatic and coagulation evidence, and changes several available inputs and helper flags. These are construct and source-feature differences, not cosmetic renaming.

The tagger accepts three input modes. Summary-CSV mode consumes one row per stay. Canonical-pickle mode reconstructs rule features from `[ts, oc, ts_ids]`, including GCS components, first-24-hour urine output, rolling urine rates when weight is available, identifier normalization, and selected measurement aliases. Raw-concept mode aggregates admission, diagnosis, vital, laboratory, urine, vasopressor, ventilation, infection, and biomarker sources when supplied. Positive binary helpers and non-missing numeric comparisons provide evidence; absent numeric evidence does not make a rule positive. Input mode and extraction completeness therefore affect which clauses can fire.

MIMIC rules use the same broad score-or-sufficient-condition structure while exploiting richer care-process evidence. Examples include infection anchors for inflammation/sepsis, vasopressor and hypoperfusion evidence for shock, respiratory-support evidence for respiratory failure, dialysis evidence for renal dysfunction, and biomarker or rhythm evidence for cardiac strain. Table 5.2 gives the active dataset-specific definitions. The clinical sources above provide partial grounding, but the chronic-burden, combined hepatic/coagulation, metabolic, and cardiac constructs require qualified review and remain project-specific proxies.

Table 5.2: Operational MIMIC-III rule-derived proxy-state definitions. Available evidence depends on summary-CSV, canonical-pickle, or raw-concept input mode.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_CHRONIC_BURDEN`: chronic burden or baseline vulnerability.	Age or deidentified-old flag; comorbidity count or Elixhauser score; chronic-disease helpers; malignancy/immunosuppression; admission type.	Positive when at least 2 of age ≥ 75 or old-age flag, comorbidity count ≥ 2 or Elixhauser ≥ 5, chronic organ/ICD evidence, malignancy or immunosuppression, and emergency/urgent admission are present.	Helper availability is input-mode dependent; admission urgency is part of this project-specific baseline construct.
`LAT_INFLAMMATION_SEPSIS`: inflammation, suspected infection, or sepsis-like burden.	Temperature and WBC extremes; maximum lactate; culture/suspected-infection and antibiotic/suspected-infection helpers.	Score ≥ 2 across temperature ≥ 38.3 or < 36, WBC > 12 or < 4, lactate > 2.0, culture evidence, and antibiotic evidence; at least one culture, antibiotic, temperature, or WBC anchor is required.	Not formal Sepsis-3; culture and antibiotic helpers reflect care processes as well as disease evidence.
`LAT_GLOBAL_SEVERITY`: multi-domain acute physiologic severity.	Seven domains: pressure/vasopressors; oxygenation/ventilation; GCS/RASS; creatinine/urine; lactate/pH/bicarbonate; temperature/WBC; bilirubin/platelets/INR.	Positive for at least 3 abnormal domains. Main cutoffs are MAP < 70 or SBP ≤ 100; SpO2 < 92, PF ≤ 300, or ventilation; GCS < 15 or RASS $\leq -3/\geq 2$; creatinine ≥ 2.0 or 24-hour urine < 500; lactate > 2.0, pH < 7.30, or bicarbonate < 18; inflammatory thresholds; bilirubin ≥ 2.0, platelets < 100, or INR ≥ 1.5.	A project domain count, not SOFA; there is no single-marker sufficient condition.
`LAT_SHOCK`: shock or hemodynamic instability.	Minimum MAP/SBP and sustained-MAP helper; vasopressors; maximum lactate; 24-hour or rolling 6-hour urine; minimum pH and base excess.	Positive for score ≥ 2 across hypotension (MAP < 65, SBP ≤ 90, or sustained MAP < 70), vasopressors, lactate > 2.0, urine < 500 ml/24 h or < 0.5 ml/kg/h over 6 h, and pH < 7.30 or base excess ≤ -5; vasopressors plus hypoperfusion, acidosis, or hypotension is also explicit.	Support and urine helpers depend on extraction mode; not a formal shock diagnosis.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_RESPIRATORY_FAILURE`: gas-exchange or ventilatory failure/support.	Minimum SpO2 and PF ratio; maximum FiO2; mechanical/noninvasive ventilation; respiratory-rate extremes; maximum PaCO2 with pH.	Positive for score ≥ 2 across SpO2 < 92, PF ≤ 300, support (ventilation or FiO2 ≥ 0.5), RR ≥ 30 or ≤ 8, and PaCO2 ≥ 50 with pH < 7.30; support plus hypoxemia, low PF, or ventilatory failure is also explicit.	PF is derived from summary extrema; not adjudicated ARDS or respiratory failure.
`LAT_RENAL_DYSFUNCTION`: renal dysfunction.	Maximum and change-from-first creatinine; maximum BUN; 24-hour/rolling 6-hour urine; maximum potassium; minimum bicarbonate; dialysis/CRRT helper.	Positive for score ≥ 2 across creatinine ≥ 2.0 or delta ≥ 0.3, BUN ≥ 40, low urine, potassium ≥ 5.5 or bicarbonate < 18, and dialysis; dialysis/CRRT alone is sufficient.	Not full KDIGO staging; weight is required for normalized urine rate.
`LAT_HEPATIC_COAG_DYSFUNCTION`: combined hepatic and coagulation dysfunction.	Maximum bilirubin, AST, ALT, and INR; minimum platelets and albumin; prolonged-PT/PTT helpers.	Positive for score ≥ 2 across bilirubin ≥ 2.0, AST or ALT ≥ 120, platelets < 100, INR ≥ 1.5 or prolonged PT/PTT, and albumin < 2.5.	Combines liver injury, reserve, and coagulation evidence; not full SOFA or DIC scoring.
`LAT_NEUROLOGIC_DYSFUNCTION`: neurologic dysfunction with sedation caveat.	Minimum GCS; abnormal GCS-component helper; RASS extremes; pupil/focal-neurologic helper; sedation/intubation helper.	GCS ≤ 8 contributes 2; GCS ≤ 13, an abnormal GCS component, RASS ≤ -3 or ≥ 2, and pupil/focal abnormality each contribute 1; score ≥ 2 is positive. Source explicitly rejects an isolated one-point score when sedation/intubation is present.	Sedation is only partially handled; GCS components are mandatory in canonical-pickle construction.
`LAT_METABOLIC_DERANGEMENT`: acid–base, lactate, electrolyte, or glucose derangement.	Minima/maxima of pH, bicarbonate, potassium, sodium, and glucose; minimum base excess; maximum lactate.	Positive for at least 2 abnormal domains: pH < 7.30 or > 7.55; bicarbonate < 18 or > 35 or base excess ≤ -5; lactate > 2.0; potassium < 3.0 or ≥ 5.5; sodium < 130 or > 150; glucose < 70 or > 250.	Project-specific composite with no single-marker sufficient condition.
`LAT_CARDIAC_STRAIN`: cardiac strain or injury.	Troponin flags or fallback T/I values; CK-MB helper/value; arrhythmia or HR extremes; HR-plus-hypotension stress; cardiac unit/surgery context.	Positive for score ≥ 2 across troponin evidence (T ≥ 0.1 or I ≥ 0.4 fallback), CK-MB evidence (including a recorded value > 0), rhythm instability/HR > 150 or < 40, HR stress with hypotension, and cardiac-care context; the final condition additionally requires troponin-defined biomarker or rhythm evidence.	CK-MB contributes but cannot satisfy the anchor alone; assay units and clinical adjudication are not encoded.

The MIMIC output package contains the binary tag table, the merged features used by the rules, prevalence and co-occurrence summaries, unadjusted mortality-by-tag summaries, a JSON validation summary, and a decision-tree artifact. These files support implementation and descriptive validation only: prevalence is not clinical accuracy, risk ratios in the mortality table are not causal effects, and co-occurrence is not construct validation.

5.4 Predicted and Aggregated Proxy Labels

The same `LAT_*` interface can represent three generation mechanisms. Rule-derived labels come from the taggers, predicted labels from the four supervised time-series models in Chapter 6, and the final aggregate from deterministic voting over aligned binary files. Across all twelve archived final causal runs, the logs enumerate exactly five sources: one dataset-specific rule-derived table and predicted-label tables from GRU, GRU-D, STraTS, and TCN. The logs establish filenames and downstream handoff but do not co-archive the voter bytes, hashes, or checkpoint-to-export mappings.

Prediction exports contain `ts_id`, one probability column `<label>_prob` per target, and one corresponding binary column thresholded at 0.5. Voting consumes only the binary columns. The normalization helpers separate probability and binary fields, enforce the configured proxy order, and reject duplicate identifiers, nonbinary values, or incompatible schemas. Voter rows are aligned on the shared normalized identifier intersection, with the first sorted voter file defining output column and row order.

For record i, proxy state k, and M aligned voter files, let $L_{ik}^{(m)}$ be the binary value from voter m. The implemented rule is

$$\widetilde{L}_{ik} = \mathbb{I}\left(\sum_{m=1}^{M} L_{ik}^{(m)} \geq \left\lceil \frac{M}{2} \right\rceil\right).$$

It is equivalent to `2 * ones_count >= n_voters`, so an even-voter tie maps to 1. The orchestrator writes the resulting `ts_id`-keyed mixed-source table and passes it to mortality prediction, matching, CATE estimation, sensitivity analysis, and permutation stages.

This majority vote is a fixed analytical exposure interface, not expert consensus, five independent ground-truth labels, clinical truth, or a learned probabilistic label model. It consists of one rule-derived source and the four learned sources STraTS, GRU, GRU-D, and TCN. Those predicted voters were trained against the rule-derived targets, and the rule source also participates in the vote; correlated error can therefore persist or be amplified. Interpretation depends on stable construct definitions, aligned identifiers, binary-only inputs, and the exact five-source lineage. Aggregation is consequently subordinate to representation instantiation: it supplies one documented downstream interface without redefining or clinically validating the underlying proxy constructs.

Chapter 6

Predictive Modeling of Proxy States

This chapter evaluates *proxy recoverability*: whether deterministic rule-derived proxy labels correspond to learnable patterns in each testbed's irregular source-derived representation. Its supervision chain is LLM-assisted decision-rule design, project selection and deterministic source encoding, generation of rule-derived proxy labels, and deep-model prediction of those labels. The supervised target is the resulting rule-derived proxy-label matrix loaded from a separate CSV file keyed by `ts_id`; the inputs are irregular measurement histories plus static covariates prepared under the data contracts described in Chapter 4. The model outputs are learned annotations of analytical proxy labels, not source-observed exposures or adjudicated clinical states.

The final comparison contains four learned models: STraTS, GRU, GRU-D, and TCN. This chapter defines their representation, training, evaluation, and export mechanics; experiment selection and numerical comparisons are deferred to Chapters 8 and 9. Dataset-specific differences in the leading model characterize differences in the two resources and support dataset-specific model choice; they do not establish universal architectural superiority.

6.1 Self-Supervised STraTS Pretraining

STraTS represents an irregular time series as a set of observed measurement tokens rather than as a fully imputed grid. The implemented STraTS class constructs each token by adding three learned components: a continuous embedding of measurement time, a continuous embedding of the normalized measurement value, and an embedding of the variable identity. The token sequence is contextualized with a transformer-style block and then pooled with fusion attention to obtain a fixed-dimensional time-series representation [2]. Static covariates are combined with this representation before either the self-supervised forecasting head or the supervised proxy-label head is applied.

In the verified STraTS implementation, contextualized observation embeddings are pooled and combined with a learned embedding of static covariates. The recurrent, convolutional, and decay baselines in the final comparison do not have a supported self-supervised pretraining path in the current source.

The pretraining dataset loader reads the split-aware processed artifact and removes held-out test identifiers before constructing unsupervised examples. It builds the variable vocabulary from variables observed in the pretraining train split, derives value means and standard deviations from that train split, and saves the resulting variable list, normalization table, and maximum time horizon to `pt_saved_variables.pkl`. For PhysioNet 2012 the maximum time is 48 hours. For MIMIC-III the loader permits observations up to five days but samples at most a 24-hour input context and clamps implausibly old age values in the same way as the supervised loader.

Each self-supervised example is formed by sampling a forecast anchor from observed timestamps that are not the final timestamp and are at least 12 hours after record start. Observations up to the anchor form the historical context, truncated by the maximum observation count and, for MIMIC-III, by the 24-hour context limit. The target window extends two hours after the anchor. For each variable observed in that future window, the implementation keeps the last observed normalized value and sets the corresponding forecast mask entry to one; variables without future observations are masked out of the loss. Input times are rescaled to the interval $[-1, 1]$ using the dataset-specific maximum time.

The pretraining head maps the combined time-series and static representation to one forecast value per variable. Its objective is masked squared error over variables observed in the sampled future window, so unobserved future variables do not contribute to the loss. The pretraining evaluator materializes batches for each evaluation split, sampling three forecast batches per evaluation chunk when a split is first evaluated, and reports `loss_neg`, the negative masked loss, for checkpoint selection. The best pretraining checkpoint is written as `checkpoint_best.bin`; loss values themselves are not reported in this methods chapter.

Checkpoint-loaded supervised training is implemented by constructing the target supervised architecture, loading the supplied pretraining checkpoint, and copying overlapping tensors into the current model state before calling `load_state_dict`. When `--load_ckpt_path` is supplied, the supervised dataset also loads `pt_saved_variables.pkl` from the same directory to reuse the pretraining variable vocabulary, normalization statistics, and maximum-time metadata. This source path supports warm-started STraTS-family training, but the archived final exports still lack a complete manifest linking each exported CSV to the intended pretraining checkpoint, supervised checkpoint, split artifact, and archive-copy step.

6.2 Supervised Multi-Label Models

The supervised prediction task uses a deterministic rule-derived latent-label CSV as its target matrix. The loader accepts identifier columns named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and checks consistency when more than one accepted identifier column is present. It then filters the label table to the supervised split identifiers, raises an error for missing labels, raises an error for duplicate normalized identifiers, sorts rows to match the internal supervised identifier order, and treats every non-`ts_id` column as a proxy-label target. The number and order of supervised targets are therefore determined at runtime by the CSV schema, but their semantic status remains rule-derived rather than clinical ground truth.

All supervised backends share a multi-label output contract. During training the model returns binary cross-entropy with logits against the full target vector, using per-target positive-class weights computed from the supervised train split. During evaluation or prediction export, the same model call without labels returns sigmoid probabilities. Static covariates are normalized from training rows and combined with each model's time-series representation. In scratch training the binary head maps the combined representation directly to proxy-label logits; checkpoint-loaded STraTS training uses an intermediate forecasting head so overlapping pretraining tensors can be reused.

The STraTS supervised backend uses the irregular observation triplets described in Section 6.1: time, value, and variable embeddings, transformer contextualization, fusion attention, and static-feature combination [2].

The GRU baseline uses a dense hourly grid rather than the raw irregular token set. For each hour and variable, the loader builds value, observation-mask, and elapsed-time-since-observation channels, mean-fills unobserved values from training data, normalizes the filled values, and concatenates the three channel groups before passing them to a gated recurrent unit [19]. The final recurrent hidden state is concatenated with the static-feature embedding and projected through the shared supervised head.

The GRU-D backend uses a model-specific irregular sequence representation containing observed values, missingness masks, elapsed-time tensors, sequence lengths, and static covariates [3]. The implemented cell updates last observed values, applies learned exponential decay to inputs and hidden states as a function of elapsed time, and includes missingness masks in the recurrent gates. The representation passed to the supervised head is the final valid recurrent state for each sequence combined with the static-feature embedding.

The TCN baseline uses the same dense hourly value, observation-mask, and elapsed-time representation as the GRU baseline, but processes the sequence with temporal convolutional residual blocks [20]. The implementation permutes the dense grid into channel-first form, applies a temporal convolutional network, takes the last temporal embedding, and concatenates it with the static-feature embedding before projection to proxy-label logits.

Together, these backends compare sparse-event attention, ordinary recurrence on a prepared grid, missingness-aware recurrence with learned decay, and temporal convolution. All four have archived prediction outputs for both datasets, while checkpoint-to-export, split, and archive-copy lineage remains incomplete.

The supervised evaluator computes metrics independently for each rule-derived target column in the requested split. Target columns with only one observed class in that split are skipped. For the remaining targets, it computes AUROC, area under the precision-recall curve, and the maximum value of the pointwise minimum of precision and recall. The reported split-level values are simple averages over the non-degenerate target columns, with loss and negative loss also recorded. During supervised training, checkpoint selection uses the validation value of auprc + auroc; this validation score should not be interpreted as a test score. These metrics quantify learnability or reproducibility of rule-derived targets, not clinical construct validity or correctness of the LLM-assisted decision-rule design. A model can reproduce both useful rule structure and systematic rule error.

Training mechanics are shared across the supervised backends at the top-level script. The script constructs a deterministic seed by adding the run index to the configured seed, uses AdamW for trainable parameters, clips gradient norms at 0.3, evaluates at the configured validation schedule, and saves checkpoint_best.bin when the validation selection metric improves. If an eval_train split is evaluated, it is the first 2000 training examples in the current supervised split object, which makes it a diagnostic subset rather than a separate external evaluation set.

6.3 Prediction Export and Normalization

Prediction export is requested with --save_pred_csv_path. The export routine can run after a training run or in an export-only invocation with no training steps scheduled. Immediately before export, the script reloads checkpoint_best.bin from the current output directory only if that file exists. The exported split is selected by --predict_split, whose accepted values are train, val, test, and all; in the all case the exporter iterates over all supervised identifiers in the train, validation, and test split object. The inspected wrapper scripts and archived export logs support predict_split=all for the final prediction CSVs. The logs also record seed=2023, and archived checkpoint_best.bin files can be hashed; nevertheless, the records do not supply split identifiers, a processed-artifact hash, or an independently verified checkpoint-to-export and archive-copy manifest.

For every exported row, the exporter removes labels from the batch, obtains probability outputs from the model, thresholds each probability at 0.5, and writes a combined CSV. The first column is ts_id. It is followed by one probability column named <label>_prob for each target and then one thresholded binary prediction column named exactly as the target label. These binary columns are model-predicted proxy labels, not rule-derived labels and not clinical adjudications.

The combined prediction CSV is not the same file shape required by downstream voter aggregation. The normalization helper validates ts_id, separates probability and binary fields, and writes binary-only voter files. The majority-vote contract then requires a shared identifier and binary proxy schema in the approved order. The archived final vote combines one rule-derived table with predicted-label tables from GRU, GRU-D, STraTS, and TCN; Chapter 5 defines its semantics. This is an algorithmic handoff, not expert consensus.

Chapter 7

Causal Graph and Effect-Estimation Methodology

This chapter defines the causal and estimator interfaces packaged with each observational testbed. The project-specified DAGs expose representation-layer assumptions. Exposure-specific adjustment sets operationalize those graphs, while matching and heterogeneous-effect estimators characterize behavior conditional on the resulting interface. Transparency makes these assumptions inspectable and replaceable; it does not make the graphs correct or supply a known causal answer.

7.1 Dataset-Specific DAGs

The causal component of this project does not learn a causal graph from the observational data. It uses two LLM-assisted, project-specified directed acyclic graphs (DAGs), one for PhysioNet 2012 and one for MIMIC-III, as explicit representation-layer assumptions for selecting adjustment variables and organizing later effect-estimation outputs. A structured clinical and causal elicitation process generated graph proposals that were selected and encoded as active graph-construction source code. Work on LLM-derived soft priors provides a methodological precedent for using semantic judgments as candidate causal relations or orientation information [12], but it does not validate these local graphs or imply that its discovery algorithm was implemented here. The diagrams should therefore be read as source-encoded causal hypotheses that were not validated edge by edge and are not guaranteed complete, not as clinically validated or statistically discovered ground truth.

This separation is important because the main exposure variables in the causal stage are binary proxy states rather than randomized treatments. In the causal code, each selected proxy state is treated as a binary exposure A, the outcome is in-hospital mortality Y, and the graph is used to identify candidate pre-exposure background or latent proxy variables for adjustment. The thesis follows the graphical language of Pearl's DAG and backdoor framework while retaining the cautions around observational, proxy-defined exposures and well-defined interventions discussed by Hernan and Taubman [5, 7]. These cautions are especially relevant here because several variables named as “treatments” in the code are clinically closer to patient states or care-process proxies than to assigned therapeutic interventions.

The LLM has no estimator-time role in this chapter’s workflow. It is not called to inspect patient records, orient edges dynamically, select adjustment variables, or choose estimators during execution. The active graph scripts define the implemented nodes and edges, and the matching and CATE source code determines the implemented adjustment workflow from those encoded graphs and the columns available in each analysis dataframe.

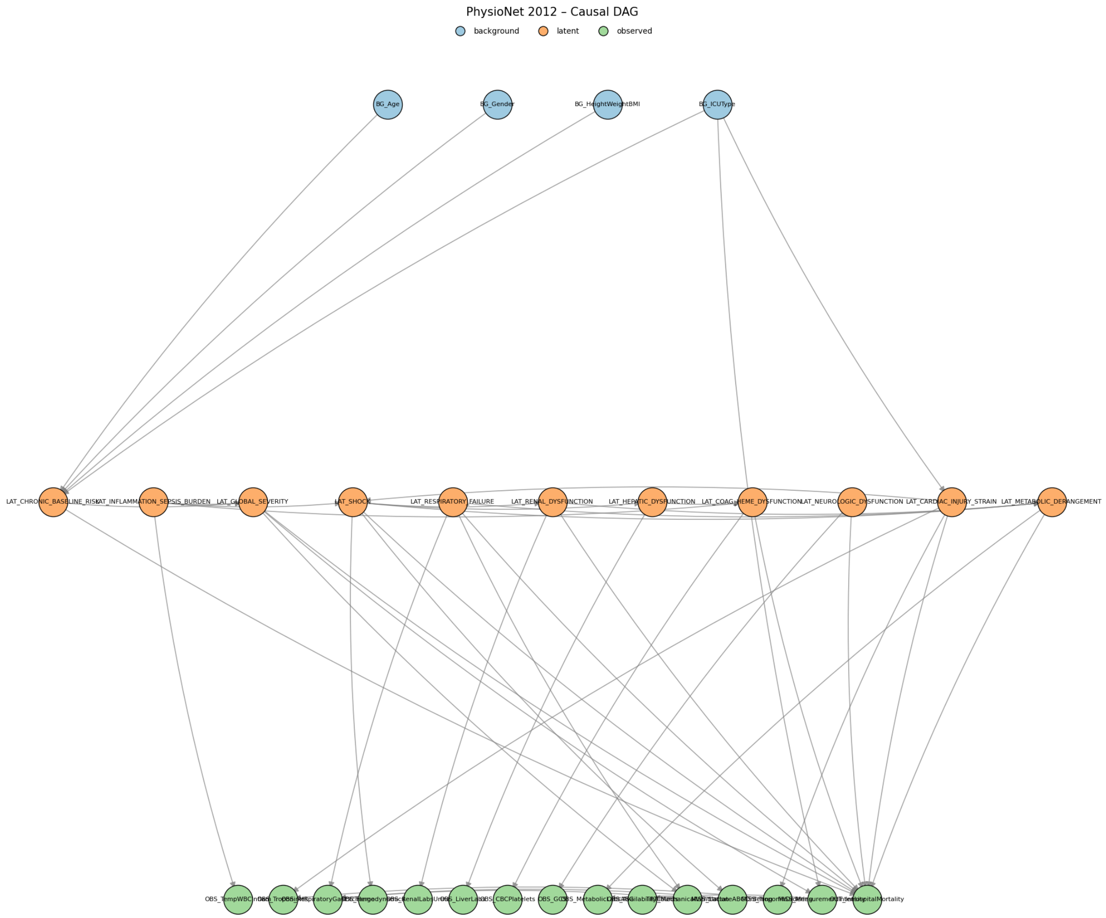

Figure 7.1: Project-specified PhysioNet 2012 DAG encoded by the active causal-pipeline graph source; arrows are assumed, not learned or clinically validated.

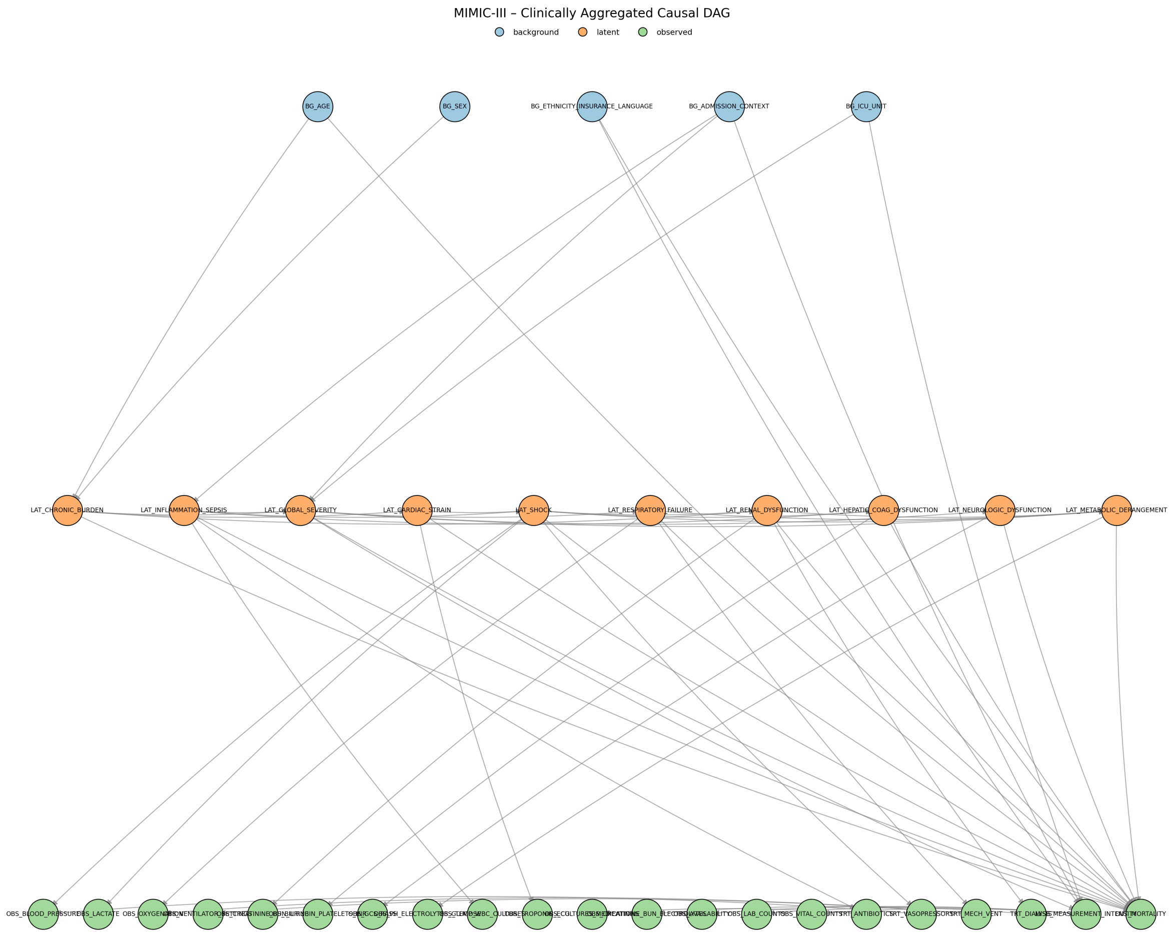

Figure 7.2: Project-specified MIMIC-III DAG encoded by the active causal-pipeline graph source; arrows are assumed, not learned or clinically validated.

Both graphs distinguish background variables, proxy states, observed measurements, care processes, missingness or measurement-intensity processes, and the mortality outcome. Only mapped background and eligible upstream proxy nodes can enter adjustment sets; measurement, care-process, missingness, and outcome nodes remain in the causal hypothesis but are excluded as adjustment candidates. MIMIC includes richer administrative and care-process structure, whereas PhysioNet uses a smaller background and observation vocabulary.

The PhysioNet graph contains 30 nodes and 45 directed edges. The implemented exposure list contains the ten non-chronic latent proxy states in the PhysioNet configuration: global severity, shock, respiratory failure, renal dysfunction, hepatic dysfunction, coagulation/hematologic dysfunction, inflammation/sepsis burden, neurologic dysfunction, cardiac injury/strain, and metabolic derangement. Chronic baseline risk is retained as an upstream baseline proxy and effect-modifier candidate rather than as an analyzed exposure.

The MIMIC graph contains 36 nodes and 57 directed edges. The implemented exposure list contains nine non-chronic latent proxy states: global severity, inflammation/sepsis, shock, respiratory failure, renal dysfunction, hepatic/coagulation dysfunction, neurologic dysfunction, cardiac strain, and metabolic derangement. Chronic burden is retained as an upstream baseline proxy and effect-modifier candidate. The MIMIC graph contains richer administrative and care-process structure than the PhysioNet graph, but the downstream adjustment implementation only uses graph nodes that can be mapped into the analysis dataframe.

Qualified clinical review of the graph arrows and a complete record of the human decisions applied to the LLM-assisted proposals remain unavailable. The graph artifacts and active source structure are recorded, but the exact producing source version, configuration, and command are not fully recovered.

The archived final run summaries and majority-vote logs establish the downstream proxy-label lineage at the stage boundary. In all twelve runs, the vote combines five enumerated inputs: one deterministic rule-derived proxy-label table and four thresholded prediction tables from GRU, GRU-D, STraTS, and TCN. Each run writes `majority_vote/latent_tags_majority_vote.csv` and passes that mixed rule-and-prediction aggregate to mortality prediction, matching, and CATE estimation. The logs preserve the external absolute filenames, but not the voter-file bytes, hashes, or checkpoint-to-export mappings; those parts of the lineage therefore remain incomplete.

7.2 Adjustment-Set Logic

Exposure-specific adjustment sets convert the project graphs into an executable analysis interface. They record what the encoded graph and available columns permit the pipeline to adjust for; they do not establish that the representation contains every confounder or that exchangeability holds.

For each dataset, treatment-like proxy state, and outcome pair, the causal pipeline derives an adjustment set from the project DAG and the columns available in the analysis dataframe.

The implementation first maps dataframe variables to graph nodes and graph nodes back to dataframe columns. Background and latent graph nodes are eligible only if they can be represented by available columns. Observed measurement nodes, care-process nodes, missingness nodes, and the outcome node are excluded from candidate adjustment sets.

Let G denote the implemented project DAG, A the binary proxy-state exposure, and Y the in-hospital mortality outcome. The code constructs a graph $G_{\underline{A}}$ by removing outgoing edges from A. This is a project helper for finding adjustment candidates; it should not be confused with the usual intervention graph notation that removes incoming edges into A. The initial candidate pool is:

$$\mathcal{C}(A,Y) = \mathcal{A}_{\text{allowed}} \cap \mathrm{An}_G(A) \cap \mathrm{An}_{G_{\underline{A}}}(Y) \setminus \mathrm{De}_G(A) \setminus \{A,Y\},$$

where $\mathcal{A}_{\text{allowed}}$ is the set of available graph nodes whose node type is background or latent. The implementation then enumerates backdoor paths between A and Y in the undirected skeleton whose first directed edge points into A. A path is treated as blocked if an adjusted non-collider is present or if an unadjusted collider has no adjusted descendant. This is an encoded graphical test over the project DAG, not proof that all real-world confounding has been controlled.

The preferred helper is the safe expanded adjustment mode. It removes encoded colliders and their descendants from the full candidate pool, then tests whether the remaining mapped variables block every encoded backdoor path. If not, a deterministic greedy fallback removes sorted candidates only while path blocking is preserved. The returned set is inclusion-minimal under that order, not necessarily unique. The accompanying identifiability flag is therefore an availability diagnostic within the implemented graph, not proof of exchangeability in the clinical population.

Table 7.1: Causal assumptions required for interpreting the estimator outputs.

assumption	methodological requirement	repository support	unresolved limitation
Graph correctness for the target question	The DAG must correctly represent the relevant causal structure for the exposure contrast and mortality outcome.	Dataset-specific graphs were developed through LLM-assisted elicitation, project selection, and active source encoding.	The arrows were not learned from data, and the human clinical review record remains incomplete.

assumption	methodological requirement	repository support	unresolved limitation
Sufficient measured adjustment	Available adjustment variables must block the relevant open backdoor paths for each proxy exposure.	The helper tests path blocking within the implemented DAG using mapped background and latent columns.	Graph variables may be unavailable, and unmeasured or misspecified confounding outside the project DAG remains possible.
Positivity or overlap	Comparable covariate strata must contain both exposed and unexposed records.	Matching outputs report match availability and match rates.	Dedicated treatment-specific overlap diagnostics and thresholds remain to be approved.
Consistency and well-defined exposure	Each binary proxy-state contrast must have a sufficiently stable meaning across records.	The pipeline applies one binary exposure column at a time.	Disease-state and severity proxies are not necessarily assignable interventions, so the causal estimand remains unresolved.
Proxy-state measurement	The binary exposure should measure the intended construct with sufficiently limited construct and classification error.	Final archived analyses use a deterministic aggregate of one rule-derived source and four predicted-label sources trained against the rule-derived targets.	No code path verifies construct validity or removes proxy-state measurement error; clinical and chart-review validation remains incomplete.
Outcome measurement	The processed `in_hospital_mortality` field must be consistently and correctly constructed from the outcome artifact.	Both estimator paths merge this processed outcome column by record identifier and require it to be present.	Source checks establish schema availability, not endpoint validity or cross-dataset construction equivalence.

assumption	methodological requirement	repository support	unresolved limitation
Temporal ordering and avoidance of post-exposure adjustment	Adjustment variables must be measured or conceptually occur before the relevant proxy-state contrast; post-exposure adjustment should be avoided.	Protection depends on the project DAG, source node classification, descendant filtering, and allowed-node filtering.	Repository graph ordering does not prove patient-record timing; actual variable timing and the correctness of the proxy-state construction window remain unverified.
Nuisance- and effect-model adequacy	LinearDML and CausalForestDML require sufficiently adequate nuisance and effect models, regularity conditions, and appropriate cross-fitting behavior for their intended interpretation.	The source configures EconML estimators, random-forest nuisance models, and estimator-provided fitting behavior.	Source configuration alone does not verify model adequacy, regularity conditions, calibration, or valid cross-fitting performance in these data.
No interference	One record's proxy-state exposure should not affect another record's outcome under the working unit-level interpretation.	Estimation treats records as separate units.	The repository does not empirically test interference or dependence across admissions, patients, units, or care processes.
Sampling and target population	The analysis sample must support the stated target population, or reweighting or another transport argument must justify a sampled target.	The optional outcome-downsampling condition is deterministic given the configured seed and is applied consistently before treatment iteration.	Original and outcome-downsampled analyses correspond to different empirical populations; no weighting or target-population correction is documented.

7.3 Matching and Heterogeneous-Effect Estimators

The causal pipeline applies two families of estimators to the mixed-source majority-vote proxy-label exposures: a transparent matched-pair baseline and model-based heterogeneous-effect estimators. Matching is a secondary descriptive and support-oriented comparison, not a definitive treatment-effect estimator. Both families use the same broad analysis frame. Patient-level aggregated proxy labels are merged with the canonical in-hospital mortality outcome and baseline covariates. For a selected exposure $A_i \in \{0, 1\}$, outcome Y_i, adjustment variables W_i, and effect modifiers X_i, the target working quantity is a contrast in predicted or matched mortality outcome between $A_i = 1$ and $A_i = 0$, conditional on the source-defined covariate information. Because the exposures are proxy states and the data are observational, often without a simple well-specified intervention, these contrasts require cautious interpretation.

Measurement error can propagate through this entire chain. An unsupported design proposal can enter a selected rule or edge; a deterministic rule can misclassify the intended construct; a deep model can reproduce that error or add prediction error; aggregation can preserve correlated error; and the resulting exposure misclassification can alter adjustment, matching, and effect estimates. An LLM-assisted graph is not intrinsically more correct than a manually proposed graph, and predictive agreement with source-encoded labels does not repair graph or exposure-measurement assumptions.

An optional analysis condition, controlled by `DOWN_SAMPLE`, down-samples mortality outcome classes rather than exposure groups. All outcome-positive rows are retained. When the outcome-negative class is larger and the positive class is nonempty, outcome-negative rows are sampled without replacement until their count equals the number of outcome-positive rows; the configured random seed controls both this sample and the subsequent row shuffle. This occurs once before treatment-specific adjustment selection and estimation, including matching and DML/PFN fitting. It is not propensity trimming, treatment balancing, matching, or a correction for confounding, and it does not improve causal identification by itself. The procedure changes the analyzed population and observed outcome prevalence. The thesis therefore treats original cohorts as primary and outcome-downsampled cohorts as robustness analyses only, with no pooling between them.

Outcome-based downsampling can change nuisance-model estimation, the distribution of covariates and proxy exposures, patient-level fitted effects, the population over which `mean_cate` is averaged, and the availability and composition of matched pairs. It also changes the sample outcome rate and therefore affects `normalized_CATE`, whose denominator is that rate. Effect summaries from original and outcome-downsampled analyses should not be compared as though they estimated the same population quantity without an approved sampling and estimand argument. Outcome-based case-control sampling does not necessarily invalidate every estimator, but interpretation here requires estimator-specific methodological justification and an explicit target-population argument.

The matching estimator is a deterministic baseline. For each proxy exposure, the pipeline drops rows with missing exposure or outcome, coerces the exposure and outcome to binary variables, imputes numeric adjustment variables with medians, and converts non-binary numeric adjustment variables into binary columns using sample-median thresholds. Already-binary variables, including ICU-type indicators, remain binary. The matcher then greedily pairs exposed and unexposed patients by Hamming distance over these binary adjustment columns, increasing the allowed distance up to the configured maximum. By default, controls are not reused. The per-pair contrast is

$$\Delta_j = Y_{j,A=1} - Y_{j,A=0},$$

and the matching summary reports the mean and standard deviation of these matched-pair differences, the number of matched pairs, the match rate, the final allowed Hamming distance, and the observed adjustment columns. This output is useful as a transparent matched descriptive contrast. It should not be labeled as an average treatment effect unless the estimand, graph assumptions, overlap, and matching design are later approved for that interpretation.

For heterogeneous-effect modeling, the pipeline includes two EconML DML estimators and an exploratory CausalPFN branch. CausalForestDML is primary, LinearDML the secondary comparator, and CausalPFN exploratory. Double machine learning estimates nuisance outcome and treatment models before estimating residualized effects, reducing first-order sensitivity to nuisance-model error under appropriate assumptions [13, 25]. LinearDML uses random-forest nuisance models with a linear final effect model; CausalForestDML uses the same nuisance family and a causal-forest final stage [14]. CausalPFN is a prior-fitted transformer for amortized in-context causal-effect estimation [26]. Its exact historical producing package version and checkpoint are unresolved, and it lacks DML-equivalent uncertainty, sensitivity, and permutation support. ICU observational-design guidance remains relevant to all three estimators [15].

The CATE pipeline saves patient-level CATE values, normalized CATE values, optional 95 percent effect-interval columns when the estimator provides them, feature-importance or coefficient diagnostics when available, model artifacts, and run-level summaries. In the current source, `mean_cate` is the arithmetic mean of patient-level CATE estimates over the model rows. It is an aggregate of the fitted model's conditional estimates and should not automatically be described as the study ATE. The normalized CATE column is computed by dividing CATE by the sample outcome rate when that rate is positive. It is an implementation-defined rescaling for comparison within this project; it is not a risk ratio, relative risk, or percent reduction.

The effect-modifier matrix X is selected from configured effect-modifier columns, typically baseline variables plus chronic and inflammation/sepsis proxy states when present. The adjustment matrix W is the observed confounder set selected from the DAG helper. The implementation can allow overlap between X and W, because configured effect modifiers are not automatically removed from the adjustment set. This is a source-supported behavior and a remaining methodological limitation.

Table 7.2: Implemented causal estimators and interpretation limits.

method	implementation summary	main outputs	interpretation boundary
Greedy Hamming matching	Binary adjustment matrix, median thresholding for non-binary covariates, median imputation, nearest available control within allowed Hamming distance, no default control reuse.	Matched pairs, pair effects, match rate, final distance, and per-treatment/global summaries.	Matched descriptive contrast; not automatically ATE or ATT.
LinearDML	EconML `LinearDML` with random-forest nuisance outcome and treatment models, binary treatment handling, configured W and X.	Patient-level CATE, optional intervals, coefficients when aligned, model artifact, global summaries.	Model-based heterogeneity estimate under project DAG, nuisance-model, overlap, and proxy-exposure assumptions.
CausalForest DML	EconML `CausalForestDML` with random-forest nuisance models and a causal-forest final stage.	Patient-level CATE, optional intervals, feature importance, model artifact, global summaries.	Flexible heterogeneity estimate; feature importance is a model diagnostic, not mechanism proof.
CausalPFN	Exploratory branch using deduplicated confounder and effect-modifier features. It writes CATE CSVs and metadata but skips EconML-specific sensitivity and permutation stages.	Patient-level CATE and normalized CATE; interval columns are unavailable and stored as missing.	Primary method source cited; exact producing package version/checkpoint, uncertainty, and comparable diagnostics remain unresolved.

Across these interfaces, model-estimated CATE summaries remain conditional on proxy measurement, temporal ordering, graph adequacy, observed adjustment, empirical support, model specification, and the estimand definition. The hierarchy remains: CausalForestDML is primary, LinearDML is the secondary comparator, and CausalPFN is exploratory and lacks equivalent archived uncertainty, sensitivity, and permutation support. Sign, rank, or magnitude agreement among fitted estimators supports converging-evidence characterization under the fixed representation and population; it does not measure causal accuracy or establish identification.

Chapter 8

Converging-Evidence Evaluation, Robustness, and Sensitivity

This chapter is the methodological center for evaluating the two observational testbeds without a causal answer key. It defines the experiment families, population hierarchy, converging-evidence axes, diagnostic framework, result-admission policy, and reproducibility boundary underlying Chapter 9. The axes are deliberately non-equivalent: each investigates a different failure mode, and their findings cannot be added together as proof of clinical correctness, causal identification, or estimator accuracy. They supplement rather than replace the project DAG, exposure definition, adjustment assumptions, empirical support, and clinical review.

8.1 Converging-Evidence Axes

8.1.1 Construct plausibility

Construct plausibility asks whether a proposed representation has a recognizable clinical rationale and an inspectable operational definition. The evidence consists of literature-grounded representation design, the preserved rationales and source links, rule inspection, and the bounded informal ICU-clinician consultation reported by the finished study. There was no blinded formal clinician-rating study, chart adjudication, inter-rater reliability analysis, or clinical construct-validation study. This axis can identify an implausible or poorly specified construct, but it cannot validate a proxy as a diagnosis, establish that a graph is correct, or show that the encoded state is causally adequate.

8.1.2 Proxy recoverability and mortality-relevant predictive information

Proxy recoverability, detailed in Chapter 6, asks whether the deterministic rule-derived labels correspond to learnable patterns in the irregular source-derived representation. Performance against those labels measures reproduction of a representation-defined target, not clinical correctness. A separate mortality-prediction task asks whether proxy representations contain information associated with source-observed in-hospital mortality. Such prognostic information can characterize a resource, but it remains associational and does not establish causal relevance, diagnostic accuracy, intervention utility, correctness of the proxy, or correctness of the DAG.

8.1.3 Cross-estimator stability

Cross-estimator stability asks whether fitted effect patterns are unique to one estimator under a fixed representation and population. Three distinct dimensions are retained: sign agreement compares direction, rank agreement compares the ordering of exposures within a resource, and magnitude agreement compares numerical scale. These dimensions are not interchangeable, and estimators may also target or approximate different quantities. Agreement supplies evidence that a pattern is not confined to one fitted estimator; it does not establish estimator accuracy, estimand equivalence, identification, or truth.

8.1.4 Clinical-literature comparison

Clinical-literature comparison provides contextual corroboration by relating a CliniCause proxy and its model-estimated mortality contrast to the nearest usable published clinical construct. Where available, the comparison records the represented clinical construct, operational definition, study population, comparison group, mortality horizon, effect scale, adjustment method, reported mortality contrast, and major comparability limitations. Candidate comparisons are interpreted only at the level supported by the finished paper and supplement.

External studies can differ in clinical definition, severity, eligibility, data source, calendar period, treatment context, observation window, mortality horizon, matching or adjustment, estimand, empirical support, and residual confounding. The comparison is therefore a way to identify broad compatibility and discrepancies. It is not a meta-analysis, calibration against causal truth, external validation of a proxy, or proof that a CliniCause estimate is correct. Numerical synthesis is reserved for Chapter 9 and is not introduced in this methods chapter.

8.2 Populations and Experiment Families

8.2.1 Predictive experiments

The predictive comparison crosses PhysioNet 2012 and MIMIC-III with four learned models: STraTS, GRU, GRU-D, and TCN. Each model predicts dataset-specific deterministic rule-derived proxy labels, not mortality or the later aggregate. The workflow includes split-aware loading, multi-label training, validation-based checkpoint selection, test evaluation, and probability/binary prediction export; STraTS additionally supports self-supervised pretraining. These experiments operationalize the proxy-recoverability axis rather than clinical validation. Training summaries and prediction exports exist for all eight dataset–model combinations. Processed-pickle hashes, split identifiers, complete producing commands, checkpoint-to-export mappings, and archive-copy histories remain incomplete, so implementation availability, artifact availability, and complete provenance are not interchangeable claims.

8.2.2 Causal experiments and population hierarchy

The causal design crosses two datasets, three estimators, and two sampling conditions, yielding twelve archived run families. CausalForestDML is primary, LinearDML secondary, and CausalPFN exploratory. Original cohorts define the primary analyses. The robustness condition retains all mortality-positive rows and samples mortality-negative rows before exposure-specific matching and estimation; it therefore changes the empirical population and outcome prevalence and is never pooled with the original cohorts.

Every final run uses a deterministic five-source aggregate: one rule-derived table plus thresholded predictions from STraTS, GRU, GRU-D, and TCN. The runtime order is graph construction, mixed-source aggregation, mortality prediction, matching, CATE estimation, saved-CATE analysis, and permutations. CausalPFN completes CATE estimation but its EconML-specific saved-analysis and permutation stages are intentionally skipped, not failed. Within each run, all prespecified dataset-specific non-chronic proxy exposures are evaluated subject to column, graph-node, binary-value, adjustment, support, and artifact checks; the exposure list is a design commitment rather than a set of causal conclusions.

8.3 Executed Settings and Provenance Status

Table 8.1 separates what the checked archive supports from what only current source defines and what remains unresolved. *Artifact-supported* denotes a checked summary, export, run summary, or diagnostic record; *log-supported* denotes an argument recorded in archived logs without a complete immutable producing manifest; *source-defined* denotes implemented behavior that must not be mistaken for a recovered historical setting. *Unresolved* and *not applicable* retain their literal meanings. This classification is necessary because a current default, a historical path, and an archived output answer different reproducibility questions.

Table 8.1: Compact specification of executed or execution-supported settings and unresolved lineage.

Analysis component	Confirmed or execution-supported settings	Unresolved provenance or interpretation boundary
Prediction scope and artifacts	*Artifact-supported*: PhysioNet 2012 and MIMIC-III crossed with STraTS, GRU, GRU-D, and TCN; all eight dataset–model combinations have checked training summaries and prediction exports. The task is multi-label prediction of dataset-specific rule-derived proxy labels.	*Unresolved*: processed-pickle hashes, split identifiers, producing source revision, and archive-copy direction. Artifact availability does not establish an independently auditable out-of-sample membership.

Analysis component	Confirmed or execution-supported settings	Unresolved provenance or interpretation boundary
Archived predictive run settings	*Artifact-supported*: checked training summaries identify run `1o10`, training fraction 0.5, validation and test evaluation, and checkpoint selection by AUPRC plus AUROC for each of the eight learned combinations. *Log-supported*: archived argument records give seed 2023 and `predict_split=all`.	*Unresolved*: the seed is not a split manifest; the raw shell command, immutable producing configuration, and checkpoint-to-export mapping are absent. `all` is the recorded export argument, not proof of a particular split membership.
Prediction export contract	*Artifact-supported*: each checked export has unique `ts_id` rows, one `<label>_prob` field and one binary `<label>` field per active target, valid probability ranges, and binary values. *Source-defined*: the binary export threshold is 0.5.	*Unresolved*: archived exports are hashable, as are candidate best checkpoints, but no immutable record proves which checkpoint produced each export.
STraTS pretraining	*Source-defined*: STraTS alone has a separate self-supervised pretraining path before supervised fine-tuning; GRU, GRU-D, and TCN do not use that path. Archived STraTS training and checkpoint artifacts exist.	*Unresolved*: pretraining-checkpoint, supervised-checkpoint, and final-export lineage is incomplete; source support does not prove the exact pretraining state used historically.
Causal run matrix and hierarchy	*Artifact-supported*: two datasets, CausalForestDML, LinearDML, and CausalPFN, and original/outcome-downsampled conditions yield twelve archived run families with stage records. Original cohorts are primary; outcome-downsampled populations are separate robustness analyses. CausalForestDML is primary, LinearDML secondary, and CausalPFN exploratory.	*Unresolved*: numbered final causal configuration files and processed-input hashes are missing. Recorded historical paths are not portable configurations, and current unnumbered configs are only candidates.
Exposure, aggregation, and support	*Artifact-supported*: every archived final run records the same five-source mixed aggregate—one rule-derived table plus binary predictions from STraTS, GRU, GRU-D, and TCN. *Source-defined*: analysis is exposure specific; graph/column checks govern exposure admission, while matching supplies descriptive support and pair-availability diagnostics before CATE interpretation.	*Unresolved*: voter bytes and hashes are not co-archived per run, and matching does not establish positivity, balance, exchangeability, or a common estimand across sampling conditions.

Analysis component	Confirmed or execution-supported settings	Unresolved provenance or interpretation boundary
Sensitivity evidence	*Artifact-supported*: CausalForestDML and LinearDML records distinguish fitting-time availability, saved-training direct values, later artifact analysis, labelled fallback reconstruction, benchmark status, partial rows, and failures. *Not applicable*: the archived CausalPFN branch has no equivalent EconML saved-analysis family.	*Unresolved*: evidence classes are not numerically interchangeable; incomplete or failed rows cannot be promoted to effects or sensitivity magnitudes, and producing environment lineage remains partial.
Permutation checks	*Artifact-supported*: validated CausalForestDML and LinearDML records contain 10 trials with seed 42 for both treatment-column and outcome permutations in each archived dataset–exposure–sampling combination. Treatment permutations alter the analyzed proxy column; outcome permutations alter mortality in a copied outcome object. *Not applicable*: CausalPFN permutations were intentionally skipped.	These checks diagnose behavior under disruption; they are not randomization tests or identification proof. Temporary trial artifacts were removed after aggregation, so only checked aggregate records remain.
Producing lineage	*Artifact-supported*: causal run summaries retain stage argv arrays, statuses, return codes, and limited interpreter paths; predictive logs retain argument records and limited device fields; result and reproducibility packages retain hashes and schemas.	*Unresolved*: missing processed-pickle hashes and split manifests; incomplete predictive producing commands and checkpoint mapping; missing numbered causal configs; incomplete source, package, hardware, and archive-copy lineage. Requirements are intended dependencies, not producing-environment proof.

8.4 Matching and Empirical Support

Matching provides a secondary descriptive comparison and indirect support evidence under its implemented binary representation. For each proxy exposure, the summaries record available pairs, match rate, final Hamming distance, a sufficient-pair flag, maximum-distance warnings, and explicit failures. Binary confounders are retained, while other numeric confounders may be median-binarized; greedy one-to-one matching then searches within an allowed distance. A larger accepted distance therefore indicates weaker exact similarity in this representation, not a calibrated distance in the original covariate space. Failure to obtain pairs is informative support evidence rather than a zero effect. Matched outcome differences remain descriptive and estimator-dependent; successful matching or a high match rate does not prove positivity, identification, or balance.

Dedicated propensity distributions, common-support plots, pre/post-match balance tables, standardized mean differences, and weighting effective sample sizes were not present in the approved archive. The available matching fields help identify unsupported comparisons but do not establish global or local positivity, covariate balance, exchangeability, or transportability [27].

8.5 Permutation or Disruption Checks

Permutation checks disrupt one relationship while retaining the remaining run structure. Treatment permutations shuffle only the currently analyzed proxy column; outcome permutations shuffle only mortality in a copied outcome object. Other proxy columns, time-series data, identifiers, graph, configuration, estimator family, and aggregation fields remain fixed as appropriate. Failures are recorded, temporary trial artifacts are removed after summary extraction, and the validated DML records contain ten trials with seed 42 for each dataset–exposure–sampling combination. These implemented disruption checks are sanity diagnostics, not formal randomization tests, p-values, calibration proof, or evidence that the DAG and exposure are valid. CausalPFN permutations are intentionally absent.

A response to disrupted exposure or outcome labels tests whether the fitted pipeline distinguishes the observed structure from deliberately broken structure under the implemented procedure. Even a clear response is a sanity check on pipeline behavior, not causal validation or proof that the original estimate is correctly identified.

8.6 Omitted-Variable Sensitivity

The DML workflow distinguishes method availability, values called during fitting, saved-training direct values, later saved-artifact recomputation, explicitly labelled fallback reconstruction, benchmark-confounder comparisons, and rendered contours. These evidence types are not numerically interchangeable. A value must retain its source and status; unavailable, partial, failed, and not applicable are statuses rather than sensitivity magnitudes. Benchmarking against an observed covariate supplies an interpretable scale but does not bound every possible unmeasured confounder. Robustness-value and omitted-variable-bias methods motivate the analysis without validating the local artifacts or identification assumptions [28, 29, 25]. No sensitivity contour is admitted as main-text numerical evidence, and CausalPFN lacks an equivalent archived sensitivity family.

This axis asks how conclusions change under a specified model of unobserved confounding. It does not establish the actual amount of confounding, prove that no confounding exists below a threshold, validate the DAG, or determine whether an estimate is true or false.

8.7 Population Perturbation

The original-versus-outcome-downsampled design studies dependence on population composition and outcome prevalence. The original cohorts remain primary, and the sampled populations remain separate: downsampling changes the empirical population, nuisance-fitting problem, support, and averaging distribution. Stability or change across those populations is evidence about this specified perturbation, not a replacement for external validation and not evidence that the original cohort is unbiased.

8.8 Provenance, Evidence Admission, and Reproducibility Boundaries

Before a value enters Chapter 9, its producing run, dataset, model or estimator, sampling condition, source path, schema, status, and population must be identified. Failed artifacts are excluded; fallback diagnostics retain their labels; source fields are interpreted at their actual granularity; and the estimand and unresolved limitations remain explicit. Directory names, copied summaries, or non-null cells alone are insufficient. This policy preserves the original-cohort and estimator hierarchy and prevents numerical availability from being mistaken for scientific admissibility.

The result manifest and compact reproducibility package record available paths, hashes, schemas, commands and log arguments, seeds, environment fields, and artifact roles. They distinguish current source behavior, archived numerical artifacts, checked result rows, reconstructed summaries, unresolved producing lineage, and full clean-checkout reproducibility. These classes support numerical transcription and execution traceability, but not clean-checkout reproduction or scientific identification. Raw and processed data and numbered final causal configurations are absent; predictive split and checkpoint lineage, source versions, environments, and archive history are incomplete; and historical paths are not portable. Requirements and current defaults describe interfaces, not necessarily the producing environment. Provenance controls what may be claimed from a testbed; provenance quality does not itself establish clinical or causal validity.

8.9 Synthesis of the Non-Equivalent Axes

The axes remain non-equivalent: plausibility, recoverability, prognostic information, estimator stability, clinical comparison, support, disruption, sensitivity, population perturbation, and provenance each diagnose a bounded aspect of the resource. As established above, none supplies clinical validation, identification, estimator accuracy, a common estimand, or a causal answer key; Chapter 9 reports their converging and discrepant findings without combining them into a score.

Chapter 9

Results

This chapter characterizes two separately instantiated observational causal-analysis testbeds rather than treating the archive as a single pipeline output. The evidence sequence covers construct plausibility, proxy recoverability, admitted mortality-relevant information, primary forest patterns, estimator sign/rank/magnitude stability, clinical-literature comparison, matching and empirical support, permutation disruption, sensitivity, and provenance. These dimensions are complementary but non-equivalent: success on one does not establish success on another, and none supplies a causal answer key.

Only results admitted through the validated result-source tables and prespecified evidence hierarchy are reported. Original-cohort analyses are primary: CausalForestDML is the primary causal-model estimator, LinearDML is the secondary comparator, and CausalPFN is exploratory. All prespecified dataset-specific proxy-state exposures are retained; none was selected according to its observed result. In all twelve archived causal runs, the implemented majority-vote stage combines one rule-derived proxy-label source with predicted-label sources from GRU, GRU-D, STraTS, and TCN, and passes the resulting aggregate downstream. Detailed robustness outputs are summarized selectively, and outcome-downsampled analyses are treated only as a robustness perspective. Every model-estimated contrast remains conditional on the causal assumptions and proxy-exposure interpretation in Chapter 7.

9.1 Analysis-Population Counts

The two instantiated testbed populations were kept separate. The original causal-analysis population contained 26,845 MIMIC-III records with nine analyzed proxy-state exposures and 7,993 PhysioNet records with ten analyzed proxy-state exposures; no dataset pooling was performed. MIMIC-III therefore contributed 18,852 more records and was approximately 3.36 times as large. These counts describe checked original-cohort causal-analysis model rows, not reconstructed raw cohorts. Prediction-export, proxy-label, majority-vote, and causal-analysis artifacts have different contracts and must not be treated as one universally identical cohort count.

9.2 Construct Plausibility

The proxy rationales and deterministic operational rules are inspectable, as are their dataset-specific names. This permits clinical and methodological critique of the representation layer. The finished study also records bounded informal feedback from ICU clinicians about the plausibility of these constructs. This is evidence of construct plausibility, not construct validation: there was no blinded formal clinician-rating study, chart adjudication, inter-rater reliability analysis, or controlled comparison among clinician-only, literature-only, LLM-assisted, hybrid, or alternative-LLM designs. No rating, reviewer-count, agreement-score, or clinical-endorsement claim is therefore made.

9.3 Proxy Recoverability

The checked archive contains eight completed dataset–model test combinations spanning the four learned models STraTS, GRU, GRU-D, and TCN (Table 9.1). Every predictive metric compares model output with deterministic rule-derived proxy-label targets produced by the source-encoded decision trees. Archived test AUROC ranged from 0.867 to 0.918. On MIMIC-III, STraTS had the smallest archived test loss and the highest archived test AUROC, AUPRC, and minRP. On PhysioNet, GRU-D led the same four archived metrics. Thus recoverability differed by dataset, and the leading model also differed, although within each dataset the leader did not depend on which of these four metrics was used. Rankings among the remaining models did vary by metric; the table therefore does not support a universal model ordering or a claim of statistical superiority.

The numerical rows are test summaries, not clinical-validation values. Summary and paired log values agreed within the validated tolerance. The archived train, validation, and test split manifest and checkpoint-to-export lineage remain incomplete. No confidence intervals or between-model significance tests were available. Agreement with a rule-derived target demonstrates learnability, not construct validity or correctness of the LLM-assisted rule design.

Table 9.1: Archived predictive test performance for the four learned models. AUROC, AUPRC, and minRP are macro-averaged across non-degenerate rule-derived proxy-label targets on the archived test split.

Dataset	Model	Test loss	Test AUROC	Test AUPRC	Test minRP
MIMIC-III	STraTS	0.348	0.905	0.869	0.795
MIMIC-III	GRU	0.386	0.881	0.840	0.770
MIMIC-III	GRU-D	0.404	0.884	0.841	0.773
MIMIC-III	TCN	0.414	0.867	0.823	0.758
PhysioNet 2012	STraTS	0.341	0.915	0.877	0.818
PhysioNet 2012	GRU	0.397	0.915	0.897	0.831

Dataset	Model	Test loss	Test AUROC	Test AUPRC	Test minRP
PhysioNet 2012	GRU-D	0.331	0.918	0.905	0.835
PhysioNet 2012	TCN	0.477	0.899	0.875	0.811

9.4 Mortality-Relevant Predictive Information

As a separate noncausal characterization axis, the checked original-cohort MIMIC artifact used ten aggregated proxy features to predict source-observed in-hospital mortality among 26,845 analysis records. With the archived stratified held-out split, test AUROC was 0.826 for class-balanced logistic regression and 0.831 for the MLP. These values match the finished-paper display after three-decimal rounding. They show that the compact MIMIC proxy representation retained mortality-relevant predictive information; they do not show that any proxy caused mortality or that the representation is clinically valid.

The analogous PhysioNet AUROC is withheld because the checked thesis artifact and finished paper use unresolved voter/cohort lineages. Until the producing cohort, voter construction, command, and split are reconciled, selecting either lineage would imply equivalent cross-resource evidence that the archive does not support.

9.5 Primary CausalForestDML Results

Tables 9.2 and 9.3 report every prespecified original-cohort majority-vote proxy-state exposure from that mixed rule-and-prediction lineage. The reported quantity is the *mean model-estimated CATE over the analyzed sample*, on the model outcome scale. It is neither a risk ratio nor an unqualified population-average causal effect. Exposure prevalence is the proportion of analyzed records carrying the corresponding binary aggregated proxy-state tag. The adjustment-set codes refer to the observed variables archived for each exposure; identifiability with the available nodes remained unresolved for every validated row.

These results do not evaluate the LLM itself. The study contains no LLM ablation, clinician comparison, prompt-sensitivity experiment, alternative-LLM replication, or graph-correctness experiment. The numerical analyses therefore cannot establish that LLM assistance improved the rules or DAGs, nor can they separate design error from rule, prediction, aggregation, or estimator error.

For MIMIC-III, all nine archived means were positive. The largest mean was for `LAT_CARDIAC_STRAIN` (0.220), followed by `LAT_INFLAMMATION_SEPSIS` (0.161). The smallest were for `LAT_METABOLIC_DERANGEMENT` (0.020) and `LAT_SHOCK` (0.021). Figure 9.1 displays this source-exact ordering. It is descriptive and does not establish clinical importance or statistical significance.

Table 9.2: Primary original-cohort MIMIC-III CausalForestDML results ($n = 26{,}845$). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
`LAT_CARDIAC_STRAIN`	0.178	0.220	M1	Identifiability unresolved
`LAT_INFLAMMATION_SEPSIS`	0.472	0.161	M0	Identifiability unresolved
`LAT_HEPATIC_COAG_DYSFUNCTION`	0.241	0.098	M2	Identifiability unresolved
`LAT_RENAL_DYSFUNCTION`	0.363	0.091	M3	Identifiability unresolved
`LAT_GLOBAL_SEVERITY`	0.719	0.062	M4	Identifiability unresolved
`LAT_RESPIRATORY_FAILURE`	0.590	0.036	M5	Identifiability unresolved
`LAT_NEUROLOGIC_DYSFUNCTION`	0.434	0.034	M5	Identifiability unresolved
`LAT_SHOCK`	0.540	0.021	M6	Identifiability unresolved
`LAT_METABOLIC_DERANGEMENT`	0.373	0.020	M7	Identifiability unresolved

The MIMIC adjustment codes are: M0, no observed adjustment variables archived; M1, age, gender, and chronic burden; M2, inflammation/sepsis and shock; M3, chronic burden and shock; M4, age, gender, chronic burden, and inflammation/sepsis; M5, age, gender, chronic burden, global severity, and inflammation/sepsis; M6, age, gender, cardiac strain, chronic burden, and inflammation/sepsis; and M7, global severity, renal dysfunction, respiratory failure, and shock. These are observed adjustment-variable records, not proof that all confounding was controlled.

For PhysioNet, nine archived means were positive and the shock mean was negative. Renal dysfunction had the largest mean (0.120), followed by cardiac injury/strain (0.112) and global severity (0.108). Coagulation/hematologic dysfunction was small and positive (0.011), while shock was -0.014. Figure 9.2 retains that negative value and the source-exact ordering; the ranking is descriptive only.

Table 9.3: Primary original-cohort PhysioNet CausalForestDML results ($n = 7{,}993$). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
`LAT_RENAL_DYSFUNCTION`	0.230	0.120	P1	Identifiability unresolved
`LAT_CARDIAC_INJURY_STRAIN`	0.355	0.112	P2	Identifiability unresolved

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
LAT_GLOBAL_SEVERITY	0.777	0.108	P3	Identifiability unresolved
LAT_HEPATIC_DYSFUNCTION	0.142	0.091	P1	Identifiability unresolved
LAT_NEUROLOGIC_DYSFUNCTION	0.659	0.082	P0	Identifiability unresolved
LAT_METABOLIC_DERANGEMENT	0.558	0.078	P4	Identifiability unresolved
LAT_INFLAMMATION_SEPSIS_BURDEN	0.563	0.072	P0	Identifiability unresolved
LAT_RESPIRATORY_FAILURE	0.614	0.065	P0	Identifiability unresolved
LAT_COAG_HEME_DYSFUNCTION	0.395	0.011	P5	Identifiability unresolved
LAT_SHOCK	0.696	−0.014	P6	Identifiability unresolved

The PhysioNet adjustment codes are: P0, no observed adjustment variables archived; P1, ICU-type indicators 1–4, cardiac injury/strain, inflammation/sepsis burden, and shock; P2, ICU-type indicators 1–4; P3, age, gender, weight, ICU-type indicators 1–4, chronic baseline risk, and inflammation/sepsis burden; P4, renal dysfunction, respiratory failure, and shock; P5, inflammation/sepsis burden and shock; and P6, ICU-type indicators 1–4, cardiac injury/strain, and inflammation/sepsis burden. The same qualification about incomplete confounding control applies.

9.6 Matching and Empirical Support

Matching was identical across the three estimator run families within a dataset and sampling condition, so Table 9.4 reports the cross-run-consistent original-cohort rows once, as supporting evidence for the primary CausalForestDML analysis. The numerical contrast is a *descriptive matched-pair outcome difference*. Treated and control record counts were unavailable in the checked table and are left unreported rather than inferred.

Eight of nine MIMIC exposures and seven of ten PhysioNet exposures produced a matched-pair summary. Matching failed for MIMIC inflammation/sepsis and for PhysioNet inflammation/sepsis burden, neurologic dysfunction, and respiratory failure because no binary matching columns remained after preprocessing. A failure is not a zero effect. Among successful rows, final allowed Hamming distance ranged from 0 to 2; increased allowed distance represents weaker exact similarity in the binary matching representation. Three successful rows were flagged as not reaching the configured sufficient-pair criterion: global severity in both datasets and PhysioNet shock.

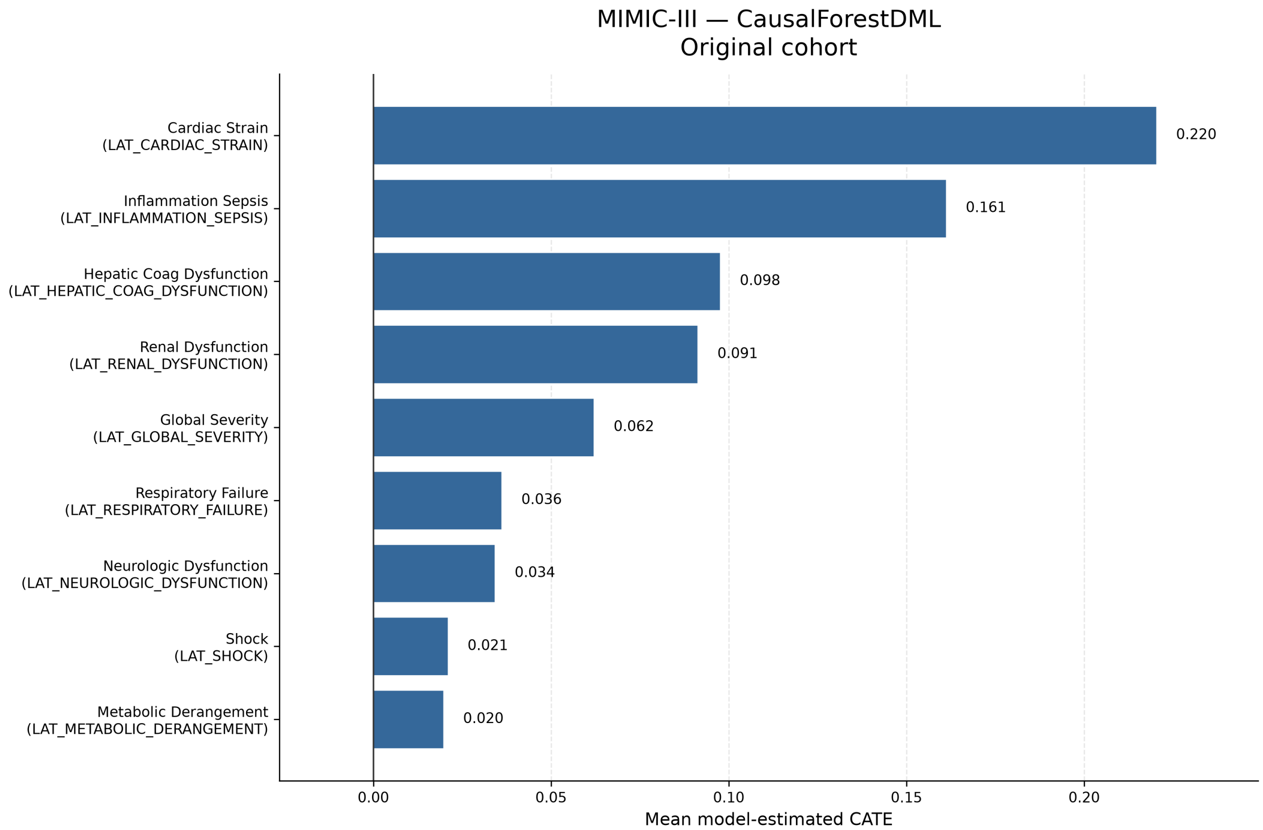

Figure 9.1: Original-cohort MIMIC-III mean model-estimated CATE values from CausalForest-DML. All nine prespecified proxy-state exposures are shown. The ordering is descriptive and does not establish statistical significance, clinical importance, or population-average causal effects.

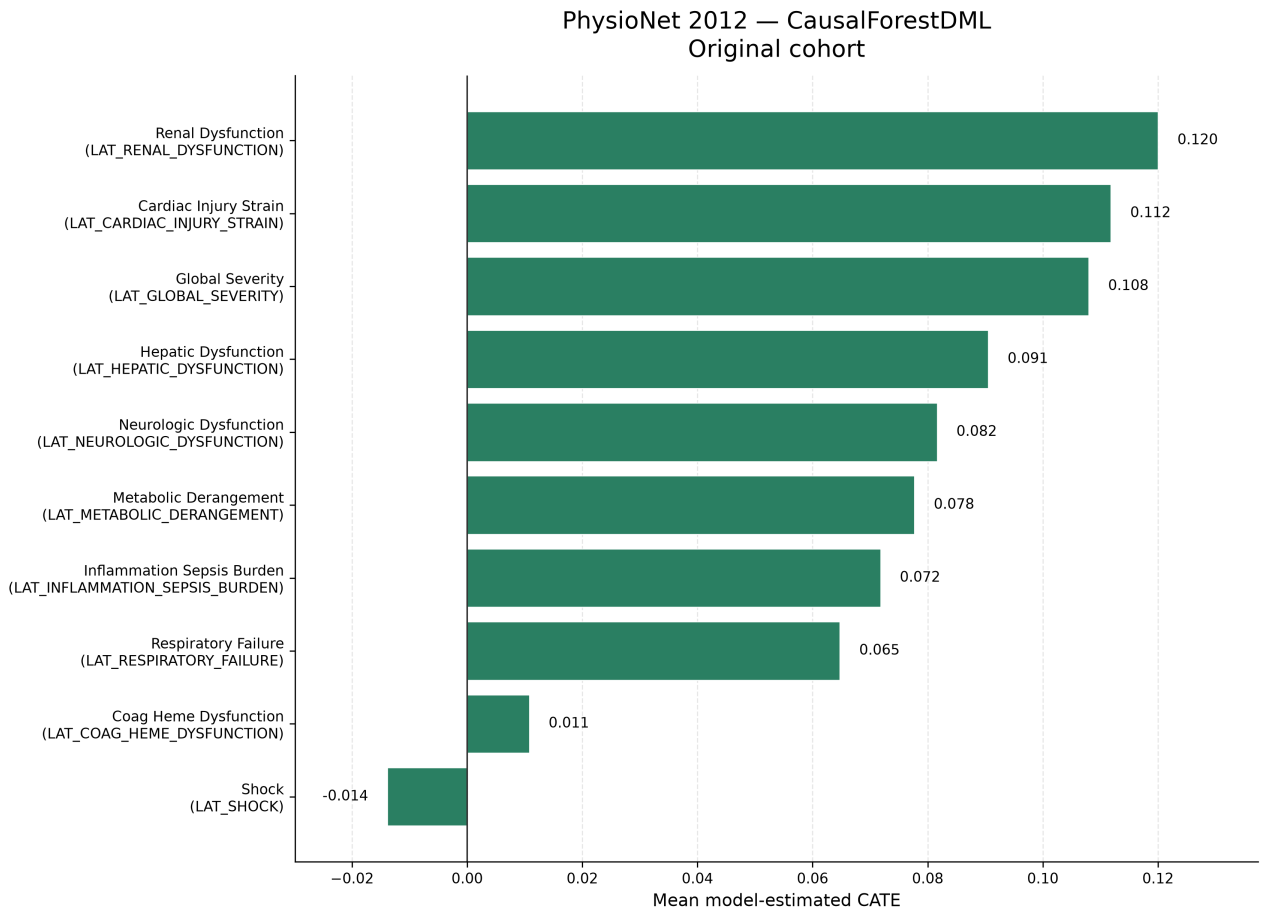

Figure 9.2: Original-cohort PhysioNet mean model-estimated CATE values from CausalForest-DML. All ten prespecified proxy-state exposures are shown. The ordering is descriptive; the negative shock estimate is retained. The figure does not establish statistical significance, clinical importance, or population-average causal effects.

Table 9.4: Original-cohort descriptive matching and empirical-support summaries. "Difference" is the descriptive matched-pair outcome difference; it is not a model-estimated CATE.

Dataset	Proxy-state exposure	Pairs	Rate	Distance	Difference	Support/status
MIMIC	`LAT_CARDIAC_STRAIN`	4,769	1.000	0	0.345	Sufficient
MIMIC	`LAT_GLOBAL_SEVERITY`	7,537	0.390	1	0.075	Weak: insufficient-pair flag
MIMIC	`LAT_HEPATIC_COAG_DYSFUNCTION`	6,478	1.000	0	0.133	Sufficient
MIMIC	`LAT_METABOLIC_DERANGEMENT`	5,061	0.506	0	0.037	Sufficient
MIMIC	`LAT_NEUROLOGIC_DYSFUNCTION`	6,275	0.539	0	0.027	Sufficient
MIMIC	`LAT_RENAL_DYSFUNCTION`	8,915	0.915	0	0.134	Sufficient
MIMIC	`LAT_RESPIRATORY_FAILURE`	10,998	0.694	2	0.106	Sufficient; distance 2
MIMIC	`LAT_SHOCK`	10,380	0.717	1	0.039	Sufficient
MIMIC	`LAT_INFLAMMATION_SEPSIS`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	2,840	1.000	0	0.134	Sufficient
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	1,763	0.558	0	0.0017	Sufficient
PhysioNet	`LAT_GLOBAL_SEVERITY`	1,782	0.287	1	0.095	Weak: insufficient-pair flag
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	1,133	1.000	0	0.137	Sufficient
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	3,378	0.757	2	0.073	Sufficient; distance 2
PhysioNet	`LAT_RENAL_DYSFUNCTION`	1,841	1.000	0	0.154	Sufficient
PhysioNet	`LAT_SHOCK`	2,433	0.438	1	0.010	Weak: insufficient-pair flag
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_RESPIRATORY_FAILURE`	–	–	–	–	Failed: no binary matching columns

For the 15 successful rows, matching and CausalForestDML shared direction in 14 cases. PhysioNet shock was the exception: its matched-pair difference was positive (0.010), while its model-estimated mean was negative (-0.014). This is a descriptive comparison, not independent causal validation. A high match rate does not establish positivity; the procedure is greedy and representation-dependent, and no dedicated propensity-overlap or balance figure was available.

9.7 LinearDML Comparison

LinearDML and CausalForestDML agreed in mean-effect direction for all 19 original-cohort dataset–exposure pairs (Table 9.5). This agreement does not imply equal magnitude or ranking. In MIMIC, CausalForestDML estimated a larger cardiac-strain mean by approximately 0.033, while LinearDML estimated a larger global-severity mean by approximately 0.018; renal and hepatic/coagulation dysfunction exchanged third and fourth positions. In PhysioNet, the largest absolute differences were for hepatic dysfunction (approximately 0.030) and renal dysfunction (approximately 0.029), and the top-ranked mean changed from renal dysfunction under CausalForestDML to cardiac injury/strain under LinearDML. LinearDML imposes a simpler final effect structure; similarity between these archived means does not show that either estimator is unbiased.

Table 9.5: Original-cohort comparison of mean model-estimated CATE from CausalForestDML and LinearDML.

Dataset	Proxy-state exposure	Forest	Linear	Same direction
MIMIC	LAT_CARDIAC_STRAIN	0.220	0.188	Yes
MIMIC	LAT_INFLAMMATION_SEPSIS	0.161	0.161	Yes
MIMIC	LAT_HEPATIC_COAG_DYSFUNCTION	0.098	0.083	Yes
MIMIC	LAT_RENAL_DYSFUNCTION	0.091	0.089	Yes
MIMIC	LAT_GLOBAL_SEVERITY	0.062	0.080	Yes
MIMIC	LAT_RESPIRATORY_FAILURE	0.036	0.039	Yes
MIMIC	LAT_NEUROLOGIC_DYSFUNCTION	0.034	0.032	Yes
MIMIC	LAT_SHOCK	0.021	0.021	Yes
MIMIC	LAT_METABOLIC_DERANGEMENT	0.020	0.018	Yes
PhysioNet	LAT_RENAL_DYSFUNCTION	0.120	0.091	Yes
PhysioNet	LAT_CARDIAC_INJURY_STRAIN	0.112	0.122	Yes
PhysioNet	LAT_GLOBAL_SEVERITY	0.108	0.107	Yes
PhysioNet	LAT_HEPATIC_DYSFUNCTION	0.091	0.060	Yes
PhysioNet	LAT_NEUROLOGIC_DYSFUNCTION	0.082	0.076	Yes
PhysioNet	LAT_METABOLIC_DERANGEMENT	0.078	0.080	Yes
PhysioNet	LAT_INFLAMMATION_SEPSIS_BURDEN	0.072	0.071	Yes
PhysioNet	LAT_RESPIRATORY_FAILURE	0.065	0.063	Yes
PhysioNet	LAT_COAG_HEME_DYSFUNCTION	0.011	0.004	Yes
PhysioNet	LAT_SHOCK	−0.014	−0.027	Yes

9.8 CausalPFN Exploratory Results

CausalPFN CATE execution completed for all 19 original-cohort comparisons. The three estimators agreed in mean-effect direction for 18 of the 19 dataset–exposure comparisons: MIMIC was concordant for 9 of 9 and PhysioNet for 9 of 10 (Table 9.6 and Figure 9.3). The sole exception was PhysioNet `LAT_SHOCK`: CausalForestDML was negative (−0.014), LinearDML was negative (−0.027), and CausalPFN was slightly positive (0.0041).

CausalPFN reproduced the prevailing direction in nearly every comparison, providing an exploratory directional comparison within this pipeline. The agreement is incomplete and does not establish estimator equivalence, superiority, causal validity, or interchangeable uncertainty. Its sensitivity and permutation stages were intentionally skipped rather than failed, and the exact producing package version and checkpoint remain unresolved even though the primary method source is now cited in Chapter 7.

Table 9.6: Original-cohort exploratory CausalPFN mean model-estimated CATE and direction agreement with both DML estimators.

Dataset	Proxy-state exposure	CausalPFN	Forest/Linear signs	All three agree
MIMIC	LAT_CARDIAC_STRAIN	0.259	+, +	Yes
MIMIC	LAT_INFLAMMATION_SEPSIS	0.149	+, +	Yes
MIMIC	LAT_HEPATIC_COAG_DYSFUNCTION	0.097	+, +	Yes
MIMIC	LAT_RENAL_DYSFUNCTION	0.087	+, +	Yes
MIMIC	LAT_GLOBAL_SEVERITY	0.073	+, +	Yes
MIMIC	LAT_RESPIRATORY_FAILURE	0.050	+, +	Yes
MIMIC	LAT_NEUROLOGIC_DYSFUNCTION	0.035	+, +	Yes
MIMIC	LAT_SHOCK	0.032	+, +	Yes

Dataset	Proxy-state exposure	CausalPFN	Forest/Linear signs	All three agree
MIMIC	`LAT_METABOLIC_DERANGEMENT`	0.031	+, +	Yes
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	0.119	+, +	Yes
PhysioNet	`LAT_RENAL_DYSFUNCTION`	0.111	+, +	Yes
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	0.107	+, +	Yes
PhysioNet	`LAT_GLOBAL_SEVERITY`	0.105	+, +	Yes
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	0.091	+, +	Yes
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	0.077	+, +	Yes
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	0.067	+, +	Yes
PhysioNet	`LAT_RESPIRATORY_FAILURE`	0.063	+, +	Yes
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	0.025	+, +	Yes
PhysioNet	`LAT_SHOCK`	0.0041	−, −	No

9.8.1 Sign, rank, and magnitude stability

Table 9.7 separates three forms of within-resource agreement. Direction is a sign comparison. Spearman's ρ compares exposure ordering. Pairwise RMSE compares the three estimators' mean-CATE magnitudes on the percentage-point scale; it measures estimator disagreement, not error against causal truth. All statistics use the full-precision original-cohort rows rather than the rounded display tables.

Table 9.7: Within-resource estimator stability for the original cohorts. Rank correlations and RMSE values are ranges across the three estimator pairs. RMSE is disagreement on the model-estimated mean-CATE scale, not estimator accuracy.

Dataset	Forest/Linear sign	All-three sign	Spearman ρ	RMSE (pp)
MIMIC-III	9/9	9/9	0.983–1.000	1.35–2.57
PhysioNet 2012	10/10	9/10	0.794–0.964	1.08–2.04

Thus, CausalForestDML and LinearDML shared direction in all 19 comparisons, whereas all three estimators shared direction in 18 of 19. Rank stability was very high in MIMIC and moderate to high in PhysioNet; magnitude disagreement was bounded but nonzero. The sole all-three sign exception remains PhysioNet shock. Agreement means that a pattern was not unique to one fitted estimator under a fixed representation and population; it does not establish equivalence, identification, or correctness.

9.9 Cross-Dataset Comparison

MIMIC-III's original causal-analysis population was substantially larger: 26,845 versus 7,993 records, a difference of 18,852 and a ratio of approximately 3.36. The larger analysis population can affect estimate stability, feasible model flexibility, and the amount of empirical support available; it does not make the MIMIC estimates automatically correct. The checked outcome rates also differed (0.121 in MIMIC and 0.142 in PhysioNet), but these rates describe the analyzed original-cohort model rows rather than equivalent raw-source cohorts.

Some proxy names suggest nominal alignment, including global severity, shock, respiratory failure, renal dysfunction, neurologic dysfunction, and metabolic derangement. Others are only

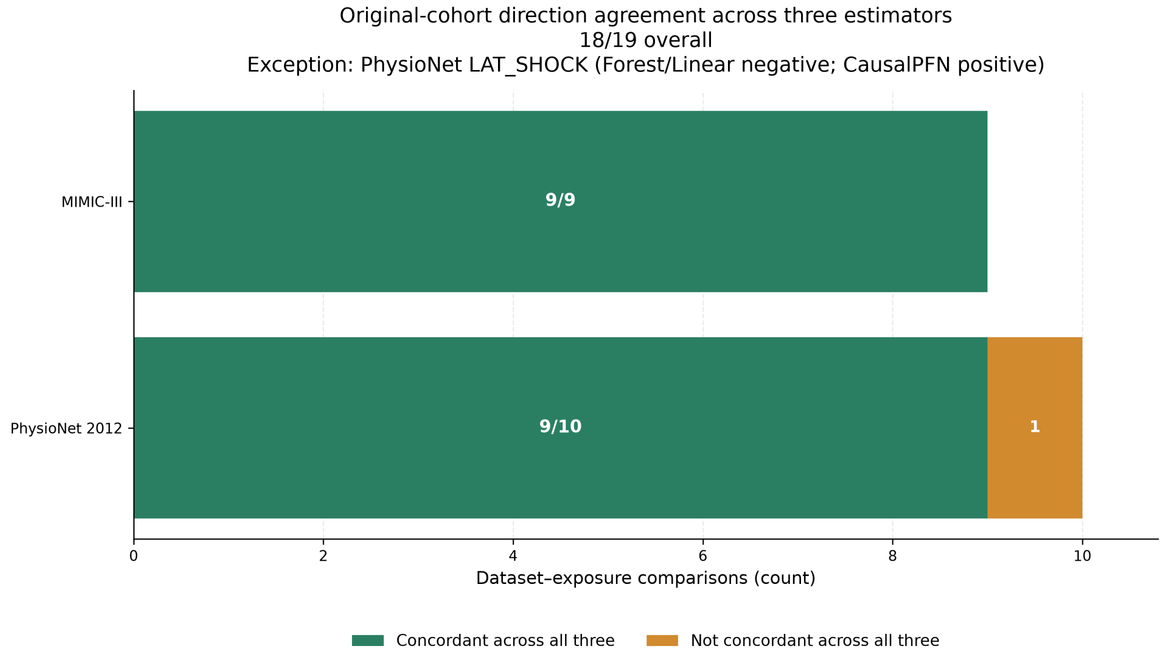

Figure 9.3: Directional concordance of the original-cohort mean model-estimated CATE across CausalForestDML, LinearDML, and exploratory CausalPFN. All three estimators shared the same direction for 18 of 19 dataset–exposure comparisons: 9 of 9 in MIMIC-III and 9 of 10 in PhysioNet. PhysioNet shock was the sole exception. Direction agreement does not imply equal magnitude, uncertainty, estimator equivalence, or causal validity.

approximately corresponding project-defined states: MIMIC combines hepatic and coagulation dysfunction while PhysioNet separates them, and the inflammation/sepsis and cardiac constructs use dataset-specific names and rules. For a descriptive cross-resource calculation, nine mappings were fixed: cardiac strain to cardiac injury/strain; inflammation/sepsis to inflammation/sepsis burden; hepatic/coagulation to hepatic dysfunction; and renal, global-severity, respiratory, neurologic, shock, and metabolic constructs to their similarly named counterparts. After averaging the three estimator means within each construct and resource, eight of the nine mapped constructs shared direction and their ranks had Spearman correlation $\rho = 0.533$. Shock was included and was the directional exception.

This moderate rank correspondence characterizes resource dependence, not failed replication or biological inconsistency. Similarly named states are not assumed to be construct-equivalent, and dataset era, measurement systems, variable availability, proxy rules, DAG structure, empirical support, and estimator behavior differ. No effects are pooled or averaged across datasets; only estimator means within a resource and mapped construct were averaged for the descriptive rank calculation. Detailed clinical interpretation follows in Chapter 10.

9.10 Clinical-Literature Comparison

Table 9.8 compares original-cohort model-estimated ranges with the nearest numerical mortality contrasts retained by the finished clinical-evidence supplement. These external studies differ from CliniCause in construct definition, severity, population, adjustment, and mortality horizon. They are contextual reference points, not ground-truth effects or a clinical validation set.

Table 9.8: Contextual clinical-literature comparison in percentage points (pp). Each CliniCause range spans CausalForestDML, LinearDML, and CausalPFN point estimates on an original cohort; it is a range across fitted estimators, not a confidence interval. MIMIC and PhysioNet remain separate. Published contrasts are heterogeneous and are not causal ground truth.

Construct	CliniCause MIMIC / PhysioNet	Nearest published contrast and design/horizon	Reading and principal limitation
Renal / AKI	8.7–9.1 / 9.1–12.0	8.1; propensity-score matched; 30-day mortality [35]	Broadly compatible; proxy and AKI definitions and horizon differ.
Hepatic / bilirubin	8.3–9.8 / 6.0–10.7	7.4; matched hospital-mortality contrast [36]	Overlapping; same MIMIC source and 2-mg/dL threshold, but an independent team and a narrower exposure.
Cardiac injury	18.8–25.9 / 11.2–12.2	17.0 unadjusted [37]	Broadly compatible in scale; populations and injury definitions differ.
Inflammation / sepsis	14.9–16.1 / 6.7–7.2	5.0 matched [38]	Lower than both CliniCause ranges; constructs, populations, and horizons differ.
Global severity	6.2–8.0 / 10.5–10.8	No defensible directly comparable numerical reference	No numerical corroboration claim is available.
Shock	2.1–3.2 / −2.7–0.4	11.1; unadjusted, other (90-day) horizon [39]	Discrepant and not commensurate; PhysioNet includes negative DML and slightly positive PFN estimates.
Coagulation / hematologic	— / 0.4–2.5	19.5 matched [40]	Discrepant; severe thrombocytopenia is narrower, and MIMIC has no separate proxy.
Respiratory	3.6–5.0 / 6.3–6.5	15.0, targeted minimum loss-based estimation [41]; other horizon	Discrepant; ARDS definitions and horizons are not equivalent to the proxy.
Neurologic	3.2–3.5 / 7.6–8.2	0.9; marginal structural model, other horizon [42]	Discrepant; delirium is not the same construct.
Metabolic	1.8–3.1 / 7.8–9.1	31.0; unadjusted hospital-mortality contrast [43]	Discrepant; lactic acidosis is more severe than the proxy.

The renal/AKI, hepatic/bilirubin, and cardiac-injury comparisons are broadly compatible or overlap in magnitude. Inflammation/sepsis, shock, coagulation/hematologic, respiratory, neurologic, and metabolic comparisons are materially discrepant, while global severity lacks a defensible numerical comparator. The bilirubin study is an independent-team analysis but uses MIMIC-III and the same 2-mg/dL threshold represented inside the broader CliniCause rule; it is not independent-population replication. These patterns identify representation and transport questions without diagnosing the cause of any discrepancy.

9.11 Robustness and Sensitivity

9.11.1 Outcome-downsampled analyses

Outcome downsampling retained every outcome-positive record and sampled outcome-negative records. It produced 6,486 MIMIC rows and 2,276 PhysioNet rows, each with a sample outcome rate of 0.500. These populations differ from the original cohorts, so their means are neither pooled nor treated as estimates of the same population quantity.

Across 57 paired original and downsampled estimator–exposure comparisons, direction was preserved in 55. CausalForestDML retained direction for all 19. LinearDML changed only for PhysioNet coagulation/hematologic dysfunction, from 0.0041 to −0.0174; CausalPFN changed only for PhysioNet shock, from 0.0041 to −0.0413. All paired means changed numerically. This does not strengthen causal evidence because the population changed.

Matching availability did not change: MIMIC again had eight successful summaries and one failure, while PhysioNet had seven successful summaries and three failures. Pair counts, match rates, distances, and descriptive differences nevertheless changed under downsampling. Detailed downsampled values remain supplementary.

9.11.2 Sensitivity, permutation, and support diagnostics

For both CausalForestDML and LinearDML, the validated sensitivity family was partial for eight of nine MIMIC exposures and seven of ten PhysioNet exposures in each sampling condition. It failed for MIMIC inflammation/sepsis and for the three PhysioNet exposures whose adjustment records were empty: inflammation/sepsis burden, neurologic dysfunction, and respiratory failure. Available rows combine saved-training-direct values with later recomputation and custom reconstruction from residual parameters; the planned real benchmark extraction is marked unimplemented by design. These source classifications prevent the values from being presented as a single estimator-native family. CausalPFN sensitivity was intentionally skipped and is not equivalent to a zero diagnostic.

Treatment- and outcome-permutation aggregate rows were available for every DML dataset–exposure–sampling combination, with ten archived trials and seed 42 in each row. In the original cohorts, an observed estimate lay more than two permutation standard deviations from its corresponding permuted mean in all 36 of 36 MIMIC DML comparisons and 34 of 40 PhysioNet DML comparisons. One comparison is one exposure under one DML estimator and one disruption type (treatment or outcome); the six nonpassing PhysioNet comparisons were concentrated in shock and coagulation/hematologic dysfunction. The strict $|z| > 2$ count is a disruption sanity diagnostic, not a p-value, formal randomization test, or identification proof. CausalPFN permutations were intentionally skipped.

Available DML sensitivity evidence shows that broad estimator agreement can coexist with fragility under specified omitted-confounding scenarios, but the archived native, reconstructed, partial, failed, skipped, and unavailable evidence classes do not support one unified numerical threshold or exposure count. Sensitivity therefore limits what may be inferred from convergence even when signs and rankings are stable.

Table 9.9 summarizes availability rather than diagnostic magnitude. Matching remains indirect, greedy, binary-representation-dependent support evidence. No dedicated overlap or propensity figure, pre/post-match balance table, or empirical proof of positivity was available. The principal warnings are the four exposure-specific matching failures per sampling-condition pair, three successful original rows with insufficient-pair flags, and the absence of an equivalent PFN diagnostic family.

Table 9.9: Robustness and diagnostic availability. Counts are exposure-level statuses, not effect estimates.

Dataset	Estimator	Sampling	Matching	Sensitivity	Permutation
MIMIC	Forest	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Original	8 available; 1 failed	Intentional skip	Intentional skip
MIMIC	Forest	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Downsampled	8 available; 1 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Original	7 available; 3 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Downsampled	7 available; 3 failed	Intentional skip	Intentional skip

9.11.3 Result provenance boundary

The checked hashes and source paths support arithmetic transcription of the admitted archive rows, but they do not remove the archive's reproducibility limitations. The `final-results/` tree is ignored or untracked in the main repository; exact numbered causal configuration files remain unavailable locally; some producing commits and archive-copy histories are incomplete; predictive split and checkpoint-to-export lineage is incomplete; and raw and processed data cannot be reproduced from the repository alone. These limitations qualify reproducibility claims, not the arithmetic correspondence between the checked rows and the tables above.

9.11.4 Converging-evidence synthesis

Taken together, the two testbeds contain inspectable constructs that have not been clinically validated; rule-derived labels that are recoverable from irregular records; broadly stable estimator signs and bounded rank/magnitude disagreement; and clinical comparisons that retain both overlap and discrepancy. Matching failures, insufficient-support flags, PhysioNet shock, heterogeneous sensitivity provenance, and incomplete computational lineage remain visible rather than being converted into favorable evidence. Convergence across some axes is therefore useful for characterizing a fixed observational resource, but provenance, support, and sensitivity prevent that convergence from being interpreted as causal or clinical truth.

Chapter 10

Discussion

This chapter interprets the two observational causal-analysis testbeds characterized in Chapter 9 within the study design and assumptions established in Chapters 3–8. The central distinction is between evidence that characterizes a fixed observational resource and evidence that would score an estimator against known causal truth. The latter is unavailable here. Prediction results, descriptive matched-pair comparisons, model-estimated contrasts, clinical reference points, and diagnostics are therefore kept non-equivalent and bounded by their sources.

10.1 Answering the Research Questions

10.1.1 Main Research Question

The main research question asked: *How can clinically grounded observational causal-analysis testbeds be constructed from source-observed irregular ICU records—without simulating patient records, treatment or exposure-assignment mechanisms, or outcomes—and what converging evidence can characterize their usefulness and limitations for evaluating causal estimators?*

CliniCause constructed two resources separately from MIMIC-III and PhysioNet 2012. Source-recorded measurements and mortality pass through explicit preprocessing and identifier contracts, while project-selected proxies, rules, cohorts, DAGs, adjustment assumptions, estimator interfaces, and provenance form an explicit representation layer. A schema-bounded LLM-assisted protocol operated only at design time; it did not access patient records, assign exposures, simulate outcomes, or estimate effects. The integrated interfaces make the resources inspectable.

Because neither resource contains a counterfactual answer key, converging, non-equivalent evidence characterizes rather than scores it. Recoverability, estimator agreement, clinical comparison, support, disruption, outcome-downsampling, sensitivity, and provenance show persistence and failure modes under a fixed representation. Agreement is not accuracy; the resources support retrospective evaluation and hypothesis generation, not clinical validation, treatment recommendation, or deployment.

10.1.2 SRQ-1—Representation Instantiation

The representation designs began with dataset schemas, the bounded literature corpus, and structured design-time LLM assistance. Project selection converted candidate ontologies, rule families, missingness considerations, and DAGs into deterministic source. The final rule code and stable `LAT_*` fields make the implemented proxy states inspectable and replaceable; they are not diagnoses or discovered biological variables. No patient record was supplied to the LLM, and the execution path used no runtime LLM call.

Instantiation remained dataset specific. MIMIC-III and PhysioNet expose different variables, frequencies, missingness processes, ontologies, thresholds, and graph structures, so similarly named proxy states are not assumed to be equivalent. The available construct evidence is plausibility only: no chart adjudication, blinded multi-clinician rating, inter-rater analysis, clinician-only comparison, or clinical validation was performed. Separate construction and the no-pooling policy preserve those differences instead of converting nominally similar labels into one common exposure.

10.1.3 SRQ-2—Contracts, Aggregation, DAG, and Provenance

The causal repository uses `[ts, oc, ts_ids]`, while the predictive workflow uses a split-aware artifact with explicit train, validation, and test identifiers. Prediction exports add paired probability and binary fields followed by binary-only voter files. Boundary checks prevent silent identifier, split, schema, and cohort conflation, although processed-pickle hashes, producing commands, and final split lineage remain incomplete.

Majority voting aligns identifiers, checks identical binary schemas, and applies the implemented tie rule to one rule-derived source and four thresholded predictions—GRU, GRU-D, STraTS, and TCN. This fixed algorithmic aggregate is not clinical consensus or a learned source-quality model; shared rule error can persist because all learned voters target the same rules and the rule source itself participates [10].

Dataset-specific source-coded DAGs define exposure-specific observed adjustment sets from available columns. They make the project's assumptions inspectable but were neither learned nor clinically validated, and a computed set does not prove exchangeability or the absence of unmeasured confounding [5]. Source paths, status classes, manifests, and checked tables expose the lineage and diagnostic gaps. These contracts, aggregation rules, graph interfaces, and provenance records are enabling mechanisms for testbed integrity; they are not the primary intellectual object and do not establish scientific validity by themselves.

10.1.4 SRQ-3—Recoverability and Mortality-Relevant Information

The four-model comparison shows that rule-derived proxy labels are recoverable from irregular source records, with archived AUROC spanning 0.867–0.918. STraTS led MIMIC and GRU-D led PhysioNet; changing lower-order rankings preclude a universal ordering. Recoverability depends on the measurements, missingness, labels, and representation, and the local tasks do not reproduce the published benchmark tasks exactly [1, 2, 3].

The admitted MIMIC-only proxy-vector analysis adds mortality-relevant predictive information: logistic regression and the MLP reached test AUROC 0.826 and 0.831, respectively, for source-observed in-hospital mortality. This is prognostic association, not causal evidence. The analogous PhysioNet number remains withheld because checked-thesis and paper voter/cohort lineages conflict. That asymmetry is a provenance limitation to preserve, not a result to reconcile rhetorically. No uncertainty intervals, paired between-model tests, or clinical validation of the targets were available.

10.1.5 SRQ-4—Converging Evidence

Estimator triangulation and diagnostics. The hierarchy uses original cohorts as primary, CausalForestDML as the primary estimator, LinearDML as secondary, CausalPFN as exploratory, and outcome-downsampled analyses as robustness evidence. The implemented representation and adjustment procedure produced positive primary forest means for all nine MIMIC exposures and nine of ten PhysioNet exposures, with shock negative. These are construct- and resource-dependent model-estimated contrasts, not evidence that a proxy caused death, that changing it would change mortality, or that a ranking identifies intervention priorities.

CausalForestDML and LinearDML agreed in direction for all 19 original-cohort dataset–exposure pairs. Across all three estimators, agreement was 18 of 19. Pairwise within-resource rank correlations were 0.983–1.000 in MIMIC and 0.794–0.964 in PhysioNet, while pairwise RMSE was 1.35–2.57 and 1.08–2.04 percentage points, respectively. These are distinct views of stability, not measures of estimator accuracy. They show that most patterns were not unique to one fitted estimator under a fixed representation and population; they do not validate the DAG, establish equal estimands or uncertainty, or rule out residual confounding.

Within this pipeline, CausalPFN was a useful complementary estimator because it reproduced the prevailing direction across nearly all prespecified comparisons despite using a different modeling approach. This result is bounded: magnitudes were not identical to DML magnitudes, and its archived sensitivity and permutation stages were intentionally absent. CausalPFN therefore remains exploratory rather than an estimator with the same diagnostic support as the DML analyses.

PhysioNet shock is the central exception rather than a detail to be averaged away. CausalForestDML and LinearDML were negative, CausalPFN was slightly positive, and the descriptive matching summary was positive. The external shock contrast is not commensurate in construct, population, or horizon. This combination warns about representation, support, estimator, and population dependence. No estimator vote resolves it, and no protective clinical effect should be inferred from the negative DML estimates.

Matching provides a descriptive baseline and empirical-support warnings. Fourteen of 15 successful original-cohort summaries shared the primary forest direction; failures, insufficient-pair flags, and varying Hamming distances remain visible. Unchanged matching availability across original and downsampled populations documents that availability pattern but proves neither positivity nor balance. Failure is missing support, not a zero estimate, and a matched-pair difference is not automatically an ATT or independent causal confirmation.

Sensitivity and permutation evidence adds layers rather than closure. At the strict $|z| > 2$ disruption threshold, all 36 MIMIC DML comparisons and 34 of 40 PhysioNet DML comparisons passed; these are sanity checks across exposure, DML estimator, and treatment/outcome permutation, not p-values or identification tests. DML sensitivity outputs were saved-training direct, reconstructed, partial, or failed depending on the exposure, and reconstructed diagnostics are not estimator-native calculations. The evidence classes therefore support the qualitative conclusion that broad agreement can coexist with omitted-confounding fragility, but not one unified numerical threshold or exposure count. CausalPFN diagnostics were intentionally skipped. Dedicated propensity, common-support, and balance plots were not archived; positivity remains unestablished [27, 28, 29]. Agreement and fragility can coexist, and sensitivity prevents estimator convergence from being treated as closure.

10.1.6 Cross-Dataset Interpretation and Robustness

The original causal-analysis populations contained 26,845 MIMIC records and 7,993 PhysioNet records, so the MIMIC analysis population was approximately 3.36 times larger. A larger analysis population can provide more information for nuisance-model fitting and modeled effect heterogeneity, and it may contribute to numerical stability. It does not demonstrate better data quality, make the MIMIC findings more valid, or erase differences in era, variables, measurement systems, missingness, proxy rules, DAGs, or patient composition. Predictive training and causal analysis also use distinct artifact contracts, so their populations should not be assumed identical merely because some row counts align.

Operation of the same representation, prediction, aggregation, graph-guided adjustment, and estimator sequence on both datasets demonstrates workflow portability. It does not make similarly named states equivalent. Across nine approximate mappings including shock, three-estimator-average ranks had moderate correspondence ($\rho = 0.533$) and eight mappings shared direction. Dataset-specific rules and graphs and MIMIC's combined domains make the no-pooling policy essential; cross-resource differences characterize contextual dependence, not pooled effects or biological inconsistency.

Outcome-based downsampling retained outcome-positive records and sampled outcome-negative records, changing both population and prevalence. Direction was preserved in 55 of 57 matched comparisons, documenting broad sign stability under this specified perturbation, but every paired mean changed and two PhysioNet comparisons changed sign. The procedure is not transport or reweighting; the means need not share a population target, and stability does not validate identification.

The clinical-literature comparison must be interpreted symmetrically. Renal/AKI, hepatic/bilirubin, and cardiac injury were broadly compatible or overlapping, while inflammation/sepsis, shock, coagulation/hematologic, respiratory, neurologic, and metabolic were discrepant; global severity had no defensible numerical comparator. External definitions, severity, populations, adjustments, and horizons differ. Agreement is contextual corroboration, not confirmation, and no external value is commensurate causal ground truth for a CliniCause proxy.

10.1.7 Relationship to Prior Work and Methodological Implications

The primary contribution is the construction and characterization of two observational causal-analysis testbeds. Integrated representation, prediction, aggregation, DAG adjustment, estimator, diagnostic, and provenance interfaces make them inspectable by preventing silent identifier, split, cohort, and construct conflation and retaining both agreement and failure modes.

CliniCause complements controlled benchmarks. Answer-key resources score error under specified mechanisms; observational testbeds retain source-recorded measurements, missingness patterns, care-process traces, covariates, and outcomes while exposing representation-dependent support, estimator, and provenance behavior without measuring causal error against truth.

This framing follows target-trial and well-defined-intervention principles [6, 7] and ICU guidance on question definition, confounding, data preparation, and reporting [15]. These sources provide context, not validation of local labels, graphs, estimates, or execution; CliniCause remains an auditable research workflow rather than a validated clinical methodology.

10.1.8 SRQ-5—Limitations and Appropriate Use

The proxy states summarize clinically inspired patterns and are analytically useful for organizing retrospective evidence. They may support hypothesis generation and help identify proxy-state contrasts that merit targeted clinical and causal investigation. High prediction performance against rule labels does not establish clinical accuracy, however, and a positive adjusted proxy-state contrast does not imply that clinicians should attempt to manipulate the proxy directly. The system did not undergo prospective validation, external clinical validation, or clinical impact analysis. Its causal quantities are not ready for bedside decision support, and no patient-level treatment recommendation follows from the results.

Principal limitations. The resources have no known causal truth and are *empirically supported with substantial unresolved boundaries.* The canonical construct, identification, support, generalizability, provenance, LLM-design, ethics, and deployment limitations follow.

10.2 Limitations and Threats to Validity

10.2.1 Construct and Measurement Validity

The principal construct limitation is that clinically inspired rules approximate conditions but were not evaluated against chart-adjudicated diagnoses. Available evidence consists of active rule code, binary artifact schemas, and limited descriptive validation summaries. This makes the constructs transparent but does not establish their sensitivity, specificity, or equivalence across datasets. Prediction can add classification error, and the mixed-source majority vote can preserve shared systematic error among models trained on the same rules. Existing mitigations are deterministic definitions, source-code traceability, dataset-specific naming, and explicit proxy terminology; the residual risk remains high because clinical review and reference-standard validation are absent.

Binary exposure states also compress severity, duration, and timing. A positive tag may combine clinically distinct trajectories, while a negative tag can reflect missing measurement rather than absence of a condition. Proxy onset relative to covariates and mortality is not fully established, so exposure misclassification and temporal ambiguity can bias both prediction and adjusted analyses. The processed outcome, `in_hospital_mortality`, is consistently required by the software, but its producing-artifact provenance is incomplete, and no competing-risk or post-discharge mortality analysis was performed.

The observation process is itself informative. Measurement frequency may reflect clinical concern, resource use, or care pathways; models that encode masks or time gaps can exploit this signal without removing its causal implications. Preprocessing and missingness-aware representation are therefore not equivalent to adjustment for the decision to measure. This concern can affect proxy construction, predictive comparisons, and effect estimation simultaneously.

10.2.2 Causal Identification and Estimand Validity

These are the central constraints on causal interpretation. Illness-state proxies are not necessarily well-defined interventions, so consistency and the meaning of contrasting exposure states may be difficult to specify. No target-trial emulation was performed, and the pipeline does not implement a method for time-varying treatment–confounder feedback. Exchangeability depends on a project-specified DAG, correct temporal ordering, and measured variables; topology alone cannot establish patient-record timing. Some archived observed adjustment sets are empty, every checked identifiability field remains unresolved, and unmeasured or misspecified confounding remains possible.

Positivity was assumed rather than established. Matching availability, failures, pair counts, and distances provide indirect evidence, but the archive lacks dedicated propensity distributions, common-support plots, and pre/post-match balance summaries. Interference is also assumed, not tested, and repeated admissions or shared ICU processes may complicate unit-level independence.

Estimand language remains deliberately restricted. The mean model-estimated CATE over the analyzed sample is an aggregate of model outputs, not automatically a population ATE. The descriptive matched-pair outcome difference is not automatically an ATT. Normalized CATE was omitted because scaling by the sample outcome rate is unstable across populations and does not create a risk ratio or another standard clinical estimand. These boundaries mitigate overstatement but do not resolve the underlying intervention-definition and identification problems.

10.2.3 Statistical, Modeling, and Support Limitations

Many displayed main estimates lack uncertainty intervals, no formal multiple-comparison strategy was documented, and this thesis did not add significance tests. Absence of an interval cannot be interpreted as either significance or insignificance. Model selection, nuisance-model adequacy, regularization, hyperparameters, and cross-fitting behavior remain assumptions whose impact was not exhaustively tested. CausalForestDML, LinearDML, and CausalPFN do not provide interchangeable uncertainty, and directional agreement can coexist with material magnitude and ordering differences.

Support evidence is incomplete and representation dependent. Some matching runs failed, some successful matches required larger Hamming distances, and some carried insufficient-pair warnings. Sensitivity outputs were partial or failed for several exposure families; reconstructed or fallback diagnostics have different status from estimator-native outputs. Permutation aggregates are limited sanity checks rather than formal inferential tests. CausalPFN lacks the corresponding archived downstream diagnostic family. The multi-estimator and diagnostic hierarchy exposes these risks, but it cannot remove them.

10.2.4 External Validity and Generalizability

MIMIC-III and PhysioNet differ in size, era, variables, measurement practices, and patient composition. Their proxy ontologies and DAGs are dataset specific, and nominally similar states are not necessarily equivalent. Numerical estimates were therefore not pooled. Operation of the workflow on both sources is positive evidence of engineering and methodological portability, especially given the inclusion of several predictive and causal model families. It is not evidence that the estimates transport to other hospitals, countries, care practices, EHR systems, or time periods.

No prospective validation, independent external clinical validation, or formal transport analysis was performed. Subgroup performance and effect stability by age, gender, ICU type, or other characteristics were not evaluated as a fairness study. Consequently, the analysis does not establish clinical generalizability or subgroup equity.

10.2.5 Reproducibility and Provenance

The validated result manifest, source tables, source paths, and checksums provide strong numerical-transcription support for the values reported in Chapter 9. They also improve artifact traceability by distinguishing canonical sources, duplicates, and excluded artifacts. This is not the same as computational rerun reproducibility. The `final-results/` archive is ignored or untracked in the main repository; exact numbered causal configurations are unavailable; result-generating source versions and copy histories are incomplete; and the nested causal repository contains local modifications.

Predictive split provenance, checkpoint-to-export mapping, and copied-output lineage remain incomplete. Raw and processed datasets are external, exact processed-pickle hashes and producing commands are unavailable, and some paths are absolute to the producing machine. Final software, environment, and hardware records are also incomplete. Thus the available records support archived numerical transcription and materially improve artifact traceability, while a clean-checkout rerun is not currently demonstrated. Scientific reproducibility—replication of findings under independently reconstructed data, definitions, and analysis—is weaker still and requires new evidence rather than additional formatting of the archive.

10.2.6 LLM-Assisted Design Layer

The LLM-assisted design layer contributed structured proposals for proxy concepts, binary decision rules, missingness considerations, and dataset-specific DAGs. Potential advantages are the systematic elicitation of domain concepts, scalable generation of candidate proxy definitions and rules, parallel dataset-specific proposals, explicit prompt artifacts that improve design provenance, and conversion of otherwise unstructured reasoning into inspectable source-coded artifacts. These advantages concern organization, traceability, and design productivity; the thesis did not compare them with clinician-only construction or measure whether LLM assistance improved clinical or causal correctness.

The main design risks include hallucinated or unsupported clinical assertions; sensitivity to prompt wording, supplied context, model version, and hosted-model updates; unknown system, browsing, and decoding settings; incomplete reproducibility of hosted-model behavior; and possible dependence on hidden training data. Project selection can introduce anchoring and confirmation bias, especially when plausible generated explanations are mistaken for independent evidence. The prompt record does not establish complete accepted/rejected decisions, and only high-level rather than line-by-line prompt-output-to-code traceability is available.

Validation gaps remain substantial. There is no documented clinician-panel adjudication of every proxy definition or graph edge, no chart-level proxy validation, no comparison of LLM-assisted with clinician-only or hybrid construction, no replication with another LLM family, and no prompt-perturbation or model-version sensitivity experiment. Error can therefore propagate from an unsupported design proposal into a deterministic label, from that label into a prediction and aggregate, and from a misspecified exposure or graph into adjustment and effect estimation. Preserved prompts and explicit source-code authority make the chain inspectable but do not validate it.

Predictive agreement with LLM-assisted rule-derived labels validates neither the clinical construct nor the LLM-generated design. Sensitivity and permutation diagnostics do not validate the proxy ontology or DAG. No claim is made that an LLM provides authoritative clinical judgment, discovered the true causal graph, or can replace a clinician or causal expert.

10.2.7 Ethical and Clinical-Deployment Considerations

Misclassification can misrepresent a record's clinical condition, amplify measurement bias, and generate misleading exposure contrasts. Groups with different testing intensity or access to measurements may be affected disproportionately. Automation bias is therefore a direct risk: users could overread model probabilities as diagnoses, proxy-state rankings as clinical priorities, adjusted estimates as treatment recommendations, or estimator agreement as causal certainty.

No dedicated fairness audit was performed. The presence of age, gender, ICU type, or other variables in an adjustment set does not constitute fairness validation, and subgroup predictive performance and effect stability remain untested. The repository does not establish the project-specific institutional approval, consent or waiver procedure, data-use agreement, or governance determination. These facts must be supplied from authoritative institutional records and must not be inferred from dataset access, de-identification, or repository contents.

The deployment boundary is unambiguous: *the system is a retrospective research workflow and is not suitable for patient-level clinical deployment or treatment recommendation in its current form.* No prospective safety, workflow, impact, or human-factors evaluation was performed. This limitation is not cured by predictive accuracy, estimator agreement, or artifact traceability. Taken together, construct measurement, intervention definition, graph validity, temporal ordering, unmeasured confounding, support, uncertainty, transport, provenance, LLM-design reproducibility, fairness, and clinical validation remain the decisive threats to stronger interpretation.

Chapter 11

Conclusions and Future Work

11.1 Summary of the Study

This thesis constructed and characterized two observational causal-analysis testbeds, instantiated separately from source-recorded MIMIC-III and PhysioNet 2012 measurements, covariates, care-process traces, and in-hospital mortality. It did not simulate patient records, exposure-assignment mechanisms, or outcomes. Instead, it made representation-defined proxy constructs, rules, cohorts, preprocessing, aggregation, empirical support, DAGs, adjustment assumptions, estimator interfaces, and provenance explicit.

Because neither resource contains a known causal answer key, characterization used converging, non-equivalent evidence rather than error against causal truth. Integration, four-model proxy prediction, five-source aggregation, matching, three-estimator comparison, diagnostics, and evidence tracking made the resources inspectable. Their proxy states remain analytical constructs rather than verified clinical conditions, and every adjusted result remains conditional on measurement, graph, intervention-definition, temporal-ordering, support, confounding, and modeling assumptions.

11.2 Main Findings and Contributions

The strongest contribution is a representation-centered framework for constructing observational causal-analysis testbeds, together with two concrete ICU instantiations whose assumptions and provenance are exposed. A shared proxy-state interface traces constructs from one rule-derived source through four predicted voters to separate dataset-specific exposure analyses, while causal and split-aware contracts keep identifiers, cohorts, and exports distinguishable. Integration remains an enabling contribution: it makes the representation, estimator interfaces, and evidence boundaries auditable, but it is not itself the principal scientific object. Checked sources and manifests improve numerical traceability without establishing clean-checkout reproduction.

The empirical predictive comparison covered STraTS, GRU, GRU-D, and TCN. STraTS led the archived proxy-label test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet

(Chapter 9, Section 9.3). The leading model was therefore resource dependent; recoverability did not establish a universal ordering or clinical validity. The separately admitted MIMIC proxy-vector result retained mortality-relevant predictive information, while the analogous PhysioNet number remained withheld because its lineage was unresolved.

In the primary original cohorts, CausalForestDML means were positive for all nine MIMIC-III exposures and nine of ten PhysioNet exposures; PhysioNet shock was negative. CausalForestDML and LinearDML shared direction in all 19 dataset–exposure comparisons, and exploratory CausalPFN agreed with both in 18 of 19. Within-resource rank correlations and pairwise RMSE showed bounded ordering and magnitude disagreement, while the nine-construct cross-resource ranking was only moderately correlated. PhysioNet shock remained the central exception across negative DML, slightly positive PFN, positive matching, and a noncommensurate external comparison.

The clinical comparison retained both broadly compatible contexts and material discrepancies. At the heuristic $|z| > 2$ disruption threshold, 36/36 MIMIC and 34/40 PhysioNet DML comparisons passed; this is not a p-value or formal randomization test. Matching failures, support warnings, outcome-downsampling changes, and heterogeneous sensitivity provenance prevented convergence from becoming closure. Agreement shows that a pattern is not unique to one fitted estimator under one resource; it does not establish that the representation, graph, estimand, or estimate is correct. The resources therefore complement controlled answer-key benchmarks while remaining unable to measure causal error against truth.

11.3 Limitations of the Conclusions

Construct validity is the first constraint: LLM-assisted, project-selected proxy rules lack chart adjudication; dataset ontologies and timing differ; prediction can add error; and mixed-source voting can preserve shared rule and measurement error. Clinical review, reference-standard evaluation, complete design-to-code decisions, and exact support for several rule families remain incomplete.

Identification and translation remain equally limited. Illness-state proxies are not necessarily well-defined interventions; the project DAGs are assumptions; temporal ordering, overlap, uncertainty, and unmeasured confounding remain unresolved; and diagnostics do not establish causal validity. The two datasets do not justify pooling or transport claims, and no prospective or external clinical validation, fairness or safety study, clinician-only comparison, alternative-LLM replication, or prompt-sensitivity analysis was performed. No clinical or causal truth, treatment recommendation, or deployment readiness follows. Ethics and governance wording and the data, configuration, split, checkpoint, source-version, environment, and archive records needed for full reproduction also remain incomplete.

11.4 Future Work

The first priority is clinical and construct validation. Future work should freeze the proxy rules and conduct blinded multi-clinician review of every proxy definition and graph edge, with reported inter-reviewer agreement. Rule-derived and predicted states should be compared with chart-adjudicated reference labels, including dataset-specific assessment of timing, missingness, thresholds, and differences among rule-derived labels, individual predicted labels, and mixed-source aggregated proxy labels. Controlled clinician-only, LLM-assisted, and hybrid construction should be compared, together with alternative LLM families, prompt perturbations, model-version changes, and decoding choices.

The second priority is a stronger observational and measurement design. A future study should specify a target-trial-aligned question with eligibility, time zero, intervention strategies, follow-up, outcome, estimand, and an explicit plan for time-varying confounding. It should add prespecified overlap and balance diagnostics, support-aware estimands, uncertainty and multiplicity procedures, and richer sensitivity analyses. Rule ablation and threshold-sensitivity experiments, formal label-noise modeling, and explicit measurement-error analyses should quantify how proxy construction, prediction, and aggregation affect downstream estimates. DAG review should proceed edge by edge and include alternative-graph and adjustment-set analyses rather than treating one encoded graph as fixed truth.

The third priority is reproducibility and external evaluation. Signed per-run manifests should link raw-data access records, processed-artifact hashes, split identifiers, checkpoints, commands, resolved configurations, software environments, source versions, and archived outputs. LLM design manifests should record complete prompts, system and decoding settings, model versions, follow-up interactions, reviewer decisions, and prompt-output-to-code mappings. Frozen definitions should then be evaluated on temporal, external, prospective, and additional held-out observational testbeds, including clinically justified subgroups. Fairness, governance, privacy, safety, workflow, and human-factors evaluation must precede any deployment consideration. Although its primary method source is now cited, CausalPFN should remain exploratory until the exact producing implementation version and checkpoint are verified and an appropriate diagnostic and uncertainty plan is available. No clinical deployment should occur before construct, causal-design, external, and prospective validation and authoritative ethics and data-governance review.

11.5 Closing Perspective

An LLM cannot replace a clinician or causal expert on the evidence presented here, and integration does not remove the need for careful observational design, diagnostics, and reproducible provenance. Transparent representation and provenance make observational testbeds inspectable. Their value lies not in supplying causal truth, but in exposing estimator stability, disagreement, representation-dependent support limitations, and sensitivity while retaining source-recorded measurements, missingness, care-process traces, covariates, and outcomes.

Appendix A

Reproducibility and Evidence Boundary

A.1 Current reproducibility interface

CliniCause now exposes a unified execution interface that can reconstruct causal-analysis resources from appropriately obtained raw inputs and execute the integrated predictive and causal workflow. The repository provides one canonical shell launcher, one project-wide dependency entry point, `requirements.txt`, dataset-isolated run directories, and explicit validation, cohort, artifact, receipt, and provenance contracts. This makes the current implementation and its reconstruction path substantially more reproducible without redistributing source ICU records.

A.2 Repository setup

On a fresh machine, clone the repository normally. The checkout contains the complete pipeline, including the `STraTS/` and `causal-irregular-time-series/` component directories:

```
git clone https://github.com/KenziVisor/CliniCause.git
cd CliniCause
```

The first component contains the temporal-model implementation; the second contains preprocessing, proxy construction, graph, and causal-analysis code. No separate repository initialization step is required.

A.3 Python 3.10 environment and dependencies

Create an isolated Conda environment, activate it, upgrade its packaging tool, and install the complete project through the root requirements file:

```
conda create -n clinicause python=3.10 -y
```

```
conda activate clinicause
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

The root file declares the parent orchestration dependencies and recursively includes the independently maintained requirement files of both component directories. It is the single supported installation interface for the current checkout; it is not an environment lock or proof of the exact software environment used to produce every archived thesis result.

A.4 Source-data access

MIMIC-III and the PhysioNet/Computing in Cardiology Challenge 2012 records are not redistributed with CliniCause. Users must obtain their own copies from PhysioNet and follow the access and license terms shown on the respective project pages. MIMIC-III is a restricted resource: access requires PhysioNet credentialing, the currently specified human-subjects training, and acceptance of its data-use agreement. The Challenge 2012 files are currently presented by PhysioNet as open access under their specified license; users must still comply with that license and verify the current project-page terms. The pipeline neither automates nor bypasses these processes, and restricted source records must remain under the authorized user's control.

A.5 Default and external data locations

After obtaining and extracting the sources in a form consistent with their official distributions, place each raw dataset root at the launcher default:

```
data/physionet2012
data/mimiciii
```

These paths denote the raw dataset roots consumed directly by the two source-specific preprocessors; they are not locations for CliniCause-derived outputs. To keep protected data outside the repository, supply absolute paths:

```
export PHYSIONET_RAW_DATA_PATH=/absolute/path/to/physionet2012
export MIMIC_RAW_DATA_PATH=/absolute/path/to/mimiciii
```

The launcher accepts absolute paths. It resolves relative values, including the defaults, from the CliniCause repository root rather than from the caller's current directory.

A.6 Constructing the reusable datasets

From the repository root and the activated environment, run:

```
STAGES=dataset-extraction bash run_clinicause.sh
```

The default dataset selector is `both`. This preset executes the implemented preprocessing, deterministic tagging, tree generation, STraTS input preparation and model stages, prediction collection and normalization, then terminates the causal program at graph construction and deterministic majority vote. Its terminal resource artifacts are the dataset-specific causal graph and the aggregated latent-tag table, accompanied by a run summary. Thus the preset reconstructs the reusable resource through the implemented aggregation boundary; it does not run the later causal estimators.

By default, output is isolated below `runs/<run-id>/`, in separate `physionet/` and `mimic/` children, with coordinator metadata under the same run root. Dataset manifests, resolved configurations, stage logs, provenance sidecars, cohort fingerprints, artifact hashes, and per-stage receipts record and validate the current execution. A generated run identifier is used unless `RUN_ID` is set.

External raw-data paths can be applied to this command without copying the records into the checkout:

```
PHYSIONET_RAW_DATA_PATH=/absolute/path/to/physionet2012 \
MIMIC_RAW_DATA_PATH=/absolute/path/to/mimiciii \
STAGES=dataset-extraction \
bash run_clinicause.sh
```

A.7 Running the complete pipeline

The canonical complete run is:

```
STAGES=all bash run_clinicause.sh
```

This retains the full integrated downstream predictive and causal workflow rather than stopping at the dataset-construction boundary. It is computationally heavier, and the temporal-model stages may require suitable accelerator resources. Optional launcher controls include `DATASET=physionet` or `DATASET=mimic` for a single source, `OUTPUT_ROOT`, `RUN_ID`, the two raw-data path variables, `STRATS_MAX_CONCURRENT`, and `PYTHON_BIN`; the two commands above remain the canonical interfaces.

A.8 Reproducibility boundary and limitations

Three claims remain distinct. The current checkout provides a unified dependency and execution interface with validated artifact contracts, and authorized users can reconstruct derived resources from separately obtained ICU inputs. Neither claim establishes exact reproduction of the archived thesis results: producing revisions, numbered configurations, splits, checkpoint-to-export links, raw/processed manifests, and complete hardware/software records remain incomplete, and runtime can vary across systems. Software reproducibility also does not establish clinical construct validity or causal identification.

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מוגש לשם קבלת תואר "מגיסטר" בפקולטה להנדסה

קובי קנזי

בהנחיית: ד"ר דן וילנצ'יק